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From Salience to Retail Herding

Origination and Confirmation Signals in Robinhood Trading, Google Search Activity, and U.S. Equity Returns

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Abstract

This thesis studies how salient market and platform signals shape retail investor attention, Robinhood herding and short-run return dynamics. Using Robintrack, TAQ retail-flow estimates, Google Search Volume Index, CRSP returns, and RavenPack news analytics for 8,597 U.S. equities from May 2018 to August 2020, the thesis examines whether the sequence from salience to information acquisition, concentrated retail participation, and return reversal can be traced across complementary empirical designs. The central contribution is to distinguish between *origination signals*, such as extreme returns, abnormal volume, news, and platform visibility, and *confirmation signals*, especially abnormal Google search volume. Origination signals make stocks salient and draw them into retail investors' consideration sets, while confirmation signals indicate that salience has translated into active information acquisition. Empirically, Robinhood buying is more concentrated than broader TAQ retail buying, with the ten most heavily bought stocks accounting for 34.5% of daily Robinhood net buying compared with 23.9% in TAQ retail data. Top Movers eligibility is associated with local increases in Robinhood participation and abnormal search activity (suggesting the validity of the origination-confirmation signals classification), and intense herding events are followed by short-run return reversals, including a 20-day buy-and-hold abnormal return of -3.71%. The evidence is consistent with temporary price pressure around concentrated retail participation, but it does not identify a single causal chain from attention to prices. The analysis is limited by the coverage of Google Trends data tilted towards larger stocks, which restricts tests of the attention-herding-return mechanism among smaller tickers where retail price influence is strongest.

Keywords: Retail Investor Attention; Robinhood; Google Trends; TAQ; Behavioral Finance; Empirical Asset Pricing

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List of Abbreviations

AR	Abnormal return
ARA	Aggregate retail attention
ASVI	Abnormal search volume index
BHAR	Buy-and-hold abnormal return
CRSP	Center for Research in Security Prices
CSS	Composite sentiment score
ESS	Event sentiment score
FE	Fixed effects
LPM	Linear probability model
OLS	Ordinary least squares
RD	Regression discontinuity
SVI	Search volume index
TAQ	Trade and quote database
WRDS	Wharton Research Data Services

I. Introduction

Over the past decades, digitalization and the rise of financial-technology brokerage platforms have transformed the experience of retail investing. Whereas earlier generations of individual investors traded through brokers or less intuitive desktop interfaces, recent fintech platforms have reduced barriers to stock-market participation by making trading cheaper, faster, and more accessible through mobile applications (Rao & Bhanotu, 2025). Robinhood is one of the most prominent examples of this shift. Its commission-free and gamified interface lowered entry barriers for a new population of young and relatively inexperienced retail investors[1]. Yet these digital interfaces have also changed the environment in which investment decisions are made. A mobile trading app does not merely provide access to markets; it filters, ranks, and displays information. This dual role, as both trading infrastructure and attention-shaping interface, raises a central question: how do mobile brokerage platforms affect retail trading behavior?

A natural starting point is the attention-based theory of retail trading. Barber and Odean (2008) argue that individual investors face fundamentally different problems when buying and selling stocks. Selling decisions are typically made over a narrow set of securities already held in the investor's portfolio, as short selling is rare or restricted among individual investors[2]. Buying decisions, by contrast, require investors to search across thousands of potential candidates. This asymmetry is economically important because attention is scarce (Kahneman, 1973, 2011). Indeed, investors cannot evaluate the full universe of tradable stocks before each purchase decision; they therefore tend to restrict their buying consideration set to stocks that have first caught their attention, such as stocks with extreme returns, unusual trading volume, or news coverage (Ardia et al., 2023; Barber & Odean, 2008; Hou et al., 2009). In this framework, attention does not necessarily determine which stock an investor ultimately buys; rather, it determines which stocks are considered in the first place.

The existing evidence suggests that Robinhood is a particularly informative setting in which to study this mechanism. As indicated above, the majority of Robinhood users are young and relatively inexperienced investors; they have therefore probably not developed sophisticated investment strategies. Since these inexperienced investors are more significantly affected by attention (Seasholes & Wu, 2007), speculate, and trade

[1] *The New York Times* reported that the average Robinhood user is 31 years old and that approximately one-half of Robinhood users have no prior investing experience; see https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html. For further details, see subsection 3.2.1.

[2] In the large discount-brokerage sample studied by Barber and Odean (2008), the mean household held a monthly average of 4.3 stocks, while the median household held 2.61 stocks. Short positions represented only 0.29% of positions, or 0.78% when value-weighted.

more frequently due to overconfidence (Barber & Odean, 2001), they likely constitute a population among which attention-induced phenomena are particularly pronounced. Moreover, Barber et al. (2022) show that Robinhood users display more concentrated buying than the broader population of retail investors, and that their herding behavior is predicted by attention measures[3]. Numerous studies also document that retail investors (in particular Robinhood users) react to visually salient platform signals (Barber et al., 2022; Chen et al., 2023; Glaze & Moss, 2026; Stein, 2020). Related work further shows that retail attention, measured through Google search activity, is associated with short-run price pressure and subsequent reversals (Da et al., 2011), while recent evidence links heightened social media attention on the subreddit `r/wallstreetbets` to Robinhood trading behavior (De Giorgi et al., 2026; Eaton et al., 2022; Fedyk, 2021; Münster et al., 2024).

Building on these premises, this thesis contributes to the literature by studying the internal transmission mechanism of attention-induced retail trading. Rather than treating investor attention as a single reduced-form object, I distinguish empirically between *origination signals* and *confirmation signals*. Origination signals, such as extreme returns or high platform visibility, are salience cues that can draw a stock into retail investors' consideration sets. Confirmation signals, such as abnormal Google search volume, reveal that this initial salience has translated into actual attention and active information acquisition. The central question of this thesis is whether this sequence can be traced empirically: from salient signals, to search attention, to concentrated Robinhood participation, and finally to short-run return reversals consistent with temporary price pressure.

To address this matter, the thesis leverages a triangulation of different empirical analyses and combines five complementary data sources. The Robintrack website[4] provides data on aggregate Robinhood users holding positions at the stock–day level from May 2018 to August 2020. The Center for Research in Security Prices (CRSP) supplies returns, prices, trading volume, and market capitalization, while TAQ data provides an external benchmark for broader retail order flow through the retail–trade classification of Boehmer et al. (2021). Google Trends search volume index is leveraged to measure active information acquisition through daily ticker–stock search intensity, and RavenPack to control for the contemporaneous news and sentiment environment as in De Giorgi et al. (2026). Because these data sources do not have identical coverage due to differences in availability and retrieval difficulties, the analysis is conducted on nested samples rather than on a single balanced panel. The *baseline trading sample* merges Robintrack, CRSP, and TAQ and

[3] "Herding" is defined as a stock-day observation where stock i on day t experiences an extremely large and positive retail order imbalance.

[4] Robintrack is a third-party website that scraped Robinhood's public API during its existence (from May 2018 to August 2020) to provide data on the number of Robinhood users holding each stock for a given timestamp. The data are still freely available at https://robintrack.net/.

is used for the main trading and return analyses. The *attention sample* further requires reliable Google Trends coverage and has therefore been restricted to stocks belonging to the Russell 3000 index to exploit the dataset construction of deHaan et al. (2024). This restriction is materially significant: the attention sample is mechanically tilted toward larger and more actively followed stocks, making it well suited to study the transmission of salience into search attention, but less suited to capture the full return consequences of herding among smaller and less liquid securities.

The empirical strategy follows the sequence implied by the mechanism. I first compare the concentration of Robinhood user changes with TAQ retail net buying to assess whether Robinhood users indeed trade in a narrower common range of stocks than the broader retail population. I then estimate stock and date fixed-effects regressions to examine whether lagged salience and attention variables are associated with next-day Robinhood user flows and with the probability of discrete herding events while controlling for flow persistence, abnormal volume, extreme returns, and the news environment. Herding events are defined, as in Barber et al. (2022), as stock–day observations where a stock experiences an extremely large and positive percentage change in the number of Robinhood users holding it[5]. The results support the view that Robinhood users display particularly concentrated attention-induced trading. Robinhood buying is more concentrated than TAQ retail buying: the ten most heavily bought stocks account for 34.51% of daily Robinhood net buying, compared with 23.93% in TAQ retail data, and the difference is statistically significant. I also find that buying is more concentrated than selling among Robinhood users, aligned with the attention-based theory of retail investing and empirical results in Barber et al. (2022). Importantly, lagged extreme returns and Robinhood user changes are strongly associated with next-day Robinhood user flows. The relation between abnormal Google search volume and Robinhood trading is more nuanced. In user-flow regressions, lagged abnormal search volume is positively associated with next-day Robinhood inflows only before controlling for lagged user flows. Once flow persistence is included, the relationship turns negative, while coefficients on lagged Robinhood user changes are large and positive across specifications. The initial relation therefore appears to reflect persistent short-run trading dynamics rather than an independent build-up of future retail demand. At the same time, lagged abnormal search volume remains positively associated with the probability of a discrete herding event; especially for small-cap stocks where attention–driven trading is likely more influential (De Giorgi et al., 2026; Fedyk, 2021; Stein, 2020). This distinction points towards the first descriptive evidence that Google search attention is best interpreted as a confirmation signal of already–triggered salience: it is informative about the attention episodes around which Robinhood participation becomes concentrated, but it is not a simple distant predictor of average next-day inflows. This interpretation also corresponds

[5] For more details on the herding event definition, refer to section 3.5

to the nature of the search decision itself. A retail investor who searches for a specific stock ticker has, by construction, already selected that stock as an object of attention. Google search activity therefore appears more plausibly as part of the immediate response to salience than as a slow-moving predictor of future retail demand.

Subsequently, to characterize the timing of the attention process, I examine event-time dynamics around large search-attention bursts and extreme overnight return events. Consistent with the interpretation that Google search volume is a confirmation signal of already–triggered salience, search attention spikes contemporaneously with extreme-return events, and Robinhood user flows spike contemporaneously with large search-attention bursts. This timing pattern is again more coherent with Google search volume being part of the immediate attention response than with it being a distant leading predictor of retail demand.

As indicated previously, individual investors face attention scarcity (Kahneman, 1973) and might therefore rely on heuristics (Tversky & Kahneman, 1974) to determine their investment decisions in an environment characterized by information overload and uncertainty. Robinhood's mobile app guides this behavioral process by designing an interface that reduces the cognitive effort required to identify and process salient information. Consequently, Robinhood users are likely to rely on intuitive thinking (System 1) rather than critical and deliberative thinking (System 2) (Kahneman, 2011). Robinhood's Top Movers list is a prominent example of a platform feature that highlights visually salient information and therefore encourages automatic and emotional thinking (Barber et al., 2022). This list is updated throughout the day and displays the 20 stocks with the largest price movements, regardless of their direction (uncommon since the majority of alternative investment platforms display top gainers and top losers separately, while top gainers are typically more prominently featured). I therefore study the Top Movers list as a case study of a platform feature that potentially shapes attention and trading behavior, in line with Barber et al. (2022). Inspired by Stein (2020), I also report complementary evidence on the Top 100 List, a continuously updated list of the 100 stocks with the largest number of Robinhood users holding them. I use the institutional features of these lists to design quasi-experimental tests of the attention-transmission mechanism through a regression-discontinuity approach. Aligned with the literature, I find that Robinhood users respond strongly to stocks with extreme price movements (Ardia et al., 2023; Barber & Odean, 2008; Reyes, 2019; Welch, 2022), and their response to extreme gainers and losers is more symmetric than the corresponding response in TAQ retail order flow, consistent with the mixed-signed design of the Top Movers list (Barber et al., 2022). Top Movers eligibility is also associated with a local increase in contemporaneous abnormal Google search activity. This pattern is central for the thesis because it links platform-based salience to active external information acquisition and suggests that an origination signal can trigger a subsequent confirmation signal.

In the return analysis, Robinhood herding events occur after sharp price increases and are followed by economically meaningful reversals. Post-event buy-and-hold abnormal returns are negative after both five and twenty trading days, and a calendar-time portfolio that shorts herding-event stocks for five trading days earns positive abnormal returns after controlling for Fama and French (2015) five factors and a momentum factor (Carhart, 1997). Comparable return dynamics also arise when herding events are defined from TAQ retail trading activity, although the reversal is less severe than in the Robinhood-based sample. The exhibits are therefore aligned with the view that extreme returns might generate salience and concentrated retail participation, which can in turn be associated with temporary upward price pressure (Barber et al., 2022; Da et al., 2011).

These findings yield three contributions. First, the thesis contributes conceptually by separating the attention process into origination and confirmation stages. Extreme returns and platform visibility are treated as signals that originate salience, whereas abnormal Google search volume is interpreted as evidence that salience has translated into active information acquisition: a confirmation signal. Second, the thesis contributes empirically by documenting the timing of this process: search attention and Robinhood participation rise sharply around the same salience episodes, suggesting that search is part of the immediate attention response rather than a slow-moving predictor of retail demand. Third, the thesis connects this attention-transmission mechanism to asset-pricing outcomes. The post-herding return reversals, the calendar-time portfolio evidence, and the TAQ robustness exercise are jointly consistent with the view that concentrated retail participation can be associated with temporary price pressure (Barber et al., 2022; Da et al., 2011). However, because the attention analysis is restricted to Russell 3000 stocks and therefore tilted toward larger securities, the return analysis cannot fully assess the return consequences of confirmation signals during herding episodes.

The remainder of the thesis proceeds as follows. Chapter 2 provides a literature review. Chapter 3 describes the data, variable construction, and nested sample structure. Chapter 4 studies concentration in retail trading and the predictability of Robinhood user flows and herding events. Chapter 5 examines the timing of the attention process. Chapter 6 analyzes in quasi-experimental settings the impact of platform-based salience on attention and trading behavior. Chapter 7 studies return dynamics around Robinhood and TAQ herding events and Chapter 8 concludes.

II. Literature Review

This section reviews the literature in five steps. First, I discuss the theory of limited attention and attention-induced buying, which provides the foundation for treating salient signals as determinants of retail investors' consideration sets. Second, I review Google search volume as a revealed-attention measure and discuss its measurement limitations. Third, I examine the literature on Robinhood and fintech retail investors. Fourth, I discuss platform salience and app-based information display. Finally, I review evidence on retail herding, temporary price pressure, and return reversals. The review highlights a gap in the existing literature: attention proxies are often treated as interchangeable, whereas this thesis distinguishes between signals that originate salience and signals that confirm active information acquisition.

2.1. Limited Attention & Attention-induced Trading

In *Attention and Effort*, Kahneman (1973, p. 3) defines *selective attention* as the process by which "an organism appears to control the choice of stimuli that will be allowed, in turn, to control its behavior". Studying the determinants of stimuli that capture attention, and the conditions under which other stimuli are ignored, is crucial to understand the behavior of economic agents. Kahneman (1973) further argues that attention varies with effort and arousal levels, but is intrinsically limited. In this context, salient and prominent stimuli will exert more influence on behavior. Odean (1998, p. 1889) indeed notes that "people systematically underweight abstract, statistical, and highly relevant information, and overweight salient, anecdotal, and extreme information".

The financial economics literature builds directly on this insight. Hirshleifer and Teoh (2003) show that limited investor attention makes the form of financial disclosure economically relevant: information that is salient, explicitly recognized, or easier to process can receive greater weight than economically equivalent information that is less prominent or disclosed only indirectly. Merton (1987) provides a related asset-pricing foundation by relaxing the assumption that all investors know and consider all securities. In his investor-recognition model, visibility affects the set of investors who hold a stock and can therefore influence required returns. Peng and Xiong (2006) extend the attention mechanism by modelling attention as a scarce processing resource allocated across layers of information. Limited attention induces *category learning*: investors process broad market- or sector-level information more intensively than firm-specific signals.[1]. These papers

[1] As an example, Peng and Xiong (2006, p. 565) uses the 2000 dot-com bubble to illustrate how investors' attention was disproportionately allocated to the technology sector, at the expense of firm-specific

jointly imply that investors do not process the full information set available in financial markets. Instead, visibility, categorization, and ease of processing shape which securities are considered.

Barber and Odean (2008) apply this logic directly to individual-investor trading. They argue that buying and selling decisions are asymmetric. Buying requires investors to search across thousands of potential securities, whereas selling is usually restricted to securities already held in the portfolio. Because investors cannot evaluate the entire investment universe before each purchase decision, attention operates as a screening mechanism that narrows the feasible choice set into a smaller consideration set. This asymmetry explains why attention-grabbing events, such as extreme returns, unusual volume, or news coverage, affect buying more strongly than selling. It also provides the theoretical basis for the concept of *origination signals*: simple and salient cues that make a stock visible enough to enter retail investors' consideration sets.

2.2. Proxies for Investor Attention

To examine the influence of attention on investor behavior, researchers have used different proxies to measure attention: extreme returns (Ardia et al., 2023; Barber & Odean, 2008), trading volume (Barber & Odean, 2008; Gervais et al., 2001; Hou et al., 2009), media coverage (Barber & Odean, 2008; Tetlock, 2011), price thresholds (Seasholes & Wu, 2007), or advertising expenditure (Lou, 2014). Da et al. (2011, p. 1462) however suggested that these proxies are *indirect* measures of attention, i.e. "make the critical assumption that if a stock's return or turnover was extreme or its name was mentioned in the news media, then investors should have paid attention to it". Following this critique, they propose to use the Google search volume index (SVI) as a more *direct* or *revealed* measure of investor attention: "if you search for a stock in Google, you are undoubtedly paying attention to it" (Da et al., 2011, p. 1462)[2]. The SVI is a normalized measure of search popularity for a given term on Google. Da et al. (2011) show that high abnormal SVI is associated with short-term contemporaneous buying pressure and positive returns, but also with subsequent reversals. More recently, Da et al. (2025) further document that aggregate retail attention (ARA), measured through value-weighted abnormal Google search volume for individual stocks, negatively predicts one-week-ahead market returns. These findings make Google search activity a natural proxy for active information acquisition and link retail attention to temporary price pressure.

Despite being a relevant proxy for investor attention, deHaan et al. (2024) note that the SVI is not without limitations and subject to measurement error. To mitigate these issues,

information: "firms that had changed to dot.com names without any fundamental changes in strategies earned significant abnormal returns around their name-change announcements".

[2] These time-series data are made available by Google Trends at https://trends.google.com/trends/.

they propose to use the following search syntax: "[stock ticker] stock" (e.g. "AAPL stock") instead of the stock ticker alone (e.g. "AAPL") or the company name (e.g. "Apple Inc."). This thesis follows deHaan et al. (2024)'s recommendation and uses their constructed SVI with the syntax "[stock ticker] stock" as a proxy for investor attention[3]. While it is not the focus of the present analysis, it should also be noted that social media platforms, such as Reddit and its infamous subreddit `r/wallstreetbets`, have become increasingly studied as proxies of revealed attention from retail investors (De Giorgi et al., 2026; Fedyk, 2021; Münster et al., 2024).

The present thesis consequently considers extreme returns, abnormal volume, news, and platform rankings as potential sources of attention; Google search volume is instead interpreted as evidence that investors have actively acquired information about a stock. The empirical question is therefore not only whether SVI predicts trading, but whether it behaves as an origination signal or as a confirmation signal of already-triggered salience. To the best of my knowledge, the literature has not yet examined this question directly.

2.3. Robinhood & Robintrack

Measuring investor attention is conceptually distinct from observing the trading behavior that such attention induces. The emergence of fintech brokerage platforms, and Robinhood in particular, has provided researchers with a rare opportunity to match attention proxies to direct measures of retail participation. A central data source in this literature is `Robintrack`, which recorded, at high frequency, the aggregate number of Robinhood users holding each covered security between May 2018 and August 2020[4]. By making retail ownership dynamics observable at the stock-day and intraday level, these data have become a canonical empirical basis for studying retail investors, and attention-induced trading. Welch (2022) leverages `Robintrack` data to show that when Robinhood investors are studied as an aggregate holdings-based crowd portfolio rather than as isolated speculative trades, their behavior from 2018 to 2020 appears more resilient and economically sensible than the popular "dumb retail crowd" narrative suggests: they provided liquidity during the COVID crash, mostly held liquid high-volume stocks rather than only meme-like names, and their consensus portfolio did not underperform. Eaton et al. (2022) study brokerage platform outages at Robinhood and general brokers and contrast the views of Welch (2022).

[3] This is one of the reasons why the thesis leverages the two nested samples mentioned in the introduction. Collecting the SVI for the full sample of stocks (i.e. all stocks listed on the NYSE, NASDAQ, and AMEX) would be prohibitively costly in terms of time and resources.

[4] `Robintrack` is an independent website (https://robintrack.net/) that collected publicly available information from Robinhood's API and reported, for each listed security, the number of Robinhood accounts holding that security. The data were observed at intervals of roughly 45 minutes and therefore capture changes in the extensive margin of Robinhood ownership rather than signed trades, order flow, or position sizes. Robinhood restricted access to these data in August 2020, after which `Robintrack` ceased updating its database.

They report that Robinhood outages are associated with lower return volatility, while general broker outages are associated with increased return volatility. Their findings also indicate that Robinhood outages are correlated with reduced market order imbalance and enhanced liquidity. These results suggest that Robinhood investors are less experienced than the broader retail population, and engage in attention-driven trading. Ozik et al. (2021) also rely on `Robintrack` data to document that the COVID-19 pandemic led to a significant increase in retail participation. As stated above, `Robintrack` data have also been extensively used to measure the relation between social media attention and retail trading (Cookson et al., 2024; De Giorgi et al., 2026; Fedyk, 2021; Münster et al., 2024). Overall, the literature motivates the use of Robinhood and `Robintrack` data as a sound empirical basis for studying retail attention and trading.

2.4. The Role of Information Salience

Central to the notion of attention is the related concept of salience; Barber et al. (2022) have exploited both the `Robintrack` data and the features of the Robinhood app to study the role of information salience in driving retail attention and trading. In particular, they report that app features such as the "Top Movers" list capture individual investors' attention and influence their trading decisions. Similarly, Stein (2020) show that the "100 Most Popular" list on the Robinhood app, which ranks stocks by the number of Robinhood users holding them, also guides retail attention and might generate temporary price pressure. In related research, Glaze and Moss (2026) confirm that various visualizations in the Robinhood app influence users' trading patterns and notice that Robinhood users display contrarian behaviors in response to these visual cues. This echoes Pagano et al. (2021), who observe that Robinhood users shifted from momentum to contrarian trading strategies at the onset of the COVID-19 pandemic. Finally, in different settings than Robinhood, Chen et al. (2023) exploit a particular display feature of common trading platforms in China to show that the structure of information presentation affects attention and returns. Taken together, these papers show that retail attention is not only drawn by firm-level events such as returns or news. It can also be shaped by the architecture through which platforms rank, visualize, and display securities. This observation motivates the Top Movers and Top 100 analyses in this thesis and supports the interpretation of platform visibility as an origination signal.

2.5. Herding, Price Pressure & Reversal

If salient signals coordinate the attention of many retail investors at the same time, attention-induced buying can become concentrated in a small number of securities. Barber et al. (2022) document precisely this pattern among Robinhood users. They show that Robinhood buying is more concentrated than broader retail buying, that herding episodes are predicted by attention proxies such as SVI and platform features, and that these episodes are followed by negative abnormal returns. Ardia et al. (2023) provide complementary intraday evidence, showing that Robinhood users react rapidly to extreme price movements and are particularly eager to open positions in stocks which experienced large negative returns. These findings link the attention-based buying hypothesis of Barber and Odean (2008) to discrete episodes of coordinated retail participation.

This concentration of demand can have asset-pricing consequences. If attention-induced buying reflects temporary demand pressure rather than permanent information about fundamentals, prices should rise around the herding episode and partially reverse afterward. This prediction is consistent with Da et al. (2011), who show that abnormal Google search volume is associated with short-run price increases and subsequent reversals, and with Barber et al. (2022), who document negative abnormal returns after intense Robinhood buying. Tetlock (2011) provides a related mechanism by showing that individual investors may overreact to stale but salient information, generating temporary price movements that later reverse. More generally, Coval and Stafford (2007) show that price pressure from constrained trading can create temporary deviations from fundamental values. These papers motivate the return analysis in this thesis, which asks whether Robinhood herding events are followed by short-run return reversals consistent with temporary price pressure.

2.6. Empirical Predictions

Taken together, the literature leads to five empirical predictions. First, because attention constrains retail stock selection, Robinhood trading should be concentrated in a small set of securities and relatively more concentrated than broader retail trading if Robinhood users are relatively less experienced and more prone to attention-induced trading. Second, salience signals such as extreme returns, abnormal volume, news, and platform visibility should be associated with Robinhood user flows and herding events. Third, if Google search volume reflects active information acquisition, abnormal SVI should be associated with herding, but its timing should reveal whether it behaves as an origination signal or as a confirmation signal. Fourth, if platform visibility manufactures attention, Top Movers eligibility should be associated with local increases in Robinhood participation and abnormal Google search attention. Fifth, if concentrated Robinhood buying creates temporary price pressure, herding events should be followed by short-run return reversals. These predictions organize the empirical analysis that follows.

III. Data & Samples Construction

3.1. Overview

This chapter describes the datasets, variable definitions, and sample restrictions used in the empirical analysis. The empirical design combines five complementary data sources. `Robintrack`[1] provides intraday stock-level counts of Robinhood users holding each security. CRSP supplies daily returns, closing and opening prices, market capitalization, as well as daily trading volume[2]. TAQ data are used to construct an external benchmark for comparing Robinhood activity with broader retail order flow, based on the trade-classification procedure of Boehmer et al. (2021)[3]. Google Trends and RavenPack[4] data are used to proxy, respectively, *direct* investor attention as Da et al. (2011, 2025), and the contemporaneous news or sentiment environment as De Giorgi et al. (2026).

Furthermore, due to data availability, the analysis is conducted on a sequence of nested samples rather than on a single balanced panel. The broadest sample is the *baseline trading sample*, constructed by merging the daily `Robintrack` variables with CRSP and TAQ by ticker and date. This sample is used to describe the evolution of Robinhood participation and validate `Robintrack`-based activity against TAQ retail order flow. The second sample is the *attention sample*, obtained by restricting the baseline sample to stock-day observations for which a Google Trends measure of search intensity can be reliably assigned. Finally, for specifications that condition on the information environment, I augment the attention sample with ticker–day measures derived from RavenPack.

The distinction across samples is economically important and materially affects the interpretation of the results. The baseline sample is designed to preserve comparability with Barber et al. (2022). By contrast, the attention sample is tailored to the analysis of search activity and is necessarily narrower because exhaustive Google Trends coverage is difficult to obtain and sensitive to the precise construction of the search variable[5]. Importantly,

[1] The `Robintrack` dataset is freely available at https://robintrack.net. `Robintrack` exploited Robinhood's public API to record the number of users holding each stock at various times during the day.

[2] More precisely, I rely on the CRSP daily stock file, as made available by Wharton Research Data Services (WRDS).

[3] This supplementary dataset is incorporated to the analysis because it provides external validation of the `Robintrack`-based measures of retail trading activity, which is important given that Robinhood users represent only a subset of the overall retail investor population (Barber et al., 2022).

[4] RavenPack News Analytics is a data analytics provider in financial services and offers a comprehensive database of news and sentiment indicators. Inspired by the work of De Giorgi et al. (2026), I use RavenPack data to control for the contemporaneous news and sentiment environment. The RavenPack full equity dataset is proprietary and was accessed through WRDS.

[5] Discussions with the authors of Barber et al. (2022) revealed that their solution to obtain Google Trends data for a large number of tickers through the unofficial R package `gtrendsR` was laborious and involved a solution including IP addresses rotating to avoid Google Trends' rate limits, as well as sleep intervals to

since the attention sample is tilted toward larger and more actively followed stocks, it is less suited to capturing the full return consequences of herding among smaller and less liquid securities.

3.2. Robinhood & Robintrack

3.2.1. Robinhood Markets Inc.

Robinhood Markets, Inc. is a U.S. financial services company headquartered in Menlo Park, California. Founded in 2013 by Vladimir Tenev and Baiju Bhatt, Robinhood states that its mission is "democratizing finance for all"[6]. Early in its development, the firm differentiated itself by offering commission-free trading[7] through a gamified mobile app and web interface. This model quickly gained traction, particularly among younger, less experienced, and smaller retail investors attracted by its low barriers to entry and ease of use. *The New York Times* reported that the average Robinhood user is 31 years old, has likely no prior trading experience[8], and, scaled by average account size, trades "nine times as many shares as E-Trade customers, and 40 times as many shares as Charles Schwab customers, in the most recent quarter"[9]. These observations resonate with Barber and Odean (2002, p. 791) who warn that "overconfident investors [young and inexperienced investors encouraged to trade by a gamified interface are likely to qualify] will trade too frequently", and earn suboptimal returns (Barber & Odean, 2001). *The New York Times* also reports on the dangerous and hazardous trading behaviors the gamified interface encourages[10]. Ozik et al. (2021, pp. 2357–2359) further show that Robinhood's user base, along with that of similar platforms, expanded sharply during the COVID-19 pandemic, as households were confined at home and experienced an increase in savings. In mid-2020, the company reported to have roughly 13 million users. Taken together, these features make Robinhood a particularly informative setting for the study of retail investor behavior.

3.2.2. `Robintrack` Data

To study the behavior of Robinhood users, I use the `Robintrack` dataset, which has become standard in the recent literature. `Robintrack` relied on Robinhood's public API to record the number of users, $N_{i,t}$, holding stock i at time t. Listing 3.1 reports a snapshot of the raw observations for Apple Inc. (AAPL).

avoid triggering Google's anti-bot mechanisms.

6 https://robinhood.com/us/en/about-us/.

7 Before its subsequent expansion into additional business lines, Robinhood's revenue model relied predominantly on payment for order flow (PFOF), under which brokers receive compensation from market makers for routing orders to them.

8 Approximately one-half of users has no prior trading experience.

9 https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html.

10 https://www.nytimes.com/2021/01/30/business/robinhood-wall-street-gamestop.html.

Listing 3.1. Original Excerpt of `Robintrack` Observations for AAPL

```
timestamp,users_holding
"2018-05-02 04:53:46",150785
"2018-05-02 06:38:58",150785
"2018-05-03 00:35:25",145510
```

The dataset spans May 2, 2018 to August 13, 2020. Data collection ends on August 13, 2020, when Robinhood discontinued public access to the API used by `Robintrack`. The raw data are affected by four disruptions. As noted above, Robinhood experienced system outages during periods of unusually high trading activity on March 2, 2020 and March 9, 2020. In addition, the `Robintrack` collection script failed on August 9, 2018, January 24-29, 2019, and January 7-15, 2020. Following Welch (2022), I begin the baseline trading sample on June 1, 2018. The resulting baseline trading sample contains observations for 554 trading days and 8,597 usable tickers[11]. To aggregate the intraday observations to a daily frequency, I follow the procedure in Barber et al. (2022) and rely on Welch (2022) for several implementation details[12]. The main variables constructed from the `Robintrack` data are *users_close*, *users_last*, *userchg*, and *userratio*. The primary variable of interest is *users_close*, defined as the last registered number of users holding a given stock after 2:00 p.m. ET but before the market close (4:00 p.m. ET). Regarding the particular timestamp of each observation, Barber et al. (2022) first raised concerns about a delay between actual observation time and retrieval time. This delay was later confirmed to be 45 minutes by Ardia et al. (2023) who discussed with Casey Primovic, the developer of `Robintrack`. Therefore, for each timestamp $t_{i,k}$ in my dataset, I subtract 45 minutes and convert it from UTC to ET to obtain the actual observation time. I then define *userchg* as the daily change in *users_close* and *userratio* as the ratio $users\_close(t)$ / $users\_close(t-1)$.

After constructing the daily variables from the `Robintrack` data, I merge them with the Center for Research in Security Prices (CRSP) daily stock file by ticker and date. To adjust CRSP returns for delisting events, I follow the procedure in Bali et al. (2016). I then augment the sample with Trade and Quote (TAQ) data from Wharton Research Data Services (WRDS) to obtain an external benchmark measure of retail trading activity, namely retail buy and sell executions. As Boehmer et al. (2021) note, the emergence of algorithmic and high-frequency trading made traditional trade-size-based classifications of retail and institutional trading increasingly unreliable, because large institutional orders were often split into many small trades. To address this problem, Boehmer et al. (2021, pp. 2254–2256) develop a classification algorithm grounded in the regulatory structure

[11] To remain aligned with Barber et al. (2022), the baseline trading sample is not restricted to ordinary common shares (share code 10 or 11).

[12] For example, Welch (2022) notes that the `Robintrack` data contain four invalid tickers (_OUT, _PRN, MTL-, and PKD~), a detail not addressed in Barber et al. (2022).

of U.S. equity markets. Their approach exploits the fact that retail marketable orders are typically executed off-exchange by wholesalers or internalizers and frequently receive small subpenny price improvement, whereas institutional orders generally do not. Accordingly, they focus on off-exchange TAQ trades reported to the FINRA TRF (exchange code "D") and classify transactions using the fractional component of the trade price relative to the nearest cent. Trades reported just above a round penny are classified as retail sells, whereas trades reported just below a round penny are classified as retail buys. More precisely, for TRF-reported trades, transactions with a fractional cent component in the interval $(0, 0.4)$ are coded as retail sells, while those in $(0.6, 1)$ are coded as retail buys; trades at round-penny prices or near half-penny prices are excluded.

Figure 1. Robinhood Users & TAQ Retail Volume. Panel A plots the total number of Robinhood users holding stocks at the market close (*users_close*) over the sample period. Panel B plots the daily total number of retail trades executed off-exchange, as classified by Boehmer et al. (2021). The black vertical line indicates the date when the COVID-19 national emergency was declared in the U.S. (March 13, 2020).

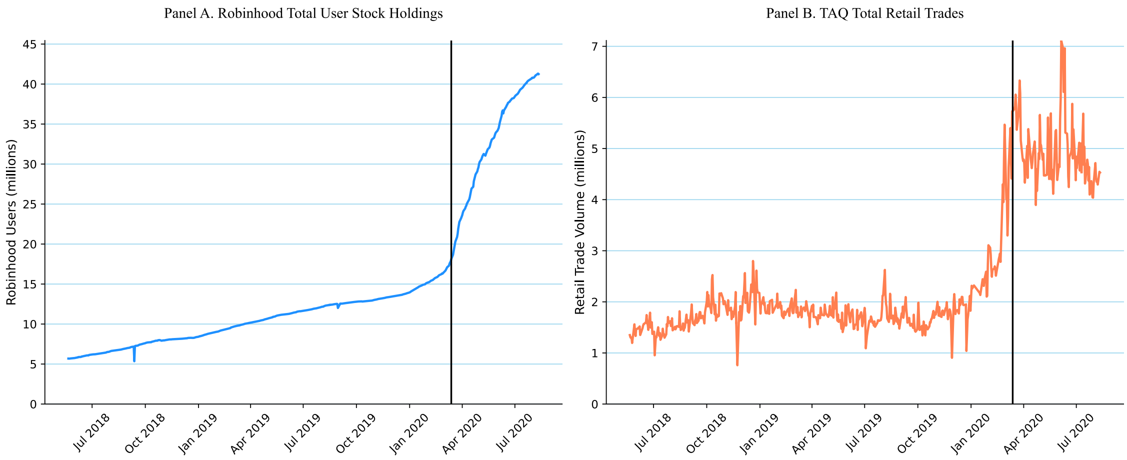
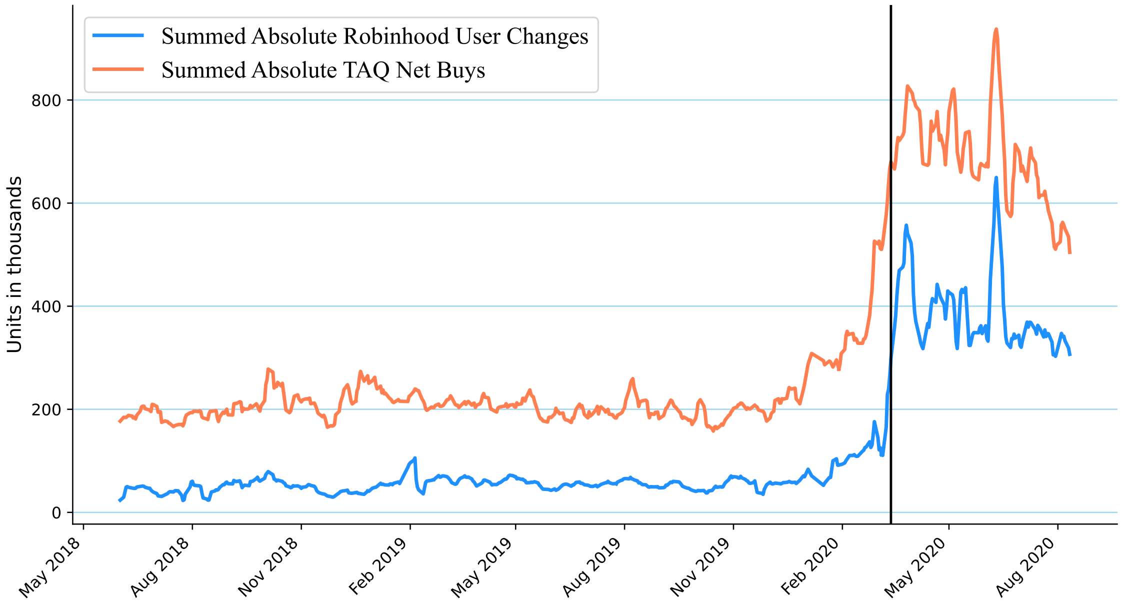


Figure 1 is consistent with Ozik et al. (2021): both aggregate Robinhood holdings and TAQ-classified retail trading activity increase sharply during the COVID-19 period. Of course, the two measures are partially overlapping, as the Robinhood user base is a subset of the overall retail investor population.

In Figure 2, I further plot, inspired by Barber et al. (2022), the 5-day moving average of the absolute daily change in the number of Robinhood users holding stocks at the market close (blue line), as well as the 5-day moving average of the absolute daily TAQ retail net buying (orange line). Again, this suggests that both measures follow similar patterns, with a pronounced increase in activity during the COVID-19 pandemic[13]. Unfortunately, the `Robintrack` data end in August 2020, so we cannot observe whether the elevated levels of

[13] No formal statistical test is conducted to compare the two time series, but Barber et al. (2022) provide empirical evidence supporting that Robinhood users proxy relatively well for the overall retail investor population.

Figure 2. Absolute Changes in Robinhood Holdings & TAQ Retail Net Buying. This figure plots the evolution over the period of the 5-day moving average of the absolute daily change in the number of Robinhood users holding stocks at the market close (*userchg*), as well as the 5-day moving average of the absolute daily TAQ retail net buying. The black vertical line also indicates the date when COVID national emergency was declared in the U.S. (March 13, 2020).



retail trading activity persisted beyond the pandemic[14].

Finally, Table 1 reports summary statistics for the baseline trading sample. Importantly, *daily_buys* and *daily_sells* count retail-classified trade executions, not share volume. The average number of Robinhood users holding a given stock is about 2,141, but the distribution is highly skewed: the median is 173, whereas the maximum is close to 1 million for the most widely held stock[15]. This pattern suggests that Robinhood holdings, and likely retail attention, are concentrated in a relatively small set of stocks, consistent with prior evidence in Barber et al. (2022), Münster et al. (2024), and Welch (2022). Table 1 further shows that the daily change in the number of Robinhood users holding a stock (*userchg*) is positive on average but economically small, with a mean of 9.72 users and a median of zero. This is consistent with the evidence in Figure 2 and with the positive mean of TAQ retail *net_buys*.

[14] Related evidence from retail options markets suggests that the pandemic period coincided with unusually elevated retail trading activity (Bryzgalova et al., 2023).

[15] The five stocks with the highest average number of Robinhood holders are Aurora Cannabis Inc. (ACB), Ford Motor Company (F), General Electric Company (GE), Apple Inc. (AAPL), and Microsoft Corporation (MSFT).

Table 1. Summary Statistics of the Baseline Trading Sample

Panel A presents summary statistics across stock-day observations. Panel B sums variables by day and then averages across days. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users\_close_t$ / $users\_close_{t-1}$), *prc* (closing price), *size* (market cap in millions), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)), and #stocks (number of stocks with users reported on `Robintrack`). This table follows the design of Barber et al. (2022, p. 3151).

Variable	*N*	Mean	SD	min	p25	p50	p75	max
Panel A: Stock-Day Observations								
users_close	3,755,888	2,142.95	15,820.03	0.00	40.00	173.00	714.00	990,444.00
users_last	3,841,684	2,140.93	15,830.70	0.00	40.00	173.00	713.00	990,587.00
userchg	3,678,526	9.72	383.71	−548,766.00	−1.00	0.00	1.00	85,193.00
userratio	3,639,029	1.01	1.18	0.00	1.00	1.00	1.00	2,087.00
prc	3,841,684	41.07	91.59	0.04	10.46	23.72	46.67	4,699.00
size ($mil.)	3,766,791	5,603.11	30,996.71	0.03	93.41	442.72	2,249.69	1,966,963.00
ret (%)	3,823,449	0.04	4.29	−100.00	−0.98	0.02	0.99	1,025.18
daily_buys	3,659,438	199.32	1,105.93	0.00	8.00	33.00	116.00	185,252.00
daily_sells	3,659,438	177.84	871.19	0.00	8.00	33.00	114.00	112,743.00
net_buys	3,659,438	21.48	365.31	−30,215.00	−7.00	0.00	10.00	86,325.00
taq_retimb	3,659,438	0.01	0.35	−1.00	−0.14	0.00	0.16	1.00
Panel B: Daily Observations (Summed Variable Averaged across Days)								
#stocks	554	6,706.95	702.91	180.00	6,197.50	6,634.00	7,458.75	7,614.00
users_close (mil.)	554	14.53	9.70	0.14	8.12	11.58	14.86	41.32
users_last (mil.)	554	14.54	9.71	0.14	8.12	11.58	14.87	41.33
userchg (000)	554	64.55	110.49	−185.82	12.92	22.09	51.80	817.19
daily_buys (000)	554	1,285.74	751.43	9.84	831.40	932.08	1,308.58	3,972.02
daily_sells (000)	554	1,146.70	598.58	9.30	783.01	879.50	1,193.29	3,200.01
net_buys (000)	554	139.04	172.09	−68.88	38.73	63.54	135.39	1,007.41

3.3. Google Trends

3.3.1. Conceptual Motivation

In the attention-induced trading literature, we often measure investor attention with *indirect* proxies, such as extreme abnormal returns (Barber & Odean, 2008), trading volume (Barber & Odean, 2008; Hou et al., 2009), news and headlines (Ballinari et al., 2022; Barber & Odean, 2008; Barber et al., 2022), or price thresholds (Kupfer & Schmidt, 2021). However, as Da et al. (2011, p. 1462) note, these "indirect" proxies do not capture investor attention but rather potential sources of attention: a "news article in the Wall Street Journal does not guarantee attention unless investors actually read it".

Interestingly, over the past decade, the availability of new data sources has enabled the use of *direct* proxies of investor attention, such as social media activity (Audrino et al., 2024; De Giorgi et al., 2026; Fedyk, 2021) or Google search activity (Da et al., 2011, 2025; deHaan et al., 2024). These proxies are referred to as "direct" because when someone searches for a stock ticker on Google, or posts about it on Reddit[16], it is more likely that they are actually paying attention to the stock, as opposed to simply being exposed to a news article or an extreme return. Hence, in line with Da et al. (2011, 2025), I use Google Trends search volume index (SVI) as the main proxy for investor attention.

[16] De Giorgi et al. (2026) uses text processing techniques to quantify the content of the subreddit `r/wallstreetbets`.

3.3.2. Variable Construction

Google provides a publicly available tool, Google Trends[17], to track the search volume intensity of specific keywords over time. The search volume index (SVI) provided by Google Trends is a normalized measure of the search interest for a given keyword. Google samples a random subset of all searches and normalizes the search volume for a keyword by the total number of searches in the specified time period and geographic region. The resulting *svi* is then scaled from 0 to 100, where 100 represents the peak search interest for that keyword during the specified time period and geographic region[18]. For an illustration of the raw *svi*, the reader can refer to Figure A.1 in the Appendix. Based on the recommendations in deHaan et al. (2024), I rely on their construction of *svi* variables, which are designed to mitigate measurement error and noise searches. For instance, Da et al. (2011) notices that the ticker "CAT" is also a common English word, which can lead to noisy search data. To address this issue, deHaan et al. (2024) constructs a more precise search term for CAT by using the syntax "CAT stock" in Google Trends. This approach helps to filter out irrelevant searches and provides a more accurate measure of investor attention for the stock[19]. It implies that the *svi* is based on ticker-stock search terms, such as "AAPL stock" rather than simply "AAPL", and is measured at the daily frequency for U.S. searches. Contrary to Barber et al. (2022) and Da et al. (2011) who rely on weekly Google Trends data, the dataset I use to build my attention sample relies on daily *svi* data, as in Da et al. (2025). Due to the difficulty of scaling Google Trends data across a large number of tickers, *svi* time series are constructed only for stocks in the Russell 3000 index[20], also following Da et al. (2011). Once I obtain the daily search volume index for each stock, I merge them with the baseline trading sample by ticker and date to construct the attention sample. This procedure results in an attention sample containing over 1 million stock-day observations for 1,979 unique tickers. I also construct the variable $asvi_{i,t-1}$, inspired by the approach in Barber et al. (2022) and outlined in Equation 3.1, to capture the lagged abnormal search interest for a stock i on the previous day $t-1$.

$$asvi_{i,t-1} = \ln\left(1 + \frac{svi_{i,t-1}}{\frac{1}{20}\sum_{\tau=2}^{21} svi_{i,t-\tau}}\right) \tag{3.1}$$

When the 20-day lagged average *svi* is zero, the *asvi* measure is undefined; such observations are excluded from specifications using *asvi*[21]. Table 2 reports summary

[17] https://trends.google.com/trends/

[18] See https://newsinitiative.withgoogle.com/resources/trainings/basics-of-google-trends/.

[19] The datasets of deHaan et al. (2024) are publicly available on their `GitHub` repository at https://github.com/robinlitjens/GoogleTickerStock-SVI/tree/main.

[20] The Russell 3000 index is a widely followed benchmark that includes the 3,000 largest U.S. stocks by market capitalization, representing approximately 98% of the investable U.S. equity market. See https://www.ftserussell.com/products/indices/russell-3000-index.

[21] These observations represent 7.15% of the attention sample. The exclusion should be kept in mind when

statistics for the attention sample similar to Table 1 and allows for comparison between the two samples.

Table 2. Summary Statistics of the Attention Sample

Panel A presents summary statistics across stock-day observations. Panel B sums variables by day and then averages across days. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users\_close_t$ / $users\_close_{t-1}$), *prc* (closing price), *size* (market capitalization), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)), #stocks (number of stocks with users reported on `Robintrack`), and *asvi* (abnormal search volume index defined in Equation 3.1).

Variable	*N*	Mean	SD	min	p25	p50	p75	max
Panel A: Stock-Day Observations								
users_close	1,037,118	3,794.27	23,947.79	1.00	96.00	306.00	1,193.00	943,931.00
users_last	1,037,118	3,797.05	23,967.66	0.00	96.00	306.00	1,193.00	944,388.00
userchg	1,015,432	17.02	334.42	−17,015.00	−2.00	0.00	2.00	51,733.00
userratio	1,015,406	1.01	0.27	0.01	1.00	1.00	1.01	136.74
prc	1,037,118	66.83	151.54	0.09	16.20	36.80	75.16	4,699.00
size ($mil.)	1,037,095	14,514.38	55,373.50	1.17	671.20	2,324.30	8,479.41	1,948,022.00
ret (%)	1,037,095	0.03	3.70	−100.00	−1.16	0.04	1.19	585.00
daily_buys	1,031,902	336.89	1,504.90	0.00	23.00	76.00	231.00	185,252.00
daily_sells	1,031,902	308.70	1,217.37	0.00	24.00	77.00	229.00	112,743.00
net_buys	1,031,902	28.18	453.17	−10,131.00	−12.00	0.00	15.00	86,325.00
taq_retimb	1,031,902	0.00	0.24	−1.00	−0.10	0.00	0.10	1.00
asvi	967,061	0.45	0.72	0.00	0.00	0.00	0.82	5.12
Panel B: Daily Observations (Summed Variables Averaged across Days)								
#stocks	554	1,872.05	116.20	11.00	1,863.00	1,878.00	1,896.00	1,917.00
users_close (mil.)	554	7.10	4.80	0.00	4.27	5.42	6.43	20.32
users_last (mil.)	554	7.11	4.80	0.00	4.27	5.42	6.43	20.32
userchg (000)	554	31.19	63.21	−64.71	3.98	8.58	19.66	467.90
daily_buys (000)	554	627.50	358.80	0.35	416.36	464.68	619.32	2,126.34
daily_sells (000)	554	575.00	280.68	0.33	407.62	455.30	598.06	1,728.91
net_buys (000)	554	52.50	91.13	−54.64	3.08	16.93	49.38	579.26

Compared with the baseline trading sample in Table 1, the attention sample in Table 2 is tilted toward larger and more actively followed stocks, consistent with its restriction to Russell 3000 firms. Relative to the baseline sample, stocks in the attention sample have substantially higher market capitalization (mean *size* of $14,514.38 million versus $5,603.11 million), are held by more Robinhood users (mean *users_close* of 3,794.27 versus 2,142.95), and exhibit greater TAQ net buying activity (mean *net_buys* of 28.18 versus 21.48). In addition, the distributions of *svi* and *asvi* are highly right-skewed, indicating that, on most stock-days, search activity is limited, whereas a small subset of stock–day observations attracts disproportionately high investor attention[22].

3.4. RavenPack News Analytics

To control for the potential confounding effect of news coverage and overall investor sentiment, I construct some controls following Audrino et al. (2020) and De Giorgi et al.

interpreting specifications using *asvi*, although it does not affect analyses based only on the broader attention sample.

[22] In the full attention sample, 34.14% of stock-day observations have $svi > 0$, 16.93% have $svi > 50$, and only 3.14% have $svi = 100$.

(2026). I augment the attention sample with news analytics data from RavenPack News Analytics, a leading provider of structured news data and sentiment analysis[23]. Inspired by the approach in De Giorgi et al. (2026), I build five control variables to capture the intensity and sentiment of news coverage for each stock-day observation. First, I construct *num_news* as the total number of press releases and news articles related to a stock on a given day and *num_news_relevant* as the number of news items for which RavenPack assigns a relevance score of at least 75 out of 100[24]. Subsequently, I construct an Event Sentiment Score *ess*, a Composite Sentiment Score *css*, and an Analyst Recommendation Change indicator *anl_chg*. These measures are aggregated over market hours to obtain stock-day level variables, which are then merged with the attention sample by ticker and date. A detailed description of the RavenPack News Analytics dataset and the construction of these variables can be found in Appendix A.1.2. Importantly, sentiment is controlled for because, as evidenced by Cookson et al. (2024), sentiment and attention contain different return-relevant information and might influence retail trading in opposite directions.

3.5. Herding Events

Table 1 and Table 3 demonstrated that daily user changes are generally small, but as presented by Barber et al. (2022, p. 3150), the data sample contains several instances of "extreme events" in which the number of users holding a stock changes drastically. These events are of particular interest, as they likely reflect episodes of intense retail attention and herding. To identify such events, I follow the procedure in Barber et al. (2022) and apply the following filters to the attention sample. First, I restrict the sample to stocks with at least 100 users on the previous day, to avoid extreme percentage changes driven by small denominators. Second, I focus on days with an increase in the number of users (i.e., $userratio > 1$). Once these filters are applied, for each trading day, I rank the stock-day observations based on *userratio* at time t and flag the top 0.5% of the distribution as herding events; denoted by the indicator *rh_herd*. I define herding events separately in the baseline trading sample and in the attention sample. The baseline herding-event sample is used for the main return analyses and preserves comparability with Barber et al. (2022). The attention herding-event sample applies the same definition after restricting to observations with Google Trends coverage and is used only in analyses involving search attention. Applying this definition to the baseline trading sample yields 5,311 herding events for 2,363 unique tickers. Applying the same definition to the attention sample yields 2,005

[23] Specifically, I use the RavenPack News Analytics dataset "full equity", as licensed by WRDS.

[24] In their user guide, RavenPack defines the relevance score as an integer score between 0-100 indicating how strongly related the mention of an entity is to the underlying news story, with higher values indicating greater relevance. For any news story that mentions an entity, RavenPack provides a relevance score. A score of 0 means the entity was passively mentioned while a score of 100 means the entity was prominent in the news story. Values above 75 are considered significantly relevant.

herding events for 869 unique tickers.

Table 3. Summary Statistics of Robinhood Herding Events

This table reports summary statistics for stock-days associated with Robinhood herding events in the baseline trading sample and in the attention sample. Herding events are defined as stock-days in the top 0.5% of the *userratio* distribution among observations with *userratio* > 1 and at least 100 users on day $t-1$. The variables are *users_close* (last observed user count for a stock prior to the 4 p.m. ET close), *users_last* (last user count of the day), *userchg* (daily change in *users_close*), *userratio* ($users\_close_t$ / $users\_close_{t-1}$), *prc* (closing price), *size* (market capitalization in millions of dollars), *ret* (daily return), *daily_buys* (number of TAQ daily retail buys), *daily_sells* (number of TAQ daily retail sells), *net_buys* (*daily_buys* − *daily_sells*), and *taq_retimb* (*net_buys*/(*daily_buys*+*daily_sells*)). Panel B also reports *asvi* (abnormal search volume index), which is available only in the attention sample.

Variable	*N*	Mean	SD	min	p25	p50	p75	max
Panel A: Herding Events in the Baseline Trading Sample								
users_close	5,311	2,477.87	7,759.79	113.00	340.00	746.00	1,846.50	156,451.00
users_last	5,311	2,549.15	7,998.70	113.00	347.00	758.00	1,878.50	156,826.00
userchg	5,301	1,065.07	3,449.76	11.00	107.00	267.00	749.00	85,193.00
userratio	5,311	1.96	1.79	1.09	1.34	1.53	1.93	47.78
prc	5,311	21.28	53.59	0.16	3.40	9.13	22.45	2,080.02
size ($mil.)	5,261	1,878.79	8,829.64	0.04	50.65	376.88	1,045.59	425,082.69
ret (%)	5,292	13.88	53.13	−91.79	−7.42	5.00	20.44	1,025.18
daily_buys	5,311	3,380.21	9,644.61	2.00	234.50	726.00	2,475.00	185,252.00
daily_sells	5,311	2,595.85	6,986.41	0.00	186.00	577.00	1,997.50	109,751.00
net_buys	5,311	784.36	3,254.39	−8,876.00	10.00	94.00	463.50	86,325.00
taq_retimb	5,311	0.11	0.17	−0.68	0.02	0.10	0.21	1.00
Panel B: Herding Events in the Attention Sample								
users_close	2,005	2,797.40	10,550.69	106.00	285.00	616.00	1,710.00	168,920.00
users_last	2,005	2,883.89	10,839.49	106.00	292.00	633.00	1,770.00	176,399.00
userchg	2,001	865.64	3,242.63	6.00	63.00	171.00	541.00	49,277.00
userratio	2,005	1.58	0.88	1.04	1.21	1.35	1.62	13.71
prc	2,005	41.79	110.68	0.23	7.43	20.10	44.40	2,990.00
size ($mil.)	2,005	5,187.49	17,919.14	2.75	429.48	1,108.86	3,579.41	425,082.69
ret (%)	2,005	2.60	25.08	−86.67	−9.93	1.06	10.77	281.27
daily_buys	2,004	2,544.34	8,465.08	1.00	232.00	615.00	1,781.75	185,252.00
daily_sells	2,004	1,912.01	5,181.31	0.00	200.00	537.50	1,547.75	98,927.00
net_buys	2,004	632.33	3,778.05	−5,413.00	−11.00	47.00	248.25	86,325.00
taq_retimb	2,004	0.08	0.17	−0.69	−0.02	0.07	0.18	1.00
asvi	1,893	0.61	0.80	0.00	0.00	0.00	1.17	4.04

Two features are particularly important for the interpretation of the empirical analysis. First, Table 3 show that herding events are associated with positive TAQ retail imbalances, therefore suggesting that concentrated retail buying is not limited to Robinhood users during these episodes. In the baseline trading sample, the average herding event is associated with an increase of about 1,065 Robinhood users and a TAQ retail imbalance of 0.11. In the attention sample, the corresponding values are about 866 users and 0.08.

Second, the comparison between the two panels highlights an important sample-selection issue. Herding events in the attention sample are related to stocks with marginally more holders (as evidenced by *users_close* and *users_last*) but less intense herding (as measured by *userratio*). However, the most prominent difference is in market capitalization (*size*). The mean market capitalization of attention-sample herding events is $5,187.49

million, compared with \$1,878.79 million in the baseline herding-event sample, and the median market capitalization is \$1,108.86 million compared with \$376.88 million. Similarly, the average event-day return is only 2.60% in the attention sample, compared with 13.88% in the baseline sample. This difference is not incidental: the Google Trends sample is restricted to stocks in the Russell 3000 for which search intensity can be reliably measured and is therefore tilting the analysis toward larger and more actively followed securities. As a result, the attention sample is well suited to studying whether herding episodes coincide with heightened search attention, but it is less well suited to measuring the full asset-pricing consequences of herding, which are likely to be strongest among smaller and less liquid stocks.

This distinction matters for the interpretation of the later return analyses. The baseline herding-event sample is the appropriate benchmark for studying short-run return reversals and preserving comparability with Barber et al. (2022). By contrast, the attention herding-event sample is used to examine the role of Google search as a confirmation signal. In Panel B, herding events display average *asvi* of 0.61, above the corresponding full-attention-sample average of 0.45. This pattern suggests that Robinhood herding events are associated with heightened search attention, but the larger-stock tilt of the attention sample implies that search-based tests should be interpreted primarily as evidence on the transmission of attention rather than as definitive evidence on the asset-pricing consequences of search-confirmed herding.

IV. Attention, Stock Selection, & Herding Predictability

This chapter examines how limited investor attention is reflected in retail stock-selection patterns and in the predictability of discrete Robinhood herding episodes. Motivated by the attention-driven buying hypothesis of Barber and Odean (2008) and Barber et al. (2022) (which emphasizes that attention should matter more for purchases, where investors search across thousands of potential candidates, than for sales, which are largely confined to previously held positions), I first document that retail trading is highly concentrated in a small subset of tickers, especially on the buy side and especially among Robinhood users.

I then estimate stock and date fixed-effects predictive regressions to assess whether lagged abnormal Google search intensity predicts next-day Robinhood user flows, and whether this relation survives once flow persistence and other salience proxies are controlled for. These regressions show that abnormal search intensity is related to user flows, but also that this relation is strongly shaped by short-run flow persistence. This evidence qualifies the interpretation of Google search volume as a clean leading indicator of retail demand.

Finally, I estimate linear probability models (LPMs) of discrete herding events. These specifications show that lagged abnormal search intensity is positively associated with the probability of a Robinhood herding event, even after controlling for lagged herding, lagged user flows, extreme returns, abnormal volume, and the news environment. I further show that this attention–herding relation is strongest among relatively smaller stocks in the attention sample, consistent with previous findings (Da et al., 2011; De Giorgi et al., 2026; Fedyk, 2021; Garcia, 2026).

4.1. Concentration of Retail Trading

When retail investors decide whether to buy or sell a stock, they face fundamentally different choice problems once one accounts for the size of the relevant stock universe. Because short selling is typically unavailable to retail investors, sales are largely confined to the subset of stocks already held in the portfolio, so that the corresponding consideration set is extremely narrow.[1] By contrast, when choosing what to buy, they must search across thousands of potential candidates. This creates a severe search problem, particularly given retail investors' limited financial literacy (Lusardi, 2015) and limited attention (Kahneman, 2011). In this setting, the literature shows that retail investors tend to restrict their buying consideration set to attention-grabbing stocks, that is, stocks made salient by news coverage, social media, or other channels (Barber & Odean, 2008; Seasholes & Wu, 2007). If retail investors actively trade only a small subset of salient stocks, one should observe a high

[1] On average, retail investors hold between three and four stocks in their portfolio (Seasholes & Wu, 2007).

degree of concentration in retail trading activity, especially on the buy side. This section therefore examines and compares the concentration of Robinhood and TAQ retail trading by following the approach of Barber et al. (2022).

To assess the concentration of retail trading activity, I construct daily rank-share series separately for Robinhood user changes and TAQ retail net buying. For each stock-day, Robinhood net flow is measured by the daily change in the number of Robinhood users holding the stock ($userchg_{i,t}$), while TAQ net flow is measured by retail net buys ($net\_buys_{i,t}$). I then treat purchases and sales separately. On the buy side, I retain only stock-days with strictly positive net flow and interpret the flow itself as the amount bought. On the sell side, I retain only stock-days with strictly negative net flow and convert these values into positive magnitudes so that they can be interpreted as the amount sold[2]. For each trading day and for each series in turn, I rank stocks in descending order of the relevant flow magnitude and compute, for every rank, the share of aggregate daily net buying or net selling accounted for by that stock. I retain the top ten ranks and discard lower-ranked observations, so that each series measures the fraction of total daily activity captured by the ten largest stock-level flows. Following this procedure, Figure 3 reports, for each rank, the time-series average of these daily shares, expressed in percentage terms. Thus, the first bar corresponds to the average fraction of daily net buying or selling attributable to the single most heavily traded stock on a given day, the second bar to the second most heavily traded stock, and so forth through rank ten[3].

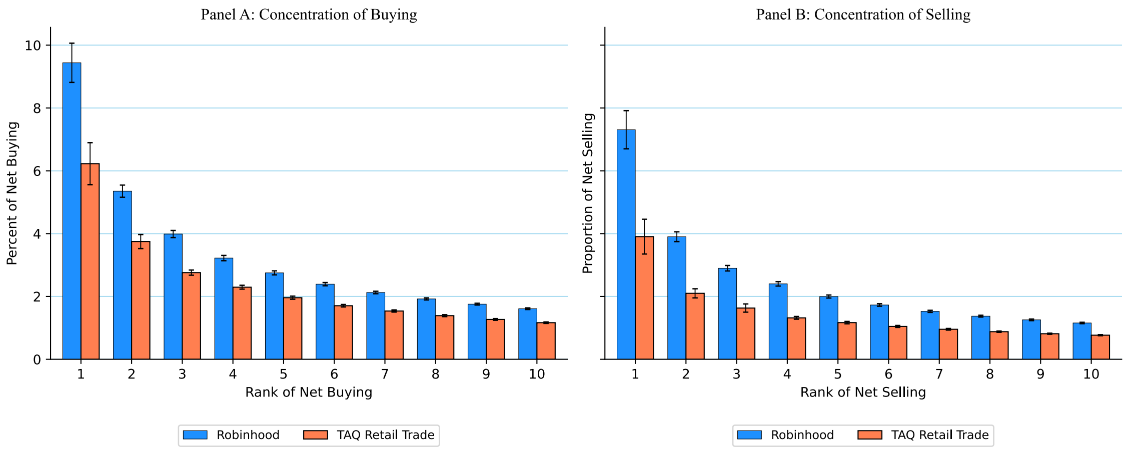
Figure 3 confirms the claims of the literature that retail trading activity is highly concentrated in a small number of stocks and more so on the buy side. On the buy side (Panel A), the most heavily traded stock accounts for 9.44% of daily net buying on Robinhood and 6.23% in TAQ retail data, while the top ten stocks together account for 34.51% and 23.93%, respectively. On the sell side, the most heavily traded stock accounts for 7.31% of daily net selling on Robinhood and 3.90% in TAQ retail data, while the top ten stocks together account for 25.55% and 14.53%, respectively. In line with Barber et al. (2022), Figure 3 also suggests that retail trading is more concentrated for Robinhood users than the broader retail investor population captured in TAQ data. This apparent material difference in concentration levels might be attributable to the facts reported in subsection 3.2.1: Robinhood users are generally younger and more inexperienced than the average retail investor[4], thus plausibly more biased towards attention-grabbing stocks (Seasholes & Wu, 2007). Overall, these findings are consistent with the view that limited attention is reflected in concentrated retail trading, particularly on the buy side, and that

[2] This separation is important to characterize how concentrated retail buying and retail selling are within each day and not to observe netting across purchases and sales.

[3] To quantify sampling uncertainty, I compute standard errors across trading days and construct 95% confidence intervals using the conventional normal approximation.

[4] https://www.nytimes.com/2020/07/08/technology/robinhood-risky-trading.html.

Figure 3. Concentration of Retail Trading Activity. This figure reports the average fraction of daily net buying and net selling attributable to the ten most heavily traded stocks on Robinhood and in TAQ retail data. The left panel focuses on net buying, while the right panel focuses on net selling. Buying is defined using positive net flows; selling is defined as the absolute value of negative net flows. The bars represent the time-series average of these daily shares, expressed in percentage terms. The error bars represent 95% confidence intervals computed across trading days.



this concentration is especially pronounced among Robinhood users.

Finally, to corroborate the rank-based evidence above, I compute, also following Barber et al. (2022), two complementary daily concentration measures for retail trading activity: the Herfindahl-Hirschman Index (HHI) and the share of aggregate activity accounted for by the ten largest stock-level flows. The HHI is defined formally in Equation 4.1.

$$\mathrm{HHI}_t = \sum_{i=1}^{N_t} s_{i,t}^2 \tag{4.1}$$

In Equation 4.1, $s_{i,t}$ denotes the share of stock i in total retail net buying or total retail net selling on day t, and N_t is the number of stocks with non-zero activity on the relevant side of the market that day. A higher HHI indicates that trading is concentrated in fewer tickers. I compute this statistic separately for Robinhood user changes and TAQ retail net buying, and separately for purchases and sales. On the buy side, I retain only stocks with strictly positive net flow and scale each stock's flow by the cross-sectional total of positive flows on that day. On the sell side, I retain only stocks with strictly negative net flow, convert these flows into positive magnitudes, and again normalize by the corresponding cross-sectional total. This yields, for each day and for each dataset, a complete cross-sectional distribution of shares from which the HHI is calculated. In parallel, I compute the cumulative share of daily activity accounted for by the ten largest stocks on each side of the market. The HHI and Top-10 measures are therefore complementary: the former summarizes the concentration of the entire cross-section, whereas the latter isolates the economic importance of the largest trading destinations.

Table 4. Concentration of Retail Buying & Selling

The sample consists of stocks for which both Robinhood user changes and TAQ retail net buying are observed. Statistics are computed from daily concentration measures and reported as time-series averages across trading days. In Panel A, *HH_buy* is the Herfindahl-Hirschman index for stocks with net buying, defined as the sum of squared shares of buying, and *Top10_buy* is the percentage of all net buying accounted for by the 10 stocks with the highest level of net buying. In Panel B, *HH_sell* is the Herfindahl-Hirschman index for stocks with net selling, defined as the sum of squared shares of selling, and *Top10_sell* is the percentage of all net selling accounted for by the 10 stocks with the highest level of net selling. HHIs are multiplied by 10,000 to express them in basis points. Standard errors, reported in parentheses, are computed from the time-series variation in the daily statistics. The difference column reports Robinhood minus TAQ on the common daily sample. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Changes	TAQ Retail Trades	Difference (RH − TAQ)
Panel A: Mean Daily Concentration Measures among Stocks with Net Buying			
HH_buy (bps)	261.414***	170.726***	90.688***
	(22.105)	(32.814)	(11.326)
Top10_buy (%)	34.512***	23.933***	10.579***
	(0.423)	(0.411)	(0.356)
Panel B: Mean Daily Concentration Measures among Stocks with Net Selling			
HH_sell (bps)	173.416***	92.123***	81.293***
	(19.703)	(26.631)	(19.566)
Top10_sell (%)	25.550***	14.531***	11.019***
	(0.380)	(0.364)	(0.386)

Table 4 reports the time-series means of these daily concentration statistics for Robinhood and TAQ, together with standard errors computed across trading days. To ensure that the comparison between the two datasets is not driven by differences in time coverage, I first construct the daily statistics separately for Robinhood and TAQ and then merge them by date, so that the difference between the two is evaluated on the common daily sample only. I report four objects for each dataset: the HHI of net buying, the fraction of net buying attributable to the ten most heavily traded stocks, the HHI of net selling, and the corresponding Top-10 share for selling. For each statistic, I also report the daily mean difference between Robinhood and TAQ and assess its statistical significance using the time-series variation of the date-by-date differences. This procedure provides a compact summary of whether the greater concentration suggested by Figure 3 is a persistent feature of the data rather than a purely visual pattern.

From Table 4, the visual patterns in Figure 3 are confirmed more formally. On the buy side, the HHI of net buying is 261.414 bps for Robinhood and 170.726 bps for TAQ, while the Top-10 share equals 34.512% and 23.933%, respectively. On the sell side, the HHI of net selling is 173.416 bps for Robinhood and 92.123 bps for TAQ, while the Top-10 share equals 25.550% for Robinhood and 14.531% for TAQ. The Robinhood–TAQ difference is statistically significant at the 1% level for all four measures. These results confirm two central features of the data: retail trading is more concentrated on the buy side than on the sell side, and Robinhood trading is systematically more concentrated than the broader retail activity captured in TAQ. This conclusion is, of course, conditional on the sample period under study, which overlaps with the COVID-19 episode and may therefore reflect a particularly unusual composition of retail market participants. Another caveat is that the Robinhood *userchg* and TAQ *net_buys* variables are not directly comparable measures of retail trading activity, so that some of the measured difference in concentration may reflect differences in the underlying activity measures rather than differences in investor behavior alone[5]. Subject to these qualifications, the evidence is consistent with Robinhood users exhibiting a stronger tendency to concentrate their trading in attention-grabbing stocks. The next sections examine whether lagged salience and attention proxies help predict the user flows and herding episodes underlying this concentration.

4.2. Determinants of User Flows

First, I examine the relationship between Robinhood daily stock-level user flows $userchg_{i,t}$ and a lagged measure of abnormal Google search attention $asvi_{i,t-1}$[6], controlling for several lagged variables within a nested linear regression framework. More formally, I estimate variations of the specification in Equation 4.2.

$$userchg_{i,t} = \alpha_i + \delta_t + \beta asvi_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \tag{4.2}$$

In Equation 4.2, α_i and δ_t are stock and date fixed effects[7], respectively, $\mathbf{X}_{i,t-1}$ is a vector of lagged control variables, and $\varepsilon_{i,t}$ is the error term. The coefficient of interest is β, which captures the relationship between search attention and user flows conditional on stock fixed effects, date fixed effects, and the lagged controls. I estimate this specification using ordinary least squares (OLS) with standard errors two-way clustered by stock and date to account for heteroskedasticity and dependence in the error term $\varepsilon_{i,t}$. The design purposefully exploits lagged variables to emphasize the role each regressor plays in *predicting* user flows, rather than describing contemporaneous comovements.

4.2.1. Control Variables

Before estimating the regressions described by Equation 4.2, I first describe more precisely the lagged control variables included in $\mathbf{X}_{i,t-1}$. To capture the strongly documented tendency of retail investors to chase past returns (Ardia et al., 2023; Barber & Odean,

[5] Barber et al. (2022) also report this caveat but an examination of the correlation between two TAQ series, daily new users for each stock and daily order imbalance for each stock, indicates that the two measures are highly correlated and do not produce material differences in concentration statistics.

[6] The abnormal search attention variable $asvi_{i,t-1}$ is defined in Equation 3.1 and captures the lagged abnormal search interest for a stock i on the previous day $t-1$.

[7] The stock fixed effects α_i control for time-invariant characteristics of stocks that may influence user flows, such as the stock's general popularity. The date fixed effects δ_t control for time-specific factors that may affect all stocks simultaneously, such as market-wide news or events.

2008; Eaton et al., 2022), I construct the binary variable $top\_20_{i,t-1}$ which equals one if stock i belongs to the set of the 20 stocks with the largest absolute return on day $t-1$, and zero otherwise. I also include—aligned with the attention literature—a lagged measure of abnormal trading volume defined in a manner analogous to $asvi_{i,t-1}$, which I denote by $abvol_{i,t-1}$ (see Equation 4.3).

$$abvol_{i,t-1} = \ln\left(1 + \frac{volume_{i,t-1}}{\frac{1}{20}\sum_{\tau=2}^{21} volume_{i,t-\tau}}\right) \tag{4.3}$$

Furthermore, to account for persistence in user flows, I include the lagged dependent variable $userchg_{i,t-1}$ as a control. Finally, I also add lagged controls from RavenPack related to news coverage and sentiment. These variables have been described in section 3.4.

4.2.2. Estimation & Results

Table 5 reports estimates of Equation 4.2 across a sequence of increasingly demanding specifications. The coefficient on $asvi_{i,t-1}$ is statistically significant throughout, but its sign reverses once the lagged dependent variable $userchg_{i,t-1}$ is introduced. This pattern suggests that abnormal search intensity is closely tied to highly persistent short-run flow dynamics. In particular, the strong positive coefficient on $userchg_{i,t-1}$ indicates substantial persistence in Robinhood user flows[8], so that the positive relation between search attention and subsequent inflows in the parsimonious specification appears largely to reflect this persistence. Conditional on past order-flow pressure, the remaining variation in $asvi_{i,t-1}$ is instead associated with lower next-day inflows, consistent with a timing effect rather than with sustained incremental buying pressure. An analogous pattern arises for $abvol_{i,t-1}$, which is strongly positive in the baseline specification but becomes economically negligible and statistically insignificant once lagged user flows are controlled for[9]. By contrast, the coefficient on $top\_20_{i,t-1}$ remains positive and highly significant across all columns, implying that stocks with extreme lagged returns are correlated with roughly 22 additional Robinhood users on the following day. This relation is economically meaningful and aligned with the view that extreme price moves make stocks salient to retail investors, thereby drawing incremental attention and participation (Barber & Odean, 2008; Barber et al., 2022; Eaton et al., 2022). The RavenPack sentiment variables, $css_{i,t}$ and $ess_{i,t}$, are also significant predictors of user flows but with opposite signs, thus warranting caution in

[8] In unreported regressions that reproduce exactly the specification in column (4) but with two additional lags of $userchg_{i,t-\tau}$ for $\tau = 5, 10$, I find that all the lags are statistically significant at the 1% level and remain economically meaningful with coefficients of 0.508, 0.182, and 0.059 for $\tau = 1, 5, 10$, respectively.

[9] In further unreported regressions that reproduce exactly the specification in column (4) but exclude the lagged dependent variable $userchg_{i,t-1}$, I find that the coefficients on both $asvi_{i,t-1}$ and $abvol_{i,t-1}$ are positive, 2.391 and 56.369 respectively, and highly significant at the 1% level, thus confirming that the reversal pattern observed in Table 5 is indeed driven by the inclusion of lagged user flows.

Table 5. Determinants of Robinhood User Flows

The table examines the determinants that predict the Robinhood $userchg_{i,t}$ variable ($users\_close_t - users\_close_{t-1}$) using linear regressions estimated by OLS with fixed effects for stock and date. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top\_20_{i,t-1}$ is an indicator that equals one if the absolute return is ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $num\_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num\_news\_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ for which RavenPack assigns a relevance score of at least 75. $ess_{i,t-1} \in [-1,1]$ is the lagged Event Sentiment Score of RavenPack. $css_{i,t-1} \in [-1,1]$ is the lagged and weighted by relevance Composite Sentiment Score of RavenPack. $anl\_chg_{i,t-1}$ captures the lagged change in analyst recommendations for stock i on day $t-1$ (see section 3.4 for more details on RavenPack variables). The sample used to estimate these regressions is the attention sample, i.e. the subset of the baseline trading sample for which Google search volume data are available. Robust standard errors reported in parentheses are clustered by day and stock. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$userchg_{i,t}$			
	(1)	(2)	(3)	(4)
$asvi_{i,t-1}$	2.406***	−0.960***	−0.951***	−0.954***
	(0.606)	(0.298)	(0.298)	(0.298)
$top\_20_{i,t-1}$	60.558***	21.762***	21.782***	21.777***
	(12.027)	(6.776)	(6.778)	(6.776)
$abvol_{i,t-1}$	56.192***	−0.008	0.068	0.182
	(8.952)	(3.662)	(3.667)	(3.664)
$userchg_{i,t-1}$		0.603***	0.603***	0.603***
		(0.046)	(0.046)	(0.046)
$num\_news_{i,t-1}$			0.0002*	0.0002*
			(0.0001)	(0.0001)
$num\_news\_relevant_{i,t-1}$			−0.0018**	−0.0019**
			(0.0009)	(0.0008)
$ess_{i,t-1}$				−9.4544***
				(2.954)
$css_{i,t-1}$				12.648**
				(6.436)
$anl\_chg_{i,t-1}$				169.70
				(152.23)
Observations	934,966	934,966	934,966	934,966
R^2	0.0032	0.3636	0.3636	0.3637
Entity FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

their interpretation[10].

Finally, the buy and sell side regressions reported in Appendix A.2 sharpen the analysis. Persistence is almost entirely a buy-side phenomenon, with $userchg_{i,t-1}$ loading strongly in the buy specification but becoming insignificant in the sell specification, whereas salience variables such as $top\_20_{i,t-1}$ and $abvol_{i,t-1}$ remain positively associated with both inflows and the magnitude of outflows, indicating that attention reallocates retail trading toward a small set of highly visible stocks. Importantly, $asvi_{i,t-1}$ positively predicts sell-side outflows, implying that high lagged search attention for a given stock is associated with more users exiting that stock the following day[11]. By contrast, statistical significance vanishes on the buy side. Overall, these results qualify previous observations in the literature (Da et al., 2011) and reveal that lagged search attention, as measured by abnormal Google search volume, is mostly contemporaneously associated with user flows, thus suggesting that it serves as a *confirmatory* proxy of the salience of a particular stock i at time t rather than indicating *anticipatory* buying pressure. However, the relatively small economic magnitude of the coefficients on $asvi_{i,t-1}$ in the fully specified regressions warrants circumspection.

4.3. Determinants of Herding Events

4.3.1. Estimation & Main Results

To complement the linear user-flow regressions, I now turn to linear probability models (LPMs) specifications of herding events[12]. These specifications address a distinct question: rather than asking whether abnormal search attention predicts the average magnitude of next-day user flows, they test whether elevated lagged search attention is associated with a higher probability that a stock enters a discrete herding state. For comparability and consistency, the design of these specifications is similar to the one of the user-flow regressions and exploits the exact same subset of the data, to the exception of the dependent variable $rh\_herd_{i,t}$ which is now defined as a binary indicator of whether stock i is in a herding state on day t[13]. More formally, I estimate variations of Equation 4.4.

[10] The time correlation between $css_{i,t}$ and $ess_{i,t}$ is 0.6766 but the panel correlation across ticker-date observations drops to 0.2143, so multicollinearity should not be a concern. However, the fact that these two sentiment measures have opposite signs and similar magnitudes suggests that they may be capturing different aspects of the news environment, or that one of them may be picking up noise rather than true sentiment. Further analysis would be needed to disentangle these relations but since the construction of these variables depends on proprietary RavenPack algorithms and do not constitute the focus of my analysis I do not investigate this issue further.

[11] See Table A.1 in section A.2 for more details on the sell-side regression results.

[12] I use LPMs to enhance the comparability with Barber et al. (2022). I also favor LPMs because they accommodate high-dimensional stock and date fixed effects in a transparent way and because the coefficients are directly interpretable as changes in probabilities. Logit or probit specifications would be possible alternatives, but their interpretation with high-dimensional fixed effects is less direct.

[13] For the precise definition of the herding state variable $rh\_herd_{i,t}$, see section 3.5.

$$rh\_herd_{i,t} = \alpha_i + \delta_t + \beta asvi_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \tag{4.4}$$

Table 6 reports estimates of Equation 4.4 across a sequence of increasingly demanding specifications. The coefficient on $asvi_{i,t-1}$ is positive and statistically significant across all specifications, indicating that a one-unit increase in $asvi_{i,t-1}$ is associated with a 0.02 percentage-point increase in the probability that stock i enters a herding state on day t, holding constant stock fixed effects, date fixed effects, and the other covariates. The economic magnitude of this relation is small but non-negligible, especially considering the unconditional mean of *rh_herd* in the regression sample (0.19%). These results are aligned with the ones observed in Barber et al. (2022) and persist even after controlling for lagged herding events, or lagged user flows[14]. The direction of the correlation should not be interpreted in opposition to the pattern described in Table 5, where the coefficient on $asvi_{i,t-1}$ becomes negative once lagged user flows are controlled for. The linear regressions and the LPMs address different empirical questions and the LPMs are especially focused on identifying the role of search attention in extreme relative changes in user flows, thereby capturing tail events.

Turning to the other covariates, the coefficient on the lagged dependent variable $rh\_herd_{i,t-1}$ follows our earlier observations and shows that a stock experiencing a herding event on day $t-1$ is 12.3 percentage points more likely to experience another herding event on day t, thus confirming the strong persistence in herding events documented in Barber et al. (2022). The rationale for this persistence could be explained by the fact that the herding event generates social media attention (De Giorgi et al., 2026), or that herding episodes cluster in time. The coefficient on $top\_20_{i,t-1}$ is also positive and significant, indicating that stocks with extreme lagged returns are more likely to experience buying pressures. The coefficient on $abvol_{i,t-1}$ also points out that stocks with elevated lagged trading volume are more likely to experience herding events. The remaining covariates are either not statistically significant or economically meaningful predictors of herding events in the fully specified model, except for $css_{i,t-1}$ which is positive and significant at the 10% level[15].

[14] Table 6 still demonstrates that the inclusion of $rh\_herd_{i,t-1}$ diminishes the magnitude of the coefficient on $asvi_{i,t-1}$, a pattern similar to the one observed in Table 5 when $userchg_{i,t-1}$ is included. Unreported results show that in the LPMs, the inclusion of $userchg_{i,t-1}$ alone does not significantly affect the coefficient on $asvi_{i,t-1}$, whereas the inclusion of $rh\_herd_{i,t-1}$ alone reduces the coefficient on $asvi_{i,t-1}$ by about 50%.

[15] This particular evidence resonates with the empirics in Chan and Fong (2004), who document that individual investors become more inclined to buy when they perceive the crowd to be optimistic. However, it diverges with Münster et al. (2024), who note that retail investors oftentimes act as contrarians by selling stocks with positive news sentiment.

Table 6. Determinants of Robinhood Herding Events

The table examines the determinants that predict the Robinhood $rh\_herd_{i,t}$ variable (constructed as in Barber et al. (2022), it is a binary indicator taking the value of 1 if a stock is in the top 0.5% of positive user change ratio on day t and at least 100 users on day $t-1$) using linear probability models estimated by OLS with fixed effects for stock and date. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top\_20_{i,t-1}$ is an indicator that equals one if the absolute return is ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $rh\_herd_{i,t-1}$ is the lagged herding indicator. $num\_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num\_news\_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ for which RavenPack assigns a relevance score of at least 75. $ess_{i,t-1} \in [-1,1]$ is the lagged Event Sentiment Score of RavenPack. $css_{i,t-1} \in [-1,1]$ is the lagged and weighted by relevance Composite Sentiment Score of RavenPack. $anl\_chg_{i,t}$ captures the lagged change in analyst recommendations for stock i on day $t-1$ (see section 3.4 for more details on RavenPack variables). The sample used to estimate these regressions is the attention sample, i.e. the subset of the baseline trading sample for which Google search volume data are available. Robust standard errors reported in parentheses are clustered by day and stock. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$rh\_herd_{i,t}$			
	(1)	(2)	(3)	(4)
$asvi_{i,t-1}$	0.0004***	0.0002***	0.0002***	0.0002***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
$top\_20_{i,t-1}$	0.0091***	0.0057***	0.0057***	0.0057***
	(0.0009)	(0.0007)	(0.0007)	(0.0007)
$abvol_{i,t-1}$	0.0115***	0.0078***	0.0078***	0.0078***
	(0.0007)	(0.0005)	(0.0005)	(0.0005)
$rh\_herd_{i,t-1}$		0.123***	0.123***	0.123***
		(0.0076)	(0.0076)	(0.0076)
$userchg_{i,t-1}$(000)		0.0010*	0.0010*	0.0010*
		(0.0005)	(0.0005)	(0.0005)
$num\_news_{i,t-1}$			0.0000	0.0000
			(0.0000)	(0.0000)
$num\_news\_relevant_{i,t-1}$			0.0000	0.0000
			(0.0000)	(0.0000)
$ess_{i,t-1}$				0.0001
				(0.0003)
$css_{i,t-1}$				0.0023*
				(0.0013)
$anl\_chg_{i,t-1}$				-0.0344
				(0.0418)
Observations	934,966	934,966	934,966	934,966
R^2	0.0061	0.0208	0.0208	0.0208
Entity FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

4.3.2. *Heterogeneity of the Attention Relation*

The attention literature has largely reported that the effect of attention on retail trading activity is heterogeneous across stocks with different market capitalizations, with contrasted views about the strength or direction of the effect across attention proxies (Ardia et al., 2023; Da et al., 2011; De Giorgi et al., 2026; Fedyk, 2021; Warkulat & Pelster, 2024). To contribute to this debate, I investigate whether the predictive relation between $asvi_{i,t-1}$ and the probability of herding events displays such behaviors. To do so, I estimate variations of Equation 4.4 where I interact $asvi_{i,t-1}$ with a binary variable $size\_under(k)_{i,t-1}$ that equals 1 if the stock's lagged market capitalization is under the k-th percentile and 0 otherwise. More formally, I estimate the specification in Equation 4.5 for $k \in \{10, 11, \ldots, 89, 90\}$.

$$\begin{aligned} rh\_herd_{i,t} &= \alpha_i + \delta_t + \beta_1 asvi_{i,t-1} \\ &+ \beta_2 asvi_{i,t-1} \times size\_under(k)_{i,t-1} + \theta \mathbf{X}_{i,t-1} + \varepsilon_{i,t} \end{aligned} \tag{4.5}$$

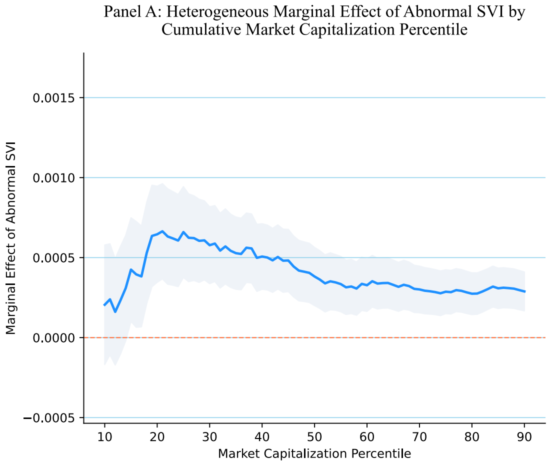
Because the regressions include stock fixed effects, the main effect of $size\_under(k)_{i,t-1}$ is absorbed whenever $size\_under(k)_{i,t-1}$ is time-invariant for a given stock i (i.e. if the stock's market capitalization is always above or always below the k-th percentile during the sample period)[16]. I therefore omit the main effect and focus on the interaction between $asvi_{i,t-1}$ and $size\_under(k)_{i,t-1}$, which remains identified by within-stock variation in abnormal search attention. The coefficient on $asvi_{i,t-1}$ captures the predictive relation between abnormal search volume and herding for stocks above the relevant size threshold, while the interaction coefficient captures how this relation differs for stocks below the threshold. Importantly, since the heterogeneity analysis is estimated in the attention sample, the lower size deciles should be interpreted as relatively small firms within the Google Trends-covered universe, not as the microcap segment of the full `Robintrack` universe. Finally, I compute the corresponding standard error using the delta method, taking into account both coefficient variances and their covariance, and construct 90% confidence intervals based on the asymptotic Gaussian approximation. Panel A of Figure 4 plots this construction.

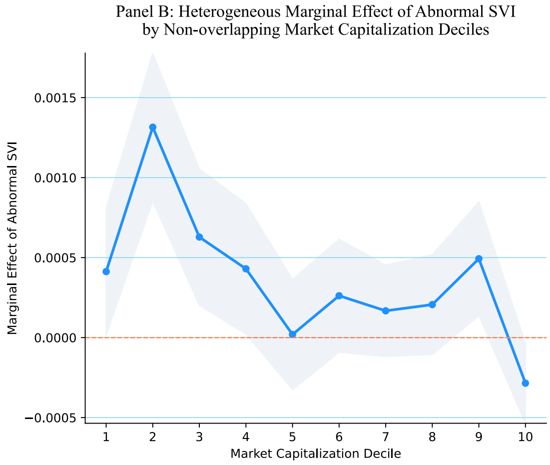
Panel A shows that when small-cap stocks are defined cumulatively, the marginal effect[17] of $asvi_{i,t-1}$ on next-day Robinhood herding is positive throughout and is strongest in the lower tail of the size distribution, peaking around the 20th to 25th market-cap percentile before gradually declining as the threshold expands to include larger firms. This pattern suggests that attention-induced herding is concentrated among relatively small stocks and

[16] In specifications that included the main effect of $size\_under(k)_{i,t-1}$, the regressors matrix was not full rank and the statistical Python package `PanelOLS` automatically dropped the main effect from the regression, thus confirming that it is not identified in the presence of stock fixed effects.

[17] In this context, the word "marginal effect" is used to refer to the partial effect of $asvi_{i,t-1}$ on the probability of herding events for stocks with market capitalization under the k-th percentile, but does not entail any causal interpretation.

Figure 4. Heterogeneity of the Attention Relation. Panel A plots the partial association between $asvi_{i,t-1}$ and the probability of herding events for stocks with market capitalization under the k-th percentile. The x-axis represents the percentile threshold k and the y-axis represents the partial association between $asvi_{i,t-1}$ and the probability of herding events for stocks with market capitalization under the k-th percentile. The shaded area represents the 90% confidence intervals for the partial association. Standard errors are computed by clustering by stock and date and using the delta method to account for the covariance between β_1 and β_2. Panel B plots the partial association between $asvi_{i,t-1}$ and the probability of herding events for each market-cap decile. The construction of the confidence intervals is analogous to the one in Panel A.





becomes diluted when progressively larger firms are added to the treated group.

To confirm this interpretation, I estimate another LPM characterized by Equation 4.6.

$$\begin{aligned} rh\_herd_{i,t} &= \alpha_i + \delta_t + \beta\, asvi_{i,t-1} \\ &+ \sum_{d \neq 10} \gamma_d \big(asvi_{i,t-1} \times 1\{decile = d\}\big) + \theta X_{i,t-1} + \varepsilon_{i,t} \end{aligned} \tag{4.6}$$

Market capitalization is partitioned into ten mutually exclusive deciles using the full cross-sectional distribution of *size*, where decile 1 contains the smallest-cap observations and decile 10 the largest-cap observations. To allow the relation between abnormal search attention and herding to vary flexibly across size groups, I interact $asvi_{i,t-1}$ with indicator variables for each market-cap decile, omitting the largest-cap decile as the reference category. For the same reasons as in the previous specifications, the main effects of the decile indicators are absorbed by stock fixed effects and thus omitted from the regression. Standard errors are two-way clustered by stock and date. Because the coefficient on $asvi_{i,t-1}$ captures the effect for the omitted reference group, namely the largest-cap decile, the marginal effect for any other decile is recovered as the sum of the coefficient on $asvi_{i,t-1}$ and the coefficient on the corresponding interaction term. I compute confidence intervals with the same method as in Panel A. This procedure yields a decile-specific profile of the sensitivity of Robinhood herding probabilities to lagged abnormal search attention across the market-cap distribution. Panel B finally confirms our interpretation of Panel A using non-overlapping market-cap deciles. The association is largest in the second decile and remains positive for several lower and middle deciles, whereas it is close to zero or negative in the largest-cap segment. Taken together, the two panels indicate that the attention-to-herding channel is not uniform across stocks, but is instead most pronounced among smaller caps. A possible explanation is that the predictive relation between search attention and herding is likely to be stronger among smaller stocks because retail order flow may represent a larger fraction of trading activity in those securities (Garcia, 2026). One possible interpretation is that smaller-cap stocks tend to be less liquid (Amihud, 2002), less institutionally followed (Merton, 1987)[18], and more prone to display lottery-like characteristics, thus attracting more retail investors (Kumar, 2009); so a given attention spike could be more likely to be associated with concentrated retail order flow and a measurable herding episode.

The evidence in this chapter supports two conclusions. First, Robinhood trading is more concentrated than broader retail trading, consistent with attention-based stock selection. Second, Google search attention does not behave as a simple leading predictor of average next-day inflows once flow persistence is taken into account. It is more informative for identifying discrete herding episodes, particularly among relatively smaller stocks. This distinction motivates the next chapter, which examines the timing of search attention around salience shocks and asks whether Google search is better interpreted as an anticipatory signal or as a confirmation signal of already-triggered attention.

[18] In Merton (1987) model, investors do not all know or follow the same set of securities. Some stocks are "neglected", in the sense that they are held in the information or consideration sets of fewer investors. Because fewer investors pay attention to them, those stocks are held less widely, their idiosyncratic risk is thus shared by a smaller pool of investors, and investors could require additional expected return as compensation for bearing that risk. The link to my attention-induced herding analysis is that a stock's investor base and visibility are themselves economically important, hence, spikes to visibility and attention should have stronger associations with their demand and pricing.

v. Timing of the Attention Process

This chapter examines the timing of Google search attention. The evidence reported in Chapter 4 so far indicates that abnormal search intensity is linked to Robinhood trading activity, yet its role appears more nuanced than that of a simple *leading* indicator of retail demand. Indeed, the patterns studied previously suggest that search attention might not primarily precede retail trading, but might instead rise contemporaneously with the salience spikes that place a stock in investors' consideration sets and ultimately culminate in herding.

Against this background, I conduct two descriptive event-study analyses of Google search attention from two different angles in order to characterize more precisely whether it behaves as an *anticipatory* signal of retail trading or rather as a *confirmatory* manifestation of heightened stock salience. Importantly, this chapter also motivates the next one, which examines platform-based salience more directly using the regression-discontinuity design around Robinhood's Top Movers eligibility threshold.

5.1. Event-Study around Search Attention Spikes

First, I examine the dynamic behavior of Robinhood user flows around unusually large attention spikes[1] by designing an event-study using a standard event-time framework in the spirit of MacKinlay (1997). Attention is measured across stock i and day t as before with $asvi_{i,t}$. Since the objective of the study is to examine the contemporaneous relation between attention and retail trading, events are defined using abnormal attention at time t. Specifically, for each stock i, I compute the empirical 99th percentile of the historical distribution of $asvi_{i,t}$, $\hat{q}_{i,0.99}$, and classify a stock-date as a high-attention event candidate whenever $asvi_{i,t} > \hat{q}_{i,0.99}$. This stock-level definition is favored over a pooled cutoff to allow for persistent heterogeneity in the unconditional level and volatility of search activity across tickers and ensures that an event corresponds to an unusually large within-stock attention burst. Because elevated attention can persist for several consecutive trading days, I define the event date as the onset of such an episode, i.e. the first day in a run of stock-dates satisfying the event-candidate criterion. I then impose a strict non-overlap restriction: once an event is retained for a given stock, no subsequent event for that stock may occur within the next $2W + 1$ trading days, where $W = 10$ denotes the half-window length. This restriction reduces contamination from adjacent retained attention episodes and makes

[1] Importantly, the word "spike" is used here in a purely descriptive sense and interpreted as a synonym of "surge" or "burst". Therefore, an attention spike is not understood as an exogenous surprise, but rather as a salient episode of elevated attention that may be driven by endogenous or exogenous factors. The event-study design is purely descriptive and does not rely on any exogeneity assumption about the timing of attention spikes.

the event-time profile easier to interpret as the path around an isolated search-attention onset. Critically, antecedent empirical evidence in Chapter 4 suggests that herding events are marked by a significant persistence, thus plausibly cluster in time. The non-overlap restriction is therefore intended to reduce the risk that the event-study design captures the path of retail trading around a single search attention spike rather than a sequence of closely spaced episodes, but is also ultimately a limitation in the sense that the possible clustering behavior of herding events is not examined. Around each retained event, I construct a balanced event-time window over trading days $\tau \in [-10, +10]$. Only complete windows are kept in the final sample, so that changes in the estimated event-time path are not driven by variation in sample composition across horizons. After imposing the non-overlap and complete-window restrictions, the final sample counts 7,554 unique events.

Subsequently, to isolate abnormal retail trading from persistent stock-specific and marketwide components, the Robinhood user-flow series is residualized with respect to stock and date fixed effects, as depicted in Equation 5.1[2].

$$userchg_{i,t} = \alpha_i + \delta_t + \varepsilon_{i,t} \tag{5.1}$$

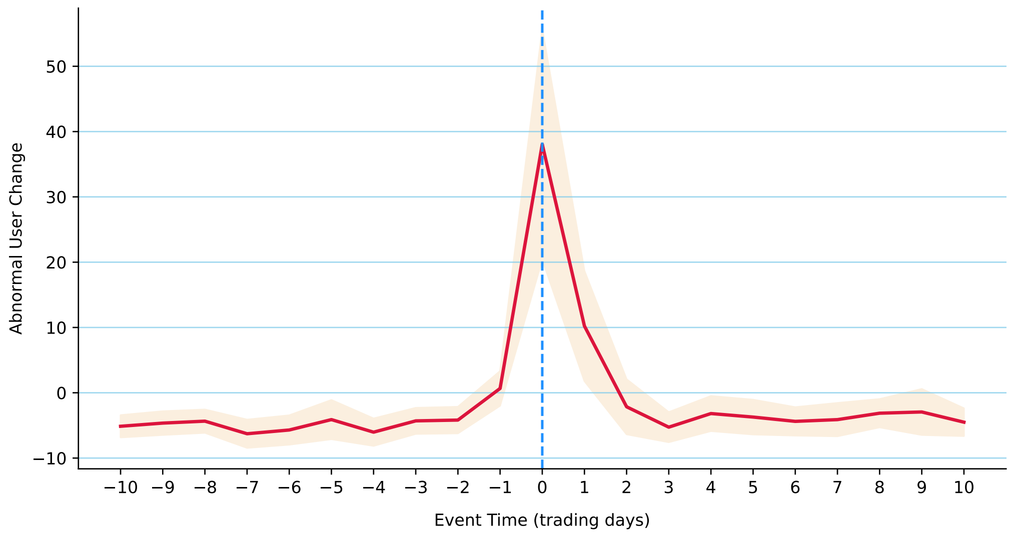
As usual, $userchg_{i,t}$ denotes the daily change in the number of Robinhood users holding stock i, α_i are stock fixed effects, and δ_t are date fixed effects. The residual $\varepsilon_{i,t}$ is interpreted as abnormal user flow, purged of time-invariant cross-sectional differences in stock popularity and of aggregate shocks common to all stocks on a given day. The event study is then conducted on these residualized flows rather than on raw user changes. The descriptive estimates reported in this section are event-time averages of the abnormal user-flow residuals computed in the full attention sample. For each relative event time $\tau \in \{-10, \dots, 10\}$, I compute the mean of $\varepsilon_{i,\tau}$, $\overline{\varepsilon}_\tau$, across all retained events, together with the corresponding cross-event standard error and confidence interval. This procedure yields a nonparametric summary of the typical path of abnormal Robinhood participation before and after a large attention spike. Because the event indicator is based on $asvi_{i,t}$, the resulting profile is informative about whether retail trading tends to rise only after attention has already become unusually high or whether abnormal flows are already present in the days preceding the attention episode[3].

Figure 5 reveals a sharp and highly localized pattern in Robinhood user flows around large search-attention bursts. In the days preceding the event, abnormal user flows are systematically negative. From $\tau = -10$ to $\tau = -2$, $\overline{\varepsilon}_\tau$ remains in a relatively narrow range

[2] I residualize only with respect to stock and date fixed effects because the objective is to describe the event-time path net of persistent stock popularity and marketwide shocks, not to estimate a structural effect of search attention on flows.

[3] The design is therefore purely descriptive, but provides a method to characterize the timing of retail participation around salient bursts of Google search attention and serves as an intuitive bridge to the causal analysis developed in the subsequent chapter.

Figure 5. Event-Time Path of Abnormal Robinhood User Flows around Search Attention Spikes. This figure depicts the average path of abnormal Robinhood user flows around large search attention bursts (the top 1% of $asvi_{i,t}$, specific to each stock). The event date ($\tau = 0$) is defined as the first day in a run of stock-dates where abnormal search volume $asvi_{i,t}$ exceeds the empirical 99th percentile of its historical distribution for stock i. The sample includes 7,554 events. Abnormal user flows are obtained as residuals from a regression of daily user changes on stock and date fixed effects. The solid line represents the average abnormal flow at each relative event time τ, while the shaded area corresponds to the 90% confidence interval based on cross-event standard errors.



between roughly −4 and −6 users per stock-day and is statistically distinguishable from zero throughout. At $\tau = -1$, the estimate becomes essentially zero and loses statistical significance. This pre-event pattern indicates that stocks experiencing an extreme spike in Google search attention are not preceded by abnormal Robinhood buying pressure over a long horizon. In that sense, the figure provides support for the view that elevated search attention does not gradually anticipate a sustained build-up in Robinhood participation.

The most notable feature of Figure 5 is the sharp jump between $\tau = -1$ and $\tau = 0$. On the event day, abnormal Robinhood user flows rise to 38.06 users, with a 90% confidence interval of $[20.41, 55.70]$, by far the largest estimate in the event window. This surge is economically and statistically large relative to the remainder of the path and suggests that the onset of an attention burst is accompanied by an abrupt and highly concentrated increase in retail participation rather than by a smooth pre-trend. At $\tau = 1$, the day after the event, abnormal user flows remain positive at 10.23 users, with a 90% confidence interval of $[1.79, 18.66]$, but the magnitude is already much smaller. Thus, the strongest flow response does not occur after the measured attention burst has materialized. This temporal ordering is economically informative. Because the event is defined using contemporaneous abnormal search intensity, the estimate at $\tau = 0$ corresponds to trading on the same day as an extreme realization of $asvi_{i,t}$. The fact that the largest abnormal inflow occurs at $\tau = 0$ therefore implies that Robinhood participation spikes, as hypothesized, *contemporaneously* with the search-attention burst itself, while the residual effect on the following day is positive but substantially attenuated.

Post-event dynamics should be interpreted with some care. Because the event-study design imposes a non-overlap restriction, the estimated path after $\tau = 0$ is conditional on the event being an isolated attention-episode onset, in the sense that no additional retained onset for the same stock may occur within the next $2W + 1$ trading days. The post-event estimates therefore do not describe the unconditional dynamics following all search-attention bursts, especially in environments where such bursts could cluster in time. Rather, they characterize the evolution of abnormal Robinhood user flows after a single retained onset that is sufficiently separated from other candidate episodes. After $\tau = 1$, abnormal user flows quickly revert and become negative again. Indeed, at $\tau = 2$, the estimates are once more significantly below zero for most horizons, generally fluctuating between -3 and -5 users per stock-day. Hence, the immediate increase in Robinhood participation around the attention spike is not followed by continued abnormal inflows. This pattern is suggestive of a short-lived burst in retail demand that rapidly dissipates and is subsequently reversed. The figure points to a particular dynamic. Large search-attention bursts are associated with a sudden and concentrated increase in Robinhood participation at the onset of the episode, followed by only a modest residual continuation on the next day and a fairly rapid normalization thereafter. The absence of a prolonged positive run-up before the event, combined with the swift decay in $\overline{\varepsilon}_\tau$ afterward, argues against interpreting Google search intensity as a slow-moving anticipatory signal of retail demand. Rather, the evidence indicates that search attention and Robinhood trading respond jointly to the same salience spike, with the strongest retail response occurring essentially simultaneously with the attention burst itself. The next event study further substantiates this claim.

5.2. Event-Study around Extreme Returns

Second, I examine the timing of abnormal Google search attention around large overnight price movements. The purpose of this complementary event study is to reverse the perspective of the previous design. Rather than asking whether Robinhood user flows rise around large attention bursts, the present exercise asks whether search attention itself spikes when a stock experiences an extreme return realization. In this way, the analysis speaks directly to whether Google search intensity behaves as an anticipatory signal of retail demand or instead as a contemporaneous manifestation of stock salience induced by sharp price movements.

To align the event definition with the institutional features of the Robinhood "Top Movers List" mechanism[4], events are defined using the cross-sectional ranking of absolute

[4] The "Top Movers List" is a feature on the Robinhood platform that highlights the top 20 stocks with the highest *absolute* overnight returns and that meet the minimum market capitalization threshold of $c = 300$ million. This feature of the Robinhood app will be developed further in the next chapter.

overnight returns within each trading day. Additionally, a stock is deemed eligible on day t if $size_{i,t} \geq c$, where the baseline cutoff is $c = 300$ million, again to mirror the eligibility criteria of the "Top Movers List" feature. Among eligible stocks, I then rank $|overnight\_ret_{i,t}|$ in descending order within each date and define an extreme-return event $E_{i,t}$ as in Equation 5.2.

$$E_{i,t} = 1\left\{rank_t\left(|overnight\_ret_{i,t}|\right) \leq 20 \ \text{ and } \ size_{i,t} \geq c\right\}. \tag{5.2}$$

Thus, an event corresponds to a stock that belongs to the daily top 20 in absolute overnight returns within the market-cap-eligible universe. I additionally partition the event sample into positive and negative overnight return events according to the sign of $overnight\_ret_{i,t}$.

As in the previous event study, I impose a non-overlap restriction in order to isolate distinct episodes. Let $W = 10$ denote the half-window length. For each stock, I retain an event at date t only if no earlier retained event for the same stock occurred within the preceding $2W$ trading days. This ensures that the final event windows, defined over $\tau \in [-10, +10]$, do not contain two retained event dates for the same stock. The restriction is useful because extreme return realizations can cluster for individual stocks during volatile periods; without such a filter, the event-time profile would be difficult to interpret as the dynamics around a single salient price burst. At the same time, as in the previous event study, the resulting post-event path is conditional on the event being sufficiently isolated from other candidate episodes. This procedure yields 6,247 unique extreme-return events, of which 3,270 are positive and 2,977 are negative.

The outcome of interest is contemporaneous abnormal search attention, $asvi_{i,t}$, i.e. at the same calendar day as the event. Around each retained event, I then build a stacked event-time panel by collecting all stock-day observations in the interval $[t-10, t+10]$ and indexing them by relative event time τ. Only complete windows are retained, so that every event contributes the full set of $2W + 1 = 21$ relative trading days to the final sample.

The first set of estimates is purely nonparametric. For each relative event time τ, I compute the cross-event average of $asvi_{i,t+\tau}$ together with its cross-event standard error. This yields the average event-time path of abnormal search attention around extreme overnight return events. The corresponding confidence intervals are based on the conventional Gaussian approximation. This descriptive profile is reported in Figure 6 Panel A.

To complement these raw event-time averages, I also estimate an event-time regression in the stacked panel (see Equation 5.3).

$$asvi_{e,\tau} = \alpha_e + \delta_t + \sum_{\substack{k=-10 \\ k\neq-1}}^{10} \beta_k \, 1\{\tau = k\} + u_{e,\tau}, \tag{5.3}$$

where e indexes retained events, α_e are event fixed effects, and δ_t are calendar-date fixed effects. The omitted category is $\tau = -1$, so each coefficient β_k measures the difference in abnormal search attention at relative event time k compared with the day immediately preceding the event. Event fixed effects absorb all time-invariant heterogeneity across retained events, while date fixed effects remove aggregate fluctuations in search intensity common to all stocks on a given calendar day. Standard errors are two-way clustered by event and calendar date. The outcomes of this regression are reported in Table 7 and Panel B of Figure 6 [5].

Figure 6. Event-Time Path of Abnormal Search Attention around Extreme Return Events. Panel A depicts the average path of abnormal Google search attention around extreme overnight return events, defined as stock-dates where the absolute overnight return ranks in the top 20 among stocks with market capitalization at the open exceeding \$300 million. Overnight returns are measured from the previous close to the current open. Panel B reports the corresponding event-time regression estimates from Equation 5.3. Both panels report 90% confidence intervals. The sample includes 6,247 non-overlapping events.

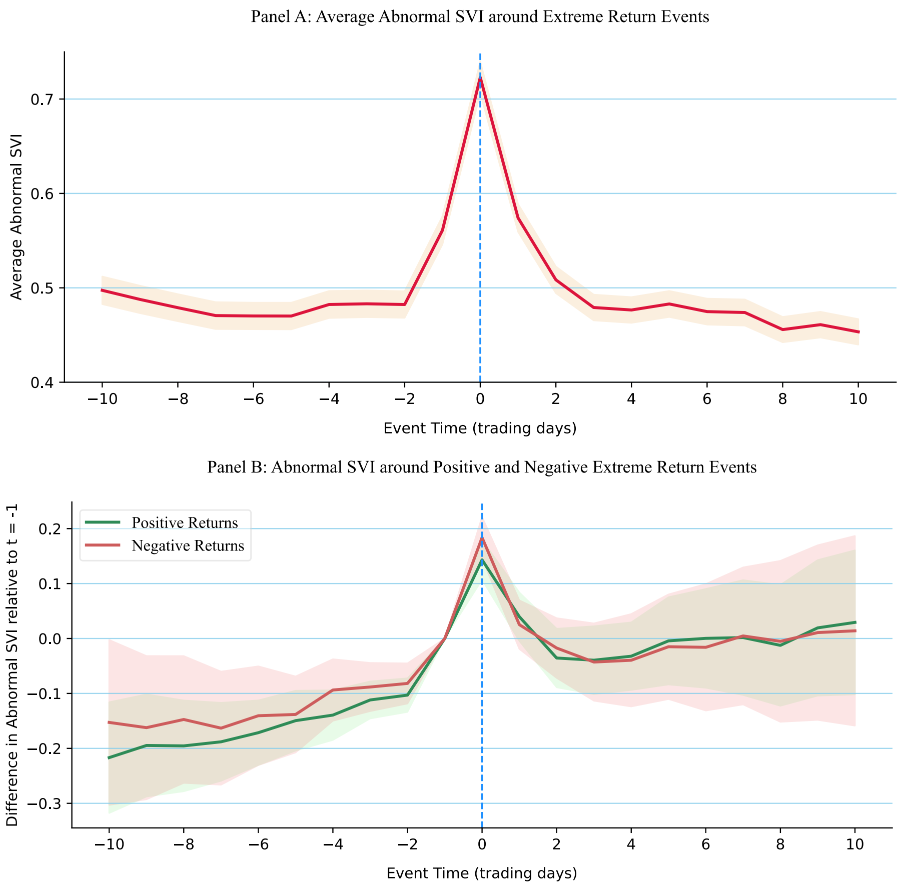


Figure 6 and Table 7 provide a consistent timing pattern. Abnormal Google search attention does not display a long anticipatory build-up before extreme overnight return realizations. In the raw event-time averages reported in Panel A, average $asvi_{i,t}$ remains remarkably stable over the pre-event window, fluctuating narrowly around 0.18 to 0.20

[5] Due to display constraints, only the coefficients for $\tau = -5$ to $\tau = 5$ are reported in Table 7, but the full set of estimates is available in Appendix A.3.

Table 7. Abnormal Search Attention around Extreme Overnight Return Events

The table reports event-time regressions of abnormal Google search intensity around extreme overnight return events. An event is defined on day t when stock i belongs to the top 20 of the cross-sectional ranking of absolute overnight returns, $|overnight\_ret_{i,t}|$, among stocks whose market capitalization at the open exceeds \$300 million. The dependent variable is $asvi_{i,t}$, the contemporaneous abnormal search volume measure aligned with the event date. Relative event-time indicators $1\{\tau = k\}$ are included for $k \in \{-10, \dots, 10\} \setminus \{-1\}$, so that the omitted category is $\tau = -1$. Accordingly, each coefficient measures the difference in abnormal search attention at relative event time k compared with the day immediately preceding the event. For brevity, only the coefficients of the relative event-time from $\tau = -5$ to $\tau = 5$ are reported. All specifications include event fixed effects and calendar-date fixed effects. Robust standard errors reported in parentheses are two-way clustered by event and calendar date. Column (1) reports the full event sample. Columns (2) and (3) split the sample according to whether the overnight return is positive or negative. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1) Overall	(2) Positive Returns	(3) Negative Returns
$\tau = -5$	-0.1559^{***}	-0.1495^{***}	-0.1382^{***}
	(0.0198)	(0.0279)	(0.0353)
$\tau = -4$	-0.1273^{***}	-0.1394^{***}	-0.0937^{***}
	(0.0163)	(0.0231)	(0.0286)
$\tau = -3$	-0.1098^{***}	-0.1119^{***}	-0.0883^{***}
	(0.0128)	(0.0172)	(0.0224)
$\tau = -2$	-0.0939^{***}	-0.1030^{***}	-0.0818^{***}
	(0.0122)	(0.0157)	(0.0187)
$\tau = 0$	0.1788^{***}	0.1432^{***}	0.1837^{***}
	(0.0144)	(0.0184)	(0.0193)
$\tau = 1$	0.0463^{***}	0.0397^{*}	0.0252
	(0.0171)	(0.0225)	(0.0226)
$\tau = 2$	−0.0027	−0.0355	−0.0174
	(0.0192)	(0.0272)	(0.0278)
$\tau = 3$	−0.0157	−0.0395	−0.0429
	(0.0239)	(0.0315)	(0.0360)
$\tau = 4$	−0.0018	−0.0322	−0.0396
	(0.0236)	(0.0314)	(0.0430)
$\tau = 5$	0.0205	−0.0043	−0.0150
	(0.0278)	(0.0406)	(0.0486)
Observations	131,187	68,670	62,517
R^2	0.0781	0.0843	0.0837
Event FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Reference Date	$\tau = -1$	$\tau = -1$	$\tau = -1$

from $\tau = -10$ to $\tau = -2$. The event-time regression sharpens this point. Relative to the day immediately preceding the event, all coefficients from $\tau = -10$ to $\tau = -2$ are negative and highly statistically significant. Because the omitted category is $\tau = -1$, these estimates imply that abnormal search attention is already slightly elevated on the day just before the extreme return event and that this increase occurs only very late in the run-up. Put differently, the evidence is inconsistent with a gradual multi-day increase in search attention before extreme price movements. Rather, attention begins to rise only shortly before the event and does so in a concentrated manner. However, due to the daily frequency of the data, it is still difficult to derive strong and definitive conclusions about the precise temporal order. Caution is required in interpreting the results. It is also possible that a fraction of these events are related to earnings announcements, which could slightly elevate search attention a couple of days before the event, i.e., just before the surprise component of the news generates extreme returns. This hypothesis is nevertheless weakened by the absence of any apparent gradual build-up in search attention.

The important result is the sharp contemporaneous spike at the event date. In the pooled specification, the coefficient at $\tau = 0$ is 0.1788 with a standard error of 0.0144, implying a highly statistically significant increase in abnormal search attention relative to $\tau = -1$. The same pattern is visible in Panel A from Figure 6, where average abnormal Google attention jumps from 0.56 at $\tau = -1$ to 0.72 at $\tau = 0$ (+28.6%), before falling back immediately thereafter. This fast "burst–fade" pattern indicates that extreme overnight returns are accompanied by a pronounced contemporaneous burst in search activity. Economically, this is the key result of the section: abnormal Google search intensity appears, once again, to behave primarily as a *confirmation* of stock salience at the time of the return spike rather than as a slow-moving signal that precedes it by a wide margin.

The post-event profile further reinforces this interpretation. In the regression, the coefficient at $\tau = 1$ remains positive in the full sample, but is already much smaller, at 0.0463, than the event-day estimate, and from $\tau = 2$ onward the coefficients are small and statistically indistinguishable from zero. The nonparametric averages convey the same message. Abnormal search attention returns to its pre-event level immediately after the event; it declines sharply from 0.72 at $\tau = 0$ to 0.57 at $\tau = 1$.

Panel B and Columns (2) and (3) of Table 7 show that this dynamic is present for both positive and negative overnight return events. In each case, pre-event coefficients are negative and statistically significant, and the event-day coefficient is strongly positive. The point estimate is larger for negative overnight returns: the coefficient at $\tau = 0$ equals 0.1837, for negative events, compared with 0.1432 for positive events. This pattern is consistent with Ardia et al. (2023), who document faster Robinhood user reactions to negative extreme returns.

Taken together, the two event studies point to a common timing pattern. The previous analysis showed that Robinhood user flows spike essentially contemporaneously with large search-attention bursts. The present analysis shows, conversely, that abnormal search attention itself spikes simultaneously with extreme overnight return events. The two findings therefore point to the same conclusion: Google search attention is best understood not as a distant leading indicator of retail trading, but as a *confirmatory* manifestation of the salience episodes around which investor attention and retail participation rise together. In this sense, attention appears to be part of the same immediate response process as herding, rather than a separate signal that gradually forecasts it well in advance. This claim will be further supported by the next chapter, which examines this salience channel more directly using variation generated by Robinhood's Top Movers eligibility threshold.

VI. Manufacturing Attention

To formulate a trading decision, retail investors process a variety of signals. However, since their attention is limited and processing a signal is cognitively costly (Kahneman, 1973), they are likely to consult only a subset of the available information. The *complexity* of the signals might well determine which ones are attended to and which ones are ignored (Oprea, 2020; Umar, 2022). Hence, one could expect that herding by retail investors will often occur in response to simple and easily digestible signals, such as an appearance on Robinhood's "Top Movers List" (Barber et al., 2022) or Robinhood's "Top 100 List" (Stein, 2020). Robinhood's Top Movers List is a continuously updated list of 20 stocks with a "market cap of more than \$300 million" and displaying the "largest price movements as measured from the previous market close price"[1]. Importantly, the list does not restrict itself to positive price movements, but also includes stocks with large negative price movements.

I exploit this institutional feature in three steps. First, I replicate the central Top Movers patterns in Barber et al. (2022): because Robinhood displays extreme gainers and extreme losers in the same menu, Robinhood users should react more symmetrically to both types of stocks than broader retail investors. Second, I leverage the \$300 million eligibility threshold to examine whether Top Movers eligibility is associated with a discontinuity in Robinhood intraday user changes among near-threshold extreme movers. Third, I use the same threshold to test whether platform salience spills over into external information acquisition, measured by abnormal Google search intensity. This final exercise links the Top Movers mechanism to the confirmation-signal interpretation developed in the previous part.

6.1. Platform Salience & Stock Selection

6.1.1. Ranking on Absolute Price Movements

The distinctive feature of Robinhood's "Top Movers" list is that it ranks stocks by the magnitude of their price movement rather than by the sign, thereby displaying extreme winners and extreme losers in a single menu. Barber et al. (2022) argue that this design choice generates a sharp comparative prediction: conditional on being a top mover, Robinhood investors should exhibit broadly similar buying responses to top gainers and top losers because both groups are made similarly *available* through the same interface. By contrast, the broader population of retail investors, whose attention is shaped by a more conventional media environment that separates winners from losers, should react more strongly to top gainers than to top losers.

[1] https://cdn.robinhood.com/disclosures/WebDisclosures.pdf.

To replicate this mechanism, I follow Barber et al. (2022) and construct, for each trading day t, the set of 20 eligible stocks whose absolute overnight return from the close of day $t-1$ to the open of day t is largest. Eligibility is defined as having market capitalization at the market open above \$300 million. Because the exact intraday ranking displayed by Robinhood is proprietary and continuously updated, the overnight return ranking is an empirical proxy for the platform ranking at the start of the trading day. Within this daily top-20 set, I assign a top-mover score $Score_{i,t} \in \{1, \ldots, 20\}$, where the stock with the largest absolute price movement receives a score of 20 and the 20th-ranked stock receives a score of 1[2]. I then measure buying activity during the trading day using two outcomes: Robinhood intraday user change and TAQ intraday retail net buying. Robinhood intraday user change is defined as the number of Robinhood users holding stock i at the last observation before 4 p.m. ET minus the number of users at the first observation after 10 a.m. ET[3]. TAQ intraday retail net buying is defined as retail-classified buys minus retail-classified sells during regular market hours. I split the top-20 set into top gainers and top losers using the indicator $\mathbf{1}\{r_{i,t} < 0\}$.

Figure 7. Intraday Retail Buying by Top Movers Ranking. This figure replicates the rank-response design of Barber et al. (2022). The panels report average Robinhood intraday (change in Robinhood users from the first timestamp after 10 a.m. ET to the last timestamp before the market close) user change and TAQ intraday retail net buying for stocks ranked by top-mover status. Panel A ranks stocks by absolute overnight returns from the previous close to the current open. Panel B ranks stocks by absolute daily returns. Rankings are computed among stocks with market capitalization above \$300 million at the market open. The error bar represents the 90% confidence interval for the mean.

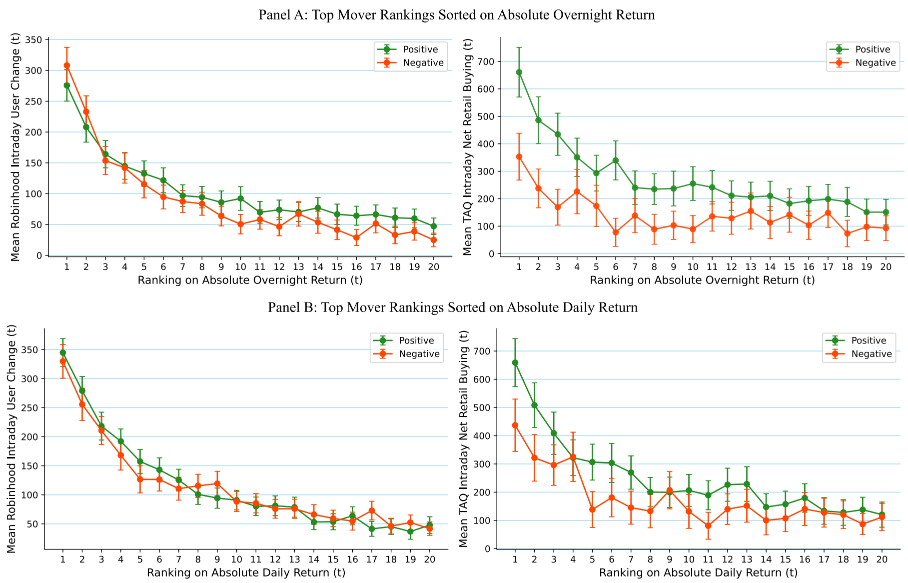


Figure 7 reports the rank-response profiles. Two patterns are relevant. First, buying

2 Because the exact intraday ranking is proprietary and continuously updated, I approximate the relevant ranking using absolute overnight returns from the previous close to the market open, as in Barber et al. (2022).

3 The first observation after 10 a.m. ET is used to reduce contamination from mechanical market-open effects, following the logic in Barber et al. (2022).

intensity increases sharply with rank extremity in both datasets, consistent with attention-based buying in the spirit of Barber and Odean (2008). Second, the gainers–losers contrast differs markedly across platforms. Among Robinhood users, the buying response to top gainers and top losers is almost identical throughout the rank distribution, whereas in TAQ data retail net buying is systematically higher for top gainers than for top losers, especially among the highest-ranked movers.

To quantify these differences, I estimate the rank-based regression used by Barber et al. (2022) and expressed in Equation 6.1.

$$y_{i,t} = \delta_t + \beta_1\, Score_{i,t} + \beta_2\, \mathbf{1}\{r_{i,t} < 0\} + \beta_3\, Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\} + \varepsilon_{i,t} \tag{6.1}$$

In Equation 6.1, $y_{i,t}$ is alternatively Robinhood intraday user change or TAQ intraday retail net buying, δ_t are day fixed effects, and the sample is restricted to the daily top-20 mover set. Standard errors are clustered by day.

Table 8. Top Movers Rank Regressions

This table estimates Equation 6.1 on the daily top-20 mover set. $Score_{i,t}$ assigns 20 to the stock with the largest absolute return and 1 to the 20th. $\mathbf{1}\{r_{i,t} < 0\}$ is an indicator for negative returns (top losers). The dependent variable is Robinhood intraday user change (RH) or TAQ intraday net retail buying (TAQ). Panel A ranks top movers by absolute overnight returns; Panel B ranks top movers by absolute daily returns. All specifications include day fixed effects. Standard errors (in parentheses) are clustered by day. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Changes	TAQ Retail Trades
Panel A: Absolute Overnight Return Rank		
$Score_{i,t}$	8.1266***	17.3109***
	(0.5181)	(1.8012)
$\mathbf{1}\{r_{i,t} < 0\}$	−16.1050**	36.7698
	(7.8472)	(25.1131)
$Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\}$	2.1649***	−9.0662***
	(0.7390)	(2.3264)
Panel B: Absolute Daily Return Rank		
$Score_{i,t}$	12.8722***	19.9989***
	(0.5578)	(1.7311)
$\mathbf{1}\{r_{i,t} < 0\}$	38.8560***	60.4367**
	(8.0687)	(25.5506)
$Score_{i,t} \times \mathbf{1}\{r_{i,t} < 0\}$	−1.9274***	−6.9353***
	(0.7288)	(2.2521)
Day FE	Yes	Yes

Table 8 reports the estimates. The coefficient on $Score_{i,t}$ is positive and highly significant for both Robinhood and TAQ, indicating that buying pressure is strongly increasing in extremity even within the top-mover set. The key distinction arises for top losers. For TAQ retail trades, the interaction term is negative and statistically significant in both panels, indicating that the rank-response profile is weaker for losers than for gainers. For

Robinhood, the corresponding interaction is positive in the overnight-ranking specification and negative but substantially smaller in the daily-ranking specification. The evidence therefore does not imply that Robinhood users treat gainers and losers identically. Rather, it indicates that the conventional gainer tilt is attenuated among Robinhood users relative to broader retail order flow, consistent with the mixed-signed design of the Top Movers list. Interestingly, these results align with the intraday observations of Ardia et al. (2023, p. 14), who find that Robinhood users are "particularly fast at opening positions in stocks that exhibit large negative price movements". This first exercise therefore suggests that the Top Movers List could be a *driver* of retail attention and trading. Indeed, its design—ranking by absolute price movements—generates a distinctive pattern of buying responses to top gainers and top losers, supporting the hypothesis that platform interfaces and information salience can shape retail trading behavior.

6.1.2. The Market-Capitalization Eligibility Threshold

The rank-based evidence in the preceding subsection suggests that the Top Movers list operates as a salience channel that reallocates retail attention. A natural next question is whether this salience is merely correlated with Robinhood buying or whether eligibility for platform visibility is associated with a local discontinuity in Robinhood buying among near-threshold extreme movers. A direct regression discontinuity design based on the boundary between the 20th and 21st mover would be ideal, but in practice the intraday ranking depends on a proprietary algorithm and is time-varying across the day[4], so that any proxy for the cutoff rank would introduce substantial measurement error. I therefore follow Barber et al. (2022) and leverage an eligibility threshold embedded in the platform rules: stocks must exceed $300 million in market capitalization to be displayed in the Top Movers list.

The RD design exploits the fact that the $300 million rule creates a sharp eligibility threshold for platform visibility. The treatment is therefore not directly observed display on the Top Movers list, but eligibility for display among stocks that are sufficiently salient to qualify as candidate top movers. The local estimand is the discontinuity in Robinhood intraday user changes at the market-capitalization threshold for near-threshold extreme overnight movers. The identifying assumption is that, absent Top Movers eligibility, potential Robinhood user changes would vary smoothly with market capitalization around the $300 million cutoff within this restricted set of candidate movers. Under this assumption, a discontinuous change in Robinhood intraday user changes at the threshold provides local evidence that platform eligibility can amplify attention among stocks that might already be salient because of extreme overnight price movements.

This interpretation is local and should not be read as an average effect of Robinhood

[4] https://cdn.robinhood.com/disclosures/WebDisclosures.pdf.

visibility for all stocks. It applies to stocks near the \$300 million threshold that also experience large overnight price movements. The restriction to extreme movers is necessary because the Top Movers list is itself based on price-movement salience, while the matching on absolute overnight returns is intended to make below-threshold controls comparable to above-threshold eligible stocks in the triggering price movement. In this sense, the design compares stocks that are similar in size and overnight-return salience but differ in eligibility for inclusion in the platform display.

To implement the design, I restrict attention to candidate top-mover stocks within a bandwidth around the cutoff. For each day t, the treated sample consists of stocks with market capitalization in $(300, 300 + b]$ million that rank in the top 20 by absolute overnight return among stocks with market capitalization above \$300 million. The control sample consists of stocks with market capitalization in $[300 - b, 300]$ million whose absolute overnight returns are comparable to those of the treated observations. Specifically, below-threshold stocks are retained if their absolute overnight return lies within the range implied by the ratio restriction

$$|R^{\text{treated}}_{\text{overnight},t}|/|R^{\text{control}}_{\text{overnight},t}| \in [0.5, 2], \tag{6.2}$$

relative to at least one treated observation on the same day. I report results for bandwidths $b \in \{50, 75, 100, 125\}$ million, mirroring Barber et al. (2022). I estimate the pooled sharp RD specification expressed in Equation 6.3.

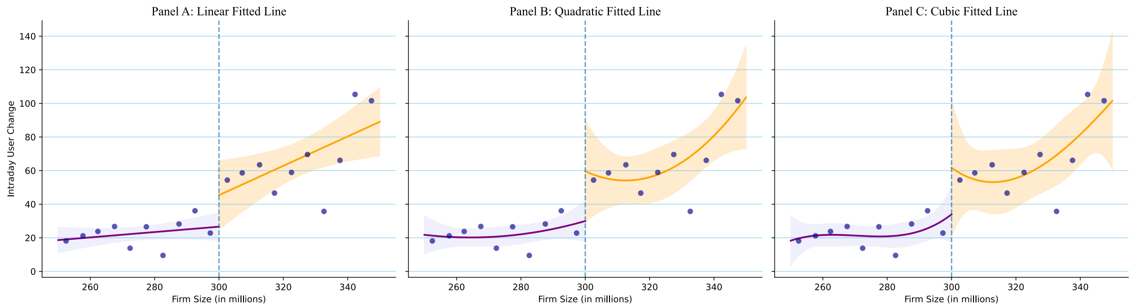
$$\begin{aligned} userchg^{\text{intra}}_{i,t} &= \beta_0 + \beta_1 \mathbf{1}\{size_{i,t} > 300\} + \sum_{n=1}^{N} \beta_2^{(n)} (size_{i,t} - 300)^n \\ &+ \sum_{n=1}^{N} \beta_3^{(n)} (size_{i,t} - 300)^n \cdot \mathbf{1}\{size_{i,t} > 300\} + \varepsilon_{i,t}, \end{aligned} \tag{6.3}$$

Where $userchg^{\text{intra}}_{i,t}$ is Robinhood intraday user change on day t and $size_{i,t}$ is market capitalization at the market open of day t, both expressed in millions. The polynomial order N is set to 1, 2, or 3. Standard errors are clustered by day[5].

Figure 8 provides graphical evidence for the baseline bandwidth ($b = 50$) under linear, quadratic, and cubic fitted functions estimated separately on each side of the cutoff. The fitted relationship between intraday user change and market cap evolves smoothly away from the cutoff, but exhibits a discrete upward shift at \$300 million, consistent with an interface-driven increase in the salience of eligible top movers. The figure should

5 For comparability with Barber et al. (2022), I do not directly implement two-way clustered standard errors, but in unreported preliminary analyses I find that the results are robust to clustering by both day and stock (i.e., no material difference).

Figure 8. Robinhood Intraday User Change around the $300 Million Market-Cap Cutoff. This figure reports binned averages of Robinhood intraday user change against market capitalization for candidate Top Movers used in the RD analysis within a $50 million bandwidth around the $300 million cutoff. The running variable is market capitalization at the market open, measured in millions and centered at $300 million. Solid curves are fitted separately on each side of the cutoff under linear (Panel A), quadratic (Panel B), and cubic (Panel C) specifications. Shaded areas denote 90% confidence intervals.



nevertheless be interpreted jointly with the regression results, because the estimated discontinuity is sensitive to the polynomial specification.

Table 9 reports the corresponding regression estimates for the various bandwidths and polynomial orders, as well as the placebo cutoff results. Across specifications, the coefficient on the above-cutoff indicator β_1 is positive, aligned with the hypothesis that the platform's eligibility rule amplifies Robinhood buying responses to extreme movers. However, the statistical significance of the discontinuity is sensitive to the choice of bandwidth and polynomial order and difficult to establish, with the most robust evidence emerging for the linear specification. On average, the point estimates suggest that crossing the $300 million threshold increases intraday user change by around 30 users, which is economically meaningful given that the average intraday user change in the sample is around 2.75 users.

6.1.3. Robustness Checks

Since the statistical significance of the discontinuity at the $300 million cutoff is sensitive to the choice of polynomial order and bandwidth, I conduct a series of robustness checks to assess the credibility of the main RD estimates.

As a falsification exercise, I also estimate the same specification around a placebo cutoff at $250 million, where no Top Movers eligibility threshold exists. The placebo estimates are positive but statistically indistinguishable from zero across all polynomial specifications: 21.45 users in the linear specification, 19.25 users in the quadratic specification, and 15.64 users in the cubic specification. The absence of statistical significance at this placebo threshold mitigates the concern that the main discontinuity at $300 million merely reflects a generic relationship between market capitalization and Robinhood user changes in this size range. At the same time, the positive point estimates indicate that the placebo exercise

Table 9. Regression Discontinuity in Robinhood Intraday User Change for Top Mover Stocks

This table estimates a sharp regression discontinuity specification around the $300 million market-cap cutoff for inclusion in Robinhood's Top Movers list. The dependent variable is Robinhood intraday user change on day t. The key regressor is an indicator that equals one if market cap at the open of day t exceeds $300 million. Polynomial functions of the centered running variable ($size_{i,t} - 300$) are included as controls, with order $N = 1, 2, 3$. Panel bandwidths vary from $50 million to $125 million. All specifications include day-level clustering of standard errors. Standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood Intraday User Change		
	(1)	(2)	(3)
Panel A: Sample Bandwidth = $50 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	18.75	115.8*	27.773
	(13.995)	(29.632)	(24.903)
Polynomial order, N	1	2	3
Observations	2,811	2,811	2,811
R^2	0.027	0.027	0.027
Panel B: Sample Bandwidth = $75 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	29.25***	19.01	21.49
	(11.184)	(15.063)	(19.533)
Polynomial order, N	1	2	3
Observations	5,058	5,058	5,058
R^2	0.032	0.032	0.033
Panel C: Sample Bandwidth = $100 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	29.683***	19.07	21.95
	(9.597)	(13.602)	(16.943)
Polynomial order, N	1	2	3
Observations	7,834	7,834	7,834
R^2	0.039	0.039	0.039
Panel D: Sample Bandwidth = $125 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	30.91***	18.75	21.02
	(8.403)	(12.888)	(15.582)
Polynomial order, N	1	2	3
Observations	11,087	11,087	11,087
R^2	0.047	0.047	0.047
Panel E: Placebo Cutoff at $250 mil. & Sample Bandwidth = $50 mil.			
$\mathbf{1}\{size_{i,t} > 250\}$	21.45	19.25	15.64
	(15.678)	(16.288)	(21.194)
Polynomial order, N	1	2	3
Observations	3,542	3,542	3,542
R^2	0.015	0.015	0.015

should be interpreted as a falsification check on statistical significance rather than as evidence of a precisely estimated zero effect.

As a covariate-balance check, I re-estimate the local RD specification using observable stock characteristics as dependent variables. The results indicate no statistically significant discontinuity in absolute overnight returns, signed overnight returns, or stock prices at the $300 million cutoff. This is important because the RD sample is constructed around extreme overnight movers, and the absence of a significant return discontinuity mitigates the concern that above- and below-cutoff stocks differ mechanically in the salience of the triggering price movement.

Finally, as a last robustness check, I estimate local-linear RD specifications using triangular-kernel weights, which assign greater weight to observations closer to the $300 million cutoff. The resulting estimates remain positive across all bandwidths and increase as the sample window expands, from 14.54 additional Robinhood users in the $50 million bandwidth to 30.24 users in the $125 million bandwidth. The estimate is imprecise in the narrowest window, but becomes statistically significant at the 10% level for the $75 million bandwidth and at the 1% level for the $100 million and $125 million bandwidths (see Table A.3 in Appendix A.4). This robustness check therefore strongly supports the baseline RD evidence: the positive discontinuity in Robinhood intraday user changes is not an artifact of the global polynomial specification, but also appears under a local-linear specification that gives more weight to observations near the cutoff.

In other economic settings, Chetty et al. (2009) show that *information salience* can materially shape individual behavior. The evidence above is consistent with that view. Robinhood's Top Movers List constitutes a salient and cognitively inexpensive signal to process (Kahneman, 1973), and thus likely increases the probability that the stocks it displays enter retail investors' consideration sets. In this sense, the interface can *manufacture attention* by heightening the salience of selected tickers and, in turn, increasing retail trading activity in those stocks. The evidence is therefore consistent with an attention-amplification mechanism: among stocks that are already salient because of extreme overnight returns, eligibility for platform display is associated with additional Robinhood participation. Still, since statistical significance varies across bandwidths and polynomial orders, this result should be interpreted as local quasi-experimental evidence rather than as a global causal effect of Robinhood visibility.

6.1.4. The Top 100 List: An Alternative Salience Channel

The preceding evidence focuses on Robinhood's Top Movers list, which makes stocks salient by displaying securities with extreme price movements. A complementary platform channel is Robinhood's "100 Most Popular" list. Unlike the Top Movers list, which is based on contemporaneous price extremity, the Top 100 list ranks securities by the number of

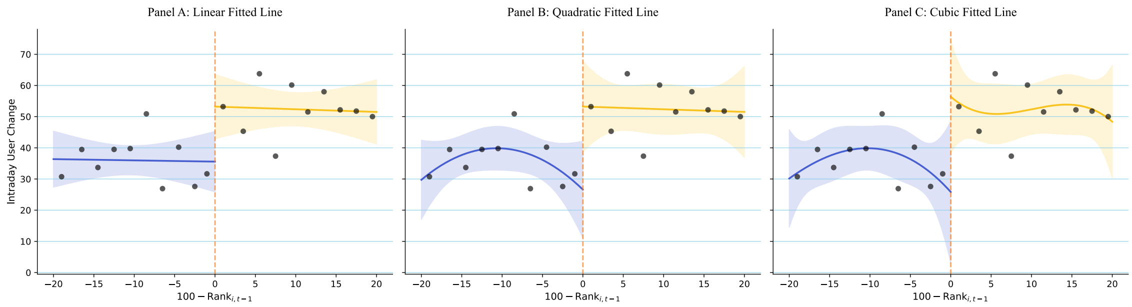
Robinhood users holding them. This distinction is useful because it isolates a different form of app-mediated salience: popularity rather than price movement. Stein (2020) emphasizes that the number of Robinhood users holding each security is not itself shown to users, but that this information is used by Robinhood to construct the list of the 100 most widely held securities. The list therefore converts an otherwise unobserved aggregate holding measure into a simple ranked interface signal. It is also displayed alongside other curated Robinhood lists, including Top Movers, which makes it a natural alternative channel through which the platform can direct retail attention. The Top 100 list therefore constitutes a useful alternative test of the above findings on the Top Movers list.

The purpose of this subsection is to generate descriptive evidence by constructing a regression discontinuity (RD) plot of Robinhood intraday user change against the relative rank of stocks on the Top 100 list (formal estimates are reserved for Appendix A.5). The RD design exploits the discontinuity induced by the Top 100 boundary itself. For each day $t-1$, stocks are ranked by the number of Robinhood users holding them at the close ($users\_close_{i,t-1}$). The treatment indicator equals one if a stock is ranked among the 100 most held stocks and zero otherwise. The running variable is the lagged centered rank, $x_{i,t-1} = 100 - Rank_{i,t-1}$, so that the cutoff is defined at $x_{i,t-1} = 0$. The dependent variable is intraday Robinhood user change at time t, $userchg^{intra}_{i,t}$, which avoids the mechanical endogeneity that would arise from using same-day user changes, since same-day user changes help determine the closing popularity rank. The identifying assumption is that, absent visibility on the Top 100 list, next-day Robinhood user changes would vary smoothly with popularity rank around the cutoff. Under this assumption, a discontinuity in $userchg^{intra}_{i,t}$ at the rank-100 cutoff visually captures the local effect of Top 100 visibility on Robinhood buying pressure. This interpretation is necessarily local: it applies only to stocks close to the rank-100 boundary and relies on the absence of discontinuities in other attention-generating variables, especially returns, and absolute returns.

Figure 9 provides visual evidence of a discrete increase in Robinhood intraday user change at the rank-100 cutoff. Across the linear, quadratic, and cubic specifications, the fitted value immediately to the right of the cutoff is above the fitted value immediately to the left, indicating that stocks barely inside the Top 100 on day $t-1$ experience higher Robinhood user accumulation on day t than stocks barely outside the list. The size of the apparent discontinuity is economically meaningful: the fitted intraday user change rises from roughly 35–40 users below the cutoff to slightly above 50 users above the cutoff. The pattern is clearest in the linear specification and remains visible under more flexible polynomial fits, although the confidence bands widen around the cutoff, especially in the quadratic and cubic panels.

The formal RD estimates reported in Table A.4 of Appendix A.5 provide similar evidence. Across bandwidths of 15 to 30 ranks and polynomial orders from one to three,

Figure 9. Robinhood Intraday User Change around Top 100 List Entry. This figure reports binned averages of Robinhood intraday user change on day t around the rank-100 cutoff of Robinhood's Top 100 list. Stocks are ranked each day $t-1$ by the number of Robinhood users holding them at the close, $users\_close_{i,t-1}$. The running variable is the lagged centered rank, $100 - Rank_{i,t-1}$, so that values greater than or equal to zero correspond to stocks ranked inside the Top 100 on day $t-1$, while negative values correspond to stocks ranked just outside the list. The sample is restricted to observations within a 20-rank bandwidth around the cutoff. Solid curves are fitted separately on each side of the cutoff under linear (Panel A), quadratic (Panel B), and cubic (Panel C) specifications. Shaded areas denote 90% confidence intervals.



the estimated discontinuity is consistently positive. The point estimates imply that stocks barely inside the Top 100 on day $t-1$ attract approximately 13 to 27 additional Robinhood intraday users on day t relative to stocks barely outside the list. Most specifications are statistically significant at the 10% level, although the estimates remain imprecise and the explanatory power is limited, as expected in a local design using high-frequency retail-flow outcomes. Balance checks further indicate no meaningful discontinuity in returns at the rank-100 cutoff, which mitigates the concern that the estimated jump in intraday user changes is driven by a mechanical relationship between popularity rank and return-based salience.[6]

Taken together, the visual and regression evidence supports the interpretation that the Top 100 list operates as an alternative platform-salience channel[7]. This finding, consistent with Stein (2020), complements the evidence on the Top Movers list and reinforces the broader claim that simple interface signals can manufacture attention by reducing the cost of stock selection and making certain securities more salient to retail investors.

6.2. The Attention Transmission Mechanism

The preceding section established that Robinhood's Top Movers interface is a salient and behaviorally consequential signal: within the displayed set, Robinhood users respond approximately symmetrically to extreme gainers and losers, and trading around the $300

[6] Below the cutoff for the bandwidth of 20, the average return is 0.124% with a standard deviation of 6.72%, while above the cutoff, the average return is 0.163% with a standard deviation of 5.13%.

[7] The result should nevertheless be interpreted locally and cautiously: the design identifies variation only for stocks close to the rank-100 boundary and relies on the smoothness of other attention-generating variables around that cutoff.

million eligibility threshold indicates that platform visibility can generate discrete increases in buy-side participation. The Top 100 list provides complementary evidence that the platform's salience effects are not limited to price-movement-based signals, but also arise from popularity-based signals. This section goes one step further and asks whether platform-induced salience also propagates *outside* the app in the form of heightened external information acquisition. This test is central to the chapter because it links the platform-salience design to the thesis's broader distinction between origination and confirmation signals. If the Top Movers list expands the set of stocks that retail investors actively consider, then it should not only affect trading, but also increase the likelihood that investors search for those stocks. Abnormal Google search volume $asvi_{i,t}$, the central metric of this thesis, provides a natural way to test this prediction. If the discontinuity in Robinhood trading at the \$300 million cutoff is driven by the platform's ability to manufacture attention, then we should also observe a discontinuity in $asvi_{i,t}$ at the same threshold, reflecting that the platform's salience induces investors to seek information about the eligible stocks. It should nevertheless be pointed out that since Robinhood's user base is only a subset of the overall retail investor population, discontinuity in $asvi_{i,t}$ at the cutoff is not guaranteed, as the platform's influence on search attention is diluted by the presence of other non-Robinhood investors who do not respond to the Top Movers list.

6.2.1. Estimation & Results

To operationalize this idea, I re-estimate the market-cap regression discontinuity design of Barber et al. (2022) but replace intraday user change with abnormal Google search intensity $asvi_{i,t}$ as the dependent variable (Da et al., 2011, 2025). The running variable is market capitalization at the market open, and the treatment indicator is eligibility for display on the Top Movers list, i.e. $\mathbf{1}\{size_{i,t} > 300\}$.

The identifying logic is the same as in the preceding RD analysis. The coefficient of interest now captures the local discontinuity in contemporaneous abnormal search attention at the \$300 million eligibility threshold among near-threshold extreme movers. The relevant identifying assumption is that, absent Top Movers eligibility, abnormal Google search intensity would vary smoothly with market capitalization around the cutoff within this restricted candidate-mover sample. As stated above, this test is demanding because Robinhood users represent only a subset of all investors whose searches contribute to Google Trends. A discontinuity in $asvi_{i,t}$ would therefore indicate that platform eligibility is associated with an information-acquisition response large enough to appear in aggregate Google search activity, despite Robinhood users being only a subset of the investors contributing to Google Trends. Specifically, on a matched top-mover sample, as in the previous RD analysis, I estimate Equation 6.4.

$$asvi_{i,t} = \beta_0 + \beta_1 \mathbf{1}\{size_{i,t} > 300\} + \sum_{n=1}^{N} \beta_2^{(n)} (size_{i,t} - 300)^n + \sum_{n=1}^{N} \beta_3^{(n)} (size_{i,t} - 300)^n \cdot \mathbf{1}\{size_{i,t} > 300\} + \varepsilon_{i,t}, \tag{6.4}$$

In Equation 6.4 $size_{i,t}$ is market capitalization in millions at the market open of day t, and $N \in \{1, 2, 3\}$ denotes linear, quadratic, and cubic specifications. Standard errors are clustered by day, reflecting that the treatment assignment and the construction of the local top-mover set are day-specific.

Figure 10. Google Abnormal Search Volume Index (ASVI) around the $300 Million Market-Cap Cutoff. This figure shows a binned scatterplot for the Google abnormal search volume index $asvi_{i,t}$ against market cap for stocks. The sample consists of Top Mover stocks used in the RD analysis within a $50 million bandwidth around the $300 million cutoff. Solid curves are fitted lines estimated separately on each side of the cutoff under linear (Panel A), quadratic (Panel B), and cubic (Panel C) specifications. Shaded areas denote 90% confidence intervals.

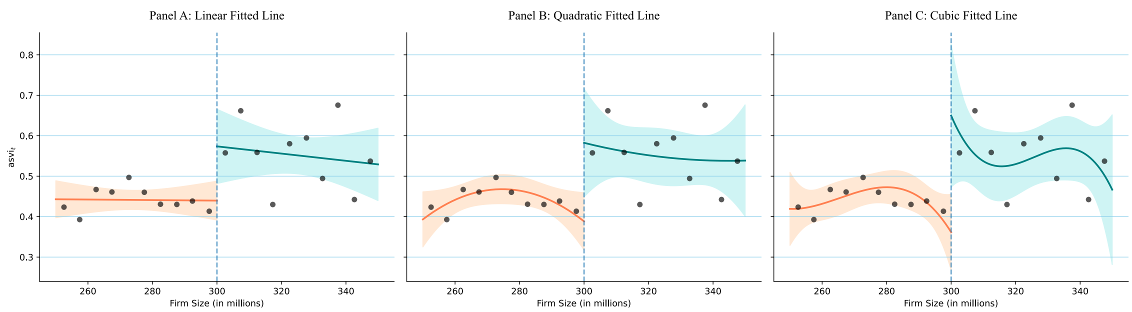


Figure 10 plots binned averages of $asvi_{i,t}$ against market capitalization within the baseline bandwidth of $50 million around the threshold, together with fitted lines estimated separately on each side of the cutoff. Across polynomial orders, the fitted relationship between abnormal search intensity and market cap is smooth away from the threshold but exhibits a visible upward shift at $300 million. This graphical discontinuity indicates that, conditional on being a top mover near the size cutoff, eligibility for the Top Movers display is associated with higher contemporaneous Google search attention.

Table 10 complements Figure 10 and reports the corresponding RD estimates. The estimated discontinuity $\hat{\beta}_1$ is positive and statistically significant across all bandwidths and polynomial specifications. In the baseline window ($50 million), the estimated discontinuity ranges from 0.134 to 0.288, depending on the polynomial order. Because $asvi_{i,t}$ is defined as a log transformation, these magnitudes correspond to economically non-trivial increases in abnormal search activity: a coefficient of 0.134 implies roughly a 14% increase relative to the counterfactual level of $1 + svi/\overline{svi}$ at the cutoff, while a coefficient of 0.288 implies an increase of about 33%.

Table 10. Regression Discontinuity in Abnormal SVI for Top Mover Stocks

This table estimates a sharp regression discontinuity specification around the \$300 million market-cap cutoff for inclusion in Robinhood's Top Movers list. The dependent variable is abnormal search volume index $asvi_{i,t}$ on day t. The key regressor is an indicator that equals one if market cap at the open of day t exceeds \$300 million. Polynomial functions of the centered running variable ($size_{i,t} - 300$) are included as controls, with order $N = 1, 2, 3$. Panel bandwidths vary from \$50 million to \$125 million. All specifications include day-level clustering of standard errors. Standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1)	(2)	(3)
Panel A: Sample Bandwidth = \$50 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	0.134**	0.194**	0.288**
	(0.064)	(0.095)	(0.126)
Polynomial order, N	1	2	3
Observations	3,066	3,066	3,066
R^2	0.004	0.005	0.006
Panel B: Sample Bandwidth = \$75 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	0.159***	0.170**	0.218**
	(0.051)	(0.076)	(0.102)
Polynomial order, N	1	2	3
Observations	5,488	5,488	5,488
R^2	0.003	0.003	0.004
Panel C: Sample Bandwidth = \$100 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	0.170***	0.148**	0.223***
	(0.043)	(0.066)	(0.086)
Polynomial order, N	1	2	3
Observations	8,093	8,093	8,093
R^2	0.003	0.003	0.003
Panel D: Sample Bandwidth = \$125 mil.			
$\mathbf{1}\{size_{i,t} > 300\}$	0.118***	0.183***	0.144*
	(0.038)	(0.060)	(0.077)
Polynomial order, N	1	2	3
Observations	11,023	11,023	11,023
R^2	0.002	0.002	0.002
Panel E: Placebo Cutoff at \$250 mil. & Sample Bandwidth = \$50 mil.			
$\mathbf{1}\{size_{i,t} > 250\}$	0.090	0.044	0.066
	(0.062)	(0.096)	(0.132)
Polynomial order, N	1	2	3
Observations	3,353	3,353	3,353
R^2	0.001	0.001	0.001

The effect remains positive and precisely estimated in wider bandwidths, suggesting that the result is not driven by a narrow local anomaly. The placebo cutoff at $250 million yields positive but statistically insignificant estimates, which mitigates the concern that the ASVI discontinuity simply reflects a generic relation between market capitalization and search activity in this size range. Finally, the local-linear triangular-kernel estimates reported in Table A.3 in Appendix A.4 provide additional support for this interpretation. When Google ASVI is used as the dependent variable, the estimated discontinuity remains positive across all bandwidths, ranging from 0.132 to 0.160, and is statistically significant throughout. Thus, the ASVI discontinuity is not an artifact of the global polynomial specification. Taken together, the placebo and local-linear results strengthen the interpretation that Top Movers eligibility is associated with a discrete increase in contemporaneous search attention among near-threshold extreme movers.

6.2.2. Origins & Confirmations of Attention

Google search is often regarded as a relatively direct measure of investor attention because it reflects an active information-acquisition decision rather than passive exposure (Da et al., 2011). Yet the present evidence indicates that search itself is subject to a prior selection stage. Retail investors face an enormous universe of potential securities, so some upstream mechanism must first determine which stocks enter the set of candidates deemed worth investigating, much as in the buying decision itself. The Top Movers and Top 100 lists are precisely such mechanisms. By curating a small subset of tickers and displaying them prominently within the trading interface, the platform lowers the effective cost of attention allocation and increases the likelihood that these stocks enter retail investors' consideration sets (Barber & Odean, 2008; Barber et al., 2022; Kahneman, 1973; Stein, 2020). Once a stock has entered that set, search naturally follows as a complementary action through which investors gather contextual information, read news, and verify recent price movements.

From this perspective, abnormal Google search intensity is best interpreted not as an autonomous driver of retail trading, but rather as a confirmatory manifestation of salience, a conclusion consistent with Chapter 5. The RD design isolates an interface-driven shift in visibility, and the fact that $asvi_{i,t}$ increases at the eligibility threshold implies that search attention is, at least in part, a *subsequent* outcome of the platform's information architecture. Taken together, these findings suggest a useful classification of signals within the attention-trading literature. *Origination signals* are signals that directly trigger attention and trading, such as inclusion in the Top Movers list or extreme returns. *Confirmation signals*, by contrast, are signals that reveal whether this temporary salience has in fact translated into investor attention. In this framework, Google search is most plausibly understood as a confirmation signal rather than an origination signal[8].

[8] In Appendix B I provide additional evidence for this interpretation.

VII. Return Dynamics around Attention-Induced Herding

The preceding chapters show that Robinhood users display more concentrated buying and selling than the broader retail investor population. They are also disproportionately drawn to attention-grabbing stocks, responsive to salient information cues (origination signals), and prone to herding. Taken together, these features suggest that Robinhood trading may have asset-pricing implications, as in Ardia et al. (2023), Barber and Odean (2008), Barber et al. (2022), and Da et al. (2011). In particular, episodes of concentrated Robinhood buying might be associated with temporary price pressure or investor overreaction to salient information, both of which predict subsequent return reversals (Barber et al., 2022; Coval & Stafford, 2007). This chapter studies return dynamics around Robinhood herding events and interprets the evidence as descriptive and predictive rather than causal.

7.1. Event Study Analysis

The first analysis studies return dynamics in event time around Robinhood herding episodes. Following the empirical approach in Barber et al. (2022), I define day 0 as the herding day and compute abnormal returns as the difference between the stock return and the CRSP value-weighted market return. In addition to daily abnormal returns, I report buy-and-hold abnormal returns (BHARs), which compare the compounded return earned by holding the event stock with the compounded market return over the same horizon. BHARs are computed separately for the pre-event and post-event windows. For the pre-event window, the BHAR from day −10 to event day τ is defined as in Equation 7.1.

$$BHAR_{i,\tau} = \prod_{t=-10}^{\tau} (1 + R_{i,t}) - \prod_{t=-10}^{\tau} (1 + R_{m,t}), \qquad \tau \leq 0. \tag{7.1}$$

For the post-event window, the same logic is applied from day 1 onward, so that post-event BHARs measure the return earned after the herding episode has occurred. Standard errors are clustered by calendar event day, which allows for arbitrary cross-sectional dependence among events occurring on the same trading day. This correction is important because many herding episodes may be concentrated on days with common marketwide or attention-related shocks. However, because longer-horizon BHARs overlap across event windows, the corresponding standard errors should still be interpreted with caution. I therefore complement this event-time evidence below with a brief calendar-time portfolio analysis.

Table 11. Event-Time Abnormal Returns

This table reports abnormal returns around Robinhood user herding events. Abnormal returns (AR) are computed as the raw return minus the CRSP value-weighted market return. Abnormal returns are averaged across all events. Buy-and-hold abnormal returns ($BHAR$) are computed as the product of one plus the stock's return through event day t less the product of one plus the market return for the same period (computed separately for the pre-event and post-event windows). Standard errors are computed by clustering on event day. *% Positive* is the percentage of returns that are positive. Herding events are defined as the top 0.5% of stocks with positive user change ratio on day 0 and a minimum of 100 users on the prior day. The sample contains 5,292 herding events. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Event Day	AR (%)	Std. Err. (%)	% Positive	BHAR (%)	Std. Err. (%)	% Positive
			Pre-Event			
-10	0.04	0.13	48%	0.04	0.13	48%
-9	−0.04	0.15	47%	0.08	0.24	47%
-8	0.34	0.26	47%	0.40	0.36	46%
-7	−0.14	0.16	46%	0.21	0.42	46%
-6	−0.25*	0.13	46%	−0.12	0.38	45%
-5	0.21	0.18	48%	0.33	0.54	46%
-4	0.33**	0.16	49%	0.53	0.50	47%
-3	0.63***	0.22	48%	1.40**	0.60	48%
-2	0.61**	0.24	50%	1.86***	0.58	49%
-1	3.82***	0.42	56%	5.89***	0.75	52%
			Event Day			
0	13.80***	0.86	64%	21.74***	1.52	58%
			Post-Event			
1	−1.11***	0.24	43%	−1.11***	0.24	43%
2	−0.77***	0.18	42%	−1.92***	0.29	40%
3	−0.34**	0.16	44%	−2.44***	0.29	39%
4	−0.26*	0.14	44%	−2.73***	0.31	38%
5	−0.31**	0.13	44%	−3.16***	0.32	38%
6	−0.23*	0.14	45%	−3.41***	0.35	38%
7	−0.17	0.13	46%	−3.63***	0.36	37%
8	0.17	0.16	46%	−3.63***	0.36	37%
9	0.33	0.23	46%	−3.54***	0.37	37%
10	0.10	0.16	45%	−3.59***	0.37	37%
11	−0.07	0.13	43%	−3.78***	0.36	37%
12	0.03	0.11	46%	−3.82***	0.37	37%
13	−0.01	0.12	45%	−3.87***	0.39	37%
14	−0.09	0.10	46%	−4.02***	0.40	37%
15	−0.01	0.10	46%	−4.03***	0.42	36%
16	0.10	0.11	47%	−3.94***	0.44	36%
17	0.02	0.13	46%	−3.81***	0.52	36%
18	0.23	0.20	47%	−3.69***	0.53	37%
19	0.14	0.12	47%	−3.66***	0.53	37%
20	0.04	0.10	47%	−3.71***	0.53	37%

7.1.1. Results in the Baseline Sample

Table 11 shows a sharp asymmetry between the return dynamics before and after herding episodes. During most of the pre-event window, abnormal returns are economically small and only weakly significant. The pattern changes in the final days before the event. Average abnormal returns rise to 0.63% on day −3, 0.61% on day −2, and 3.82% on day −1, before reaching 13.80% on the herding day itself. The corresponding pre-event BHAR reaches 21.74% by day 0. These magnitudes indicate that Robinhood herding episodes are not ordinary trading days: they occur precisely when stocks have already experienced large price movements. This accords well with the attention-based mechanism in Barber and Odean (2008) and Barber et al. (2022) and evidence from Garcia (2026), according to which retail investors are drawn disproportionately to salient stocks, especially stocks with extreme recent returns (Ardia et al., 2023)[1] or unusually high visibility. These empirical observations also reinforce this thesis' interpretation of the herding events as attention-induced phenomena, where extreme returns and platform design features ((Glaze & Moss, 2026) demonstrate the role of data visualization in shaping Robinhood users' attention) serve as origination signals that coordinate the attention of many retail traders at the same time.

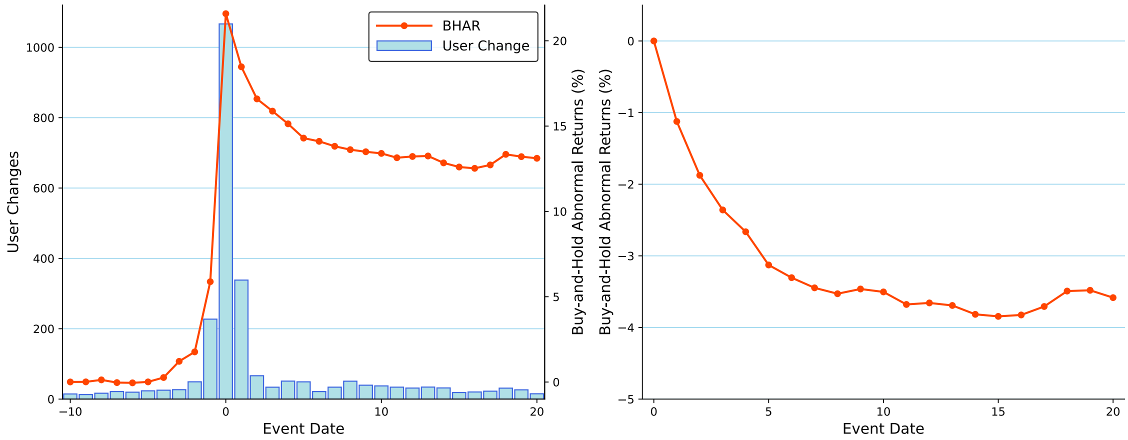
The post-event pattern is materially different. Abnormal returns become negative immediately after the herding day. The average abnormal return is −1.11% on day 1 and remains negative and statistically significant over the first six post-event trading days. The economic magnitude is substantial: the post-event BHAR is −3.16% after five trading days and −3.71% after 20 trading days. The decline is not confined to a small number of extreme observations. Only 38% of post-event BHARs are positive after five days, and only 37% remain positive after 20 days. Hence, the reversal is both economically meaningful and broad-based across herding episodes. This pattern is therefore more naturally interpreted as transitory price pressure than as a permanent information-driven revaluation. The evidence suggests that extreme returns act as origination signals that draw Robinhood users to salient stocks, after which concentrated Robinhood (retail) buying *might* temporarily and marginally elevate prices further around the herding day, with part of this effect subsequently dissipating. It corroborates Barber et al. (2022), who document negative abnormal returns following intense Robinhood buying, and Da et al. (2011), who show that retail attention predicts short-run price increases that later reverse. It also relates to Tetlock (2011), who finds that individual investors overreact to salient but partially redundant information, thereby generating temporary deviations from fundamental values.

Figure 11 visualizes the same dynamics[2]. The left panel shows that the increase in

[1] Ardia et al. (2023) show that Robinhood users respond extremely fast to origination signals, such as extreme returns. Garcia (2026) documents that these signals are particularly important when other information sources are scarce (e.g., no major news events, no earnings announcements, etc.).

[2] The design of the panels is inspired by Barber et al. (2022), who use a similar approach to illustrate the

Figure 11. Returns around Herding Events. The panel on the left depicts mean buy-and-hold returns (BHARs) and Robinhood user changes from 10 days before to 21 days after herding events. The red line represents BHARs, whereas the blue bars represent user changes. The panel on the right displays post-event mean BHARs starting from day 0. Herding events are defined as the top 0.5% of stocks with positive user change ratio on day 0 and a minimum of 100 users on the prior day.



Robinhood users is concentrated around day 0, precisely when BHARs peak. The right panel isolates the post-event horizon and makes the reversal particularly clear: post-event abnormal performance deteriorates rapidly over the first week and then stabilizes at a negative level. The figure also supports the interpretation that herding episodes are associated with a short-lived demand burst. Extreme price movements typically precede Robinhood herding by a few hours (Ardia et al., 2023) and appear to serve as origination signals. Robinhood inflows then concentrate around the herding episode and may contribute to additional, transitory price pressures, after which returns reverse as this episode-specific pressure dissipates[3].

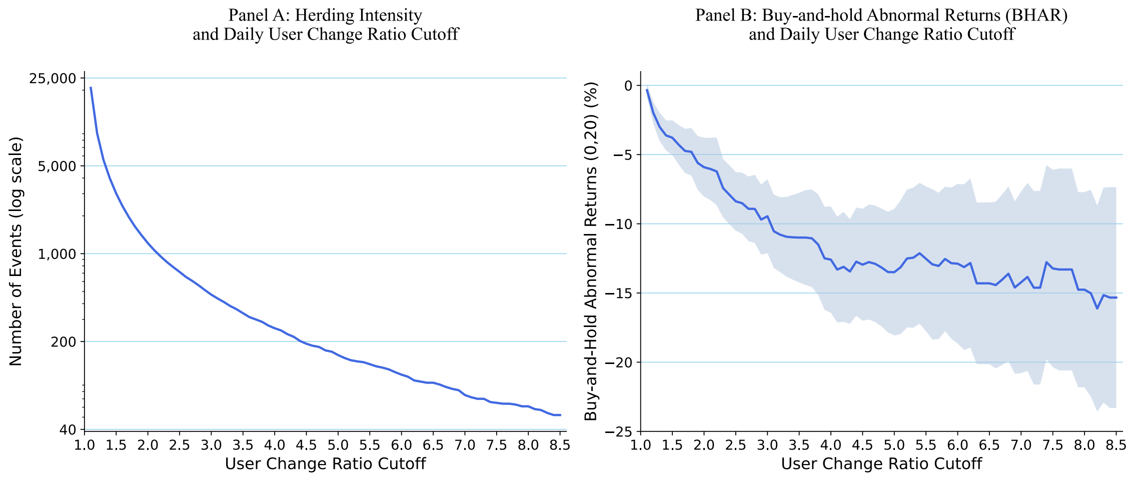
I next examine whether the magnitude of the reversal varies with the intensity of herding. Figure 12 repeats the analysis using alternative user change ratio cutoffs to define herding episodes. Panel A shows the expected trade-off: as the cutoff becomes more demanding, the number of identified episodes falls sharply. Panel B shows that the post-event reversal becomes more severe as the cutoff increases. This (monotonic) relation is important. If the reversal were entirely unrelated to retail flow intensity, one would not necessarily expect it to become more severe as the user-change-ratio cutoff increases. This pattern is therefore suggestive, although not conclusive, evidence that the intensity of retail herding is related to subsequent underperformance.

Taken together, the results support the central hypothesis of this chapter. Attention-

return dynamics around herding events.

[3] In some cases, herding events are followed by “selling–herding” events episodes (less dramatic in magnitude than buying herding events), which may be either a reaction from Robinhood users to the correction of the price by another class of investors or Robinhood users selling after the extreme returns to realize gains (Barber et al., 2022, p. 3182).

Figure 12. Price Reversal & Herding Intensity. The panels display how herding intensity and 20-day buy-and-hold returns (BHARs) vary with the daily user change ratio cutoff to identify herding episodes for stocks with at least 100 Robinhood users. Panel A plots the number of herding episodes on a log scale against the user change ratio cutoff. Panel B plots the 20-day BHARs (%) against the user change ratio cutoff. The daily user change ratio cutoff varies from 1.1 to 8.5.



induced herding is associated with large contemporaneous price increases and economically meaningful subsequent reversals. The evidence is therefore consistent with a mechanism in which extreme price movements (along the visual cues related to them on Robinhood.com) first serve as origination signals that draw Robinhood investors to salient stocks, after which concentrated Robinhood inflows may add transitory price pressure that dissipates in the subsequent reversal. At the same time, the event-study estimates should not be interpreted as fully resolving the asset-pricing implications of herding episodes. The overlapping nature of long-horizon event returns and the possibility of common shocks across events warrants further investigation.

7.1.2. Results in the Attention Sample

As a robustness check, I repeat the event-study analysis in the attention sample, that is, the subset of stocks for which Google Search Volume Index data are available. Because this sample is based on Russell 3000 constituents and is therefore mechanically tilted toward larger firms, the return effects documented above are either attenuated or, in several cases, no longer statistically significant (see Appendix C.1 for further details). It aligns with the previous chapters' evidence and the literature (De Giorgi et al., 2026; Fedyk, 2021; Garcia, 2026) showing that herding episode effects are more pronounced among smaller stocks, where attention-induced trading is also more likely to exert price pressure. The exercise highlights a central limitation of the attention sample. Google search data are available only for this restricted universe of stocks, which limits my ability to study whether herding episodes accompanied by a confirmatory attention signal have distinct asset-pricing implications.

Nevertheless, to examine whether search attention is associated with stronger return effects around Robinhood herding episodes, I conduct an unreported heterogeneity test by partitioning herding events according to whether contemporaneous abnormal Google search activity is unusually high. Specifically, I classify a herding event as high-attention if $asvi_{i,t}$ lies in the top quartile of the distribution of abnormal search volume index (ASVI) across herding events, and compare these events with the remaining normal-attention herding episodes. This exercise should be interpreted descriptively, since contemporaneous ASVI may itself respond to the same price movement, news event, or trading activity that defines the herding episode.

The unreported results indicate that high-attention herding events are substantially more extreme on the event day. Average abnormal returns equal 7.16% for high-ASVI events, compared with 1.53% for normal-ASVI events, implying a highly statistically significant difference of 5.63 percentage points. The corresponding event-day BHAR difference is similarly large, at 7.65 percentage points. Post-event return dynamics are more muted. High-ASVI events display some evidence of a sharper short-run reversal, with the BHAR difference reaching −2.08 percentage points by day 8. However, the difference largely disappears by day 20, and a joint test of equality of the post-event abnormal-return path fails to reject the null ($p = 0.26$). Contemporaneous search attention appears to corroborate particularly salient and extreme herding episodes, but the evidence does not support a statistically robust difference in the full post-event return path[4].

7.2. Calendar-Time Portfolio Analysis

7.2.1. Portfolio Construction

In the preceding event-study analysis, overlapping event windows and cross-sectional dependence issues slightly complicate the interpretation of standard errors. Therefore, to complement the event-time evidence, I conduct a brief calendar-time portfolio analysis similar to Barber et al. (2022). Importantly, this analysis should be interpreted as descriptive and predictive rather than an attempt to construct an exploitable trading strategy. The portfolio is designed to capture the return dynamics around herding episodes in a way that is consistent with the event-study evidence, but it does not account for transaction costs, liquidity constraints, or other frictions that would be relevant for actual trading.

The calendar-time portfolio is constructed from event-position returns. For each herding event in stock i on day s, I short one event position at the close of day s and follow the

[4] Again, these results should be interpreted with caution due to the bias introduced in the market capitalization distribution of the attention sample. Further research with a more comprehensive attention dataset would be needed to fully understand the role of search attention in shaping the asset-pricing consequences of herding episodes.

stock return over the next five trading days, $s+1$ to $s+5$. Overlapping herding events, including repeated events in the same stock, are treated as distinct event positions. Let us define $\mathcal{A}_t$ as the set of event positions that are active on calendar day t. I then compute the daily portfolio return as

$$R_{p,t} = -\frac{1}{|\mathcal{A}_t|} \sum_{(i,s)\in\mathcal{A}_t} R_{i,t}, \tag{7.2}$$

where $R_{i,t}$ is the realized stock return on day t for an active event position. Hence, the portfolio return is the equal-weighted average return across all active event positions on a given day. Missing stock-day returns are dropped only for the corresponding event position and day.

Subsequently, I compute the daily abnormal return of this portfolio (α) by regressing the portfolio excess return on the Fama and French (2015) five factors model as well as a momentum factor inspired by Carhart (1997)[5]:

$$\begin{aligned} R_{p,t} - R_{f,t} = \alpha + \beta_m (MKT\text{-}RF)_t + \beta_s SMB_t + \beta_h HML_t \\ + \beta_r RMW_t + \beta_c CMA_t + \beta_{mom} MOM_t + \varepsilon_t \end{aligned} \tag{7.3}$$

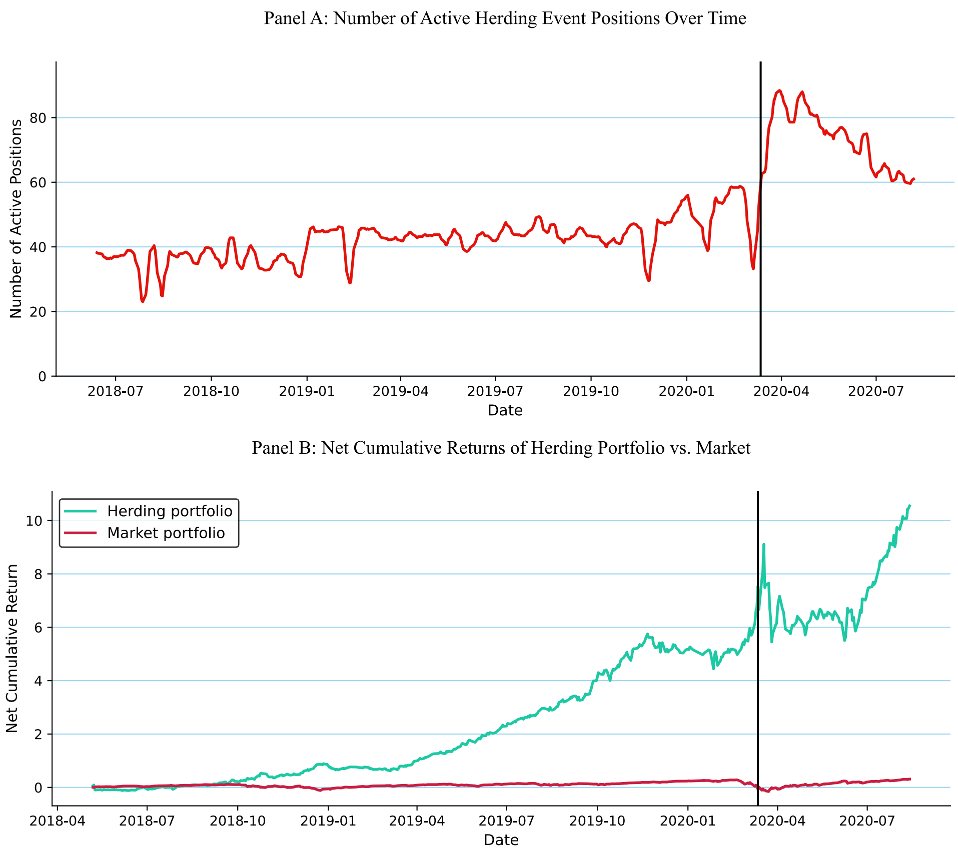
where $R_{p,t}$ is the return of the calendar-time portfolio on day t, $R_{f,t}$ is the risk-free rate, and remaining terms are the five Fama-French factors on day t as described in Fama and French (2015), as well as a momentum factor inspired by Carhart (1997). The regression is estimated using Newey-West standard errors with a lag length of 5 to account for potential autocorrelation in the portfolio returns. Before estimating the regression, I observe the number of positions in the portfolio and compare the net cumulative return of the calendar-time portfolio with the net cumulative return of the market portfolio[6] to generate a visual representation of the return dynamics.

Figure 13 visualizes the structure and returns of the calendar-time portfolio. The left panel shows that the number of active positions in the portfolio is relatively constant over time before the COVID-19 pandemic (around 40 positions), but increases sharply during the pandemic. Since the number of Robinhood users and trading activity surged during the pandemic (Ozik et al., 2021), the increase in herding events and consequently in active positions is consistent with the underlying mechanism of attention-induced herding. The right panel shows that the short strategy generates a net cumulative return

[5] The Fama-French five factors model includes the market risk premium, size, value, profitability, and investment factors (Fama & French, 2015, p. 2). These factors are commonly used to explain the cross-section of stock returns and freely available from Kenneth French's data library at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. The momentum factor is also derived from Kenneth French's data library.

[6] The net cumulative returns of the portfolios do not account for transaction costs or other frictions and are computed as the cumulative product of one plus the daily returns minus one.

Figure 13. Composition & Returns of the Herding Portfolio. Panel A plots the number of active event positions in the calendar-time portfolio. Each herding event generates a short position opened at the close of the herding day and held for five trading days. Panel B plots cumulative returns for the equal-weighted short herding portfolio and the CRSP value-weighted market portfolio, computed as cumulative products of daily returns. The black vertical line marks the U.S. COVID-19 national emergency declaration on March 13, 2020.



that significantly overperforms the market portfolio, consistent with the negative post-event returns documented in the event-study analysis. However, these returns are not risk-adjusted and transaction costs have not been accounted for, so the economic significance of this outperformance should be interpreted as descriptive evidence rather than an actual trading strategy recommendation. The regression results below provide a more rigorous assessment of the risk-adjusted performance of the calendar-time portfolio.

7.2.2. Regression Results

The regression results are reported in Table 12. The intercept (α) is positive and statistically significant across all samples, indicating that the calendar-time portfolio generates abnormal returns even after controlling for common risk factors.

Table 12. Calendar-Time Portfolio Regressions

This table reports calendar-time portfolio regressions for the short herding strategy. The dependent variable is the daily excess return of a portfolio that shorts \$1 of each herding-event stock at the close of the herding day and holds the position for five trading days. The explanatory variables are the five Fama & French (2015) factors: the market excess return ($MKT - RF$), size (SMB), value (HML), profitability (RMW), investment (CMA), and the momentum factor (MOM). The regression specification is given in Equation 7.3. The full sample covers the complete sample period. The pre-COVID sample ends before March 13, 2020, and the COVID sample begins on March 13, 2020. Standard errors, reported in parentheses, are Newey-West standard errors with five lags. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	Full Sample	Pre-COVID	COVID
α	0.0043***	0.0042***	0.0061***
	(0.001)	(0.001)	(0.002)
MKT-RF	−0.6875***	−0.4027***	−0.8292***
	(0.087)	(0.126)	(0.099)
SMB	−0.6934***	−0.4664*	−0.5354*
	(0.219)	(0.239)	(0.291)
HML	−0.4389*	0.1615	−0.8972**
	(0.263)	(0.171)	(0.366)
RMW	1.2192***	1.3960***	1.0804*
	(0.315)	(0.397)	(0.558)
CMA	0.7658**	0.5285	1.6345**
	(0.386)	(0.375)	(0.772)
MOM	0.1763	0.8195***	−0.2557
	(0.283)	(0.223)	(0.339)
Observations	562	454	108
R^2	0.356	0.158	0.670
Adjusted R^2	0.349	0.147	0.650
Newey-West lags	5	5	5

The estimated α is positive and statistically significant in all three specifications: 0.43% per day in the full sample, 0.42% per day in the pre-COVID sample, and 0.61% per day in the COVID sample. These magnitudes are economically large and indicate that the short herding portfolio earns positive risk-adjusted returns after controlling for the Fama-French five factors and the momentum factor. This evidence complements the event-study results by showing that the post-event reversal is not only visible in event time, but also translates into positive abnormal performance in a calendar-time trading strategy. The result is consistent with the interpretation that intense Robinhood buying generates temporary price pressure that subsequently reverses, as in Barber et al. (2022). The estimates should nevertheless be interpreted with caution, since the strategy abstracts from short-selling constraints, transaction costs, and implementation frictions, all of which are likely to be relevant for the stocks most affected by retail herding. The subsample estimates suggest that abnormal performance is stronger during the COVID period, when both Robinhood activity and the number of active portfolio positions increase sharply, as shown in Figure 13. This pattern is consistent with the view that the return consequences of attention-induced herding became more pronounced when retail participation was unusually high. It should also be noted that in unreported regression results, the calendar-time portfolio did not generate significant abnormal returns with observations from the attention sample ($\alpha = -0.0004$, $p = 0.685$), thereby reinforcing the outlined limitations of this dataset for studying the asset-pricing implications of Google search attention.

7.3. TAQ Herding Events

7.3.1. Alternative Herding Definition

Robinhood-based herding may differ from broader retail herding in ways that confound the return evidence. This TAQ robustness exercise tests whether the reversal pattern survives when herding is identified from the full retail population rather than from Robinhood holders alone. Specifically, I construct a TAQ-based abnormal retail trading indicator designed to capture days on which retail trading activity is unusually intense relative to the stock's own recent history. For each stock i and trading day t, I first define retail order imbalance $taq\_retimb_{i,t}$ as previously

$$taq\_retimb_{i,t} = \frac{daily\_buys_{i,t} - daily\_sells_{i,t}}{daily\_buys_{i,t} + daily\_sells_{i,t}} \tag{7.4}$$

where $daily\_buys_{i,t}$ and $daily\_sells_{i,t}$ denote the number of retail-classified buy and sell executions, respectively, as measured using the TAQ retail classification. I next construct an abnormal retail trading intensity measure. Let total retail trading volume be defined as

$$retail\_activity_{i,t} = \text{daily\_buys}_{i,t} + \text{daily\_sells}_{i,t}. \tag{7.5}$$

I then scale current retail trading activity by the stock's own lagged 20-day average retail trading volume:

$$abn\_retail\_activity_{i,t} = \frac{retail\_activity_{i,t}}{\frac{1}{20}\sum_{k=1}^{20} retail\_activity_{i,t-k}}. \tag{7.6}$$

This variable captures whether retail trading activity in stock i on day t is unusually large relative to its own recent history. I require 20 non-missing lagged trading days to compute the denominator.

Finally, I define TAQ-based herding events in two steps. First, on each trading day t, I identify stocks with directionally higher retail buying by requiring their retail order imbalance to lie above the median (50th percentile) of the daily cross-sectional distribution:

$$taq\_retimb_{i,t} > Q_{0.50,t}\left(taq\_retimb_{i,t}\right). \tag{7.7}$$

Second, among these imbalanced stocks, I rank stocks by abnormal retail trading intensity, $abn\_retail\_activity_{i,t}$, in descending order. To keep the TAQ-based event sample directly comparable to the Robinhood-based event sample, I select on each day the same number of TAQ herding events as there are Robinhood herding events on that day. Hence, if N_t^{RH} denotes the number of Robinhood herding events on day t, then the TAQ herding indicator is defined as

$$taq\_herd_{i,t} = \mathbf{1}\left\{\begin{matrix} taq\_retimb_{i,t} > Q_{0.50,t}\left(taq\_retimb_{i,t}\right) \quad \wedge \\ rank_{i,t}\left(abn\_retail\_activity_{i,t}\right) \leq N_t^{RH} \end{matrix}\right\}; \tag{7.8}$$

This construction ensures that TAQ herding events represent stocks that experience both high retail buying pressure and unusually high retail trading activity, while also holding fixed the daily number of events relative to the Robinhood-based definition[7]. The latter feature is important because it prevents differences in the event-study results from being driven mechanically by differences in the number or timing of identified events. The procedure results in a sample of 5,292 TAQ herding events and an overlap of 1,846 events (35.32%) with the Robinhood-based event sample.

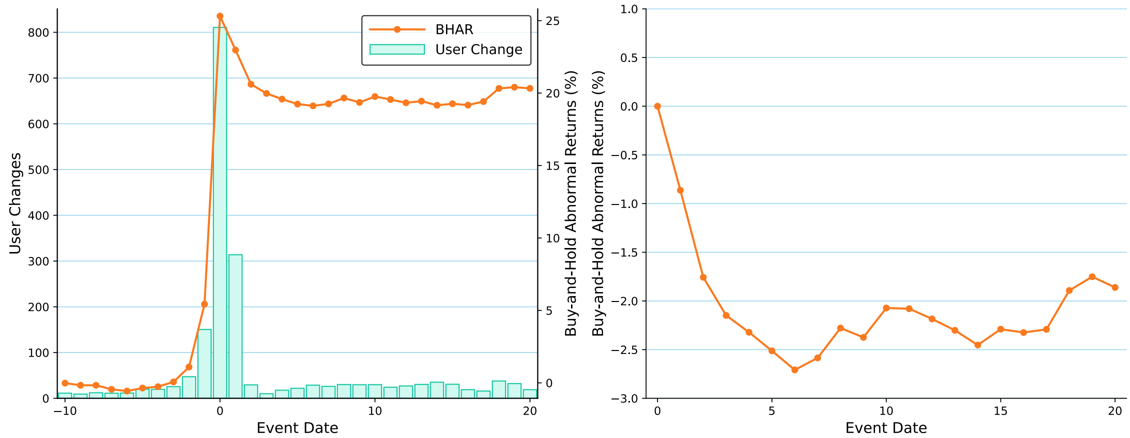
7.3.2. TAQ Herding Results

Using this alternative TAQ-based herding indicator, I repeat the same event-study analysis as in the baseline specification and the identical calendar-time portfolio analysis as in the previous sections. For brevity, I report condensed results.

First, Figure 14 shows that the return dynamics around TAQ herding events are relatively similar to those around Robinhood herding events. Abnormal returns are small and mostly insignificant during the pre-event window, but become large and significant in the final days

[7] An approach also followed by Barber et al. (2022).

Figure 14. Returns around TAQ Herding Events. The panel on the left depicts mean buy-and-hold returns (BHARs) and Robinhood user changes from 10 days before to 21 days after TAQ herding events. The red line represents BHARs, whereas the blue bars represent user changes. The panel on the right displays post-event mean BHARs starting from day 0. TAQ herding events are defined as stocks with retail order imbalance above the median and abnormal retail trading intensity in the top N_t^{RH} on day t, where N_t^{RH} is the number of Robinhood herding events on day t.



before the event, peaking at 17.50% on the event day—a larger effect than for Robinhood herding events. Post-event returns are negative and significant for several days after the event, with a BHAR of −2.56% after five trading days and −1.98% after 20 trading days. The reversal is broad-based, with only 38% of post-event BHARs remaining positive after five days and 20 days. In substance the spike in returns around TAQ herding events is even more pronounced than around Robinhood herding events, but the subsequent reversal is less severe.

Second, the calendar-time portfolio that shorts TAQ herding events generates a positive and statistically significant risk-adjusted return ($\alpha = 0.0031$, $p < 0.001$) after controlling for the Fama-French five factors and the momentum factor. The magnitude of the abnormal return is economically large, at 31 bps per day, but smaller than the abnormal return generated by the Robinhood-based calendar-time portfolio. This pattern is consistent with the event-study results showing that the return dynamics around TAQ herding events are similar but slightly less pronounced than those around Robinhood herding events (Barber et al., 2022). Taken together, the TAQ-based results indicate that similar return dynamics arise when herding episodes are defined using broader retail trading activity rather than Robinhood holdings alone. Although this does not by itself identify the attention mechanism operating outside Robinhood, it supports the external validity of the reversal pattern.

VIII. Conclusion

This thesis asked whether the sequence from salient market signals to active information acquisition, concentrated Robinhood participation, and subsequent return reversals can be traced empirically across complementary research designs. The evidence supports a qualified but affirmative answer. Robinhood user changes are more concentrated than broader TAQ retail order flow, and their herding behavior is associated with lagged salience measures such as extreme returns, and prior Robinhood user changes. Google search activity does not behave as a simple distant predictor of average future Robinhood inflows. Instead, it spikes around the same salient episodes around which Robinhood participation becomes concentrated. Near the $300 million Top Movers eligibility threshold, and among stocks with extreme price movements, platform eligibility is associated with discrete local increases in both Robinhood participation and abnormal Google search activity; suggesting that how information is presented influences decisions. The return evidence points in the same direction, but with distinct identification limits. Herding events are associated with heightened levels of stock-specific attention and followed by economically meaningful short-run reversals in the baseline sample. The calendar-time portfolio evidence is consistent with this pattern and show that attention-induced trading tend to generate predictable poor returns for the average individual investor. Taken together, these findings do not identify a single causal chain from attention to prices. They instead provide mutually consistent evidence on the formation, transmission, and asset-pricing relevance of retail attention.

The central contribution of the thesis is to distinguish between origination signals and confirmation signals. Extreme returns, abnormal trading volume, news, and platform-based visibility are best interpreted as origination signals: they make a stock salient and can place it into retail investors' consideration sets. Abnormal Google search activity, by contrast, behaves more like a confirmation signal. At the daily frequency used in this thesis, search activity appears contemporaneous with salience episodes rather than a clean leading predictor of average future Robinhood inflows. This interpretation is supported by three empirical patterns. First, the coefficient on lagged abnormal search volume in user-flow regressions changes sign once lagged Robinhood user changes are included, suggesting that search activity is closely tied to short-run flow dynamics around salient events. Second, the event-time evidence shows that abnormal search attention and Robinhood user flows rise around the moment of maximum salience rather than in a stable lead-lag sequence. Third, the regression-discontinuity estimates around the Top Movers eligibility threshold provide local quasi-experimental evidence that a platform-based origination signal is associated with external information acquisition. This is the clearest empirical link in the thesis between platform visibility and the confirmation of attention through search.

These results qualify how Google search volume should be interpreted in the retail-attention literature. Da et al. (2011) establish abnormal search volume as a revealed-attention measure and show that it predicts short-run price pressure and later reversals. The evidence in this thesis highlights a detail that has been overlooked in the literature: if an individual investor types "GPRO stock" in the Google search bar, this automatically means that GoPro's stock was already salient to them. In this respect, search activity cannot be treated as a *stimulus* (Kahneman, 1973) that initiates the attention process. Instead, it is more naturally interpreted as a *response* to a salient stimulus that has already triggered consideration. The origination–confirmation distinction therefore adds a timing interpretation to the standard attention framework: not all attention proxies capture the same stage of the decision process.

Several limitations constrain the scope of the conclusions. First, the attention sample is restricted to Russell 3000 stocks and is therefore tilted toward larger and more visible firms. This matters because retail price-pressure effects are expected to be strongest among smaller and less liquid stocks. It also prevents the thesis from testing whether the origination–confirmation framework applies in the part of the market where retail trading is most likely to be economically important. In these circumstances, the insignificant calendar-time alpha in the attention sample should be interpreted cautiously: it may reflect the larger-cap composition of the sample rather than the absence of an attention-herding mechanism. Second, the strongest quasi-experimental evidence is local. The $300 million Top Movers threshold identifies variation near a specific platform eligibility boundary and among stocks with extreme price movements; it does not establish a global causal effect of Robinhood visibility across the full cross-section. Third, the sample period runs from May 2018 to August 2020 and includes the COVID-19 episode, during which retail participation was unusually elevated. Although the pre-COVID results are broadly consistent with the main evidence, the external validity of the findings for other market regimes remains limited.

Future research could improve on these limitations in many directions. The most direct extension would be to construct reliable Google Trends measures for the full `Robintrack` universe rather than only for Russell 3000 constituents. This would allow the interaction between confirmation signals, small-cap herding, and subsequent reversals to be tested in the segment of the market where retail price pressure is most likely to be economically important. A second extension would combine the intraday resolution of `Robintrack` with higher-frequency attention proxies, if available, to study whether information acquisition precedes Robinhood flows within the trading day[1]. A third extension would test whether

[1] Currently, the Google Trends data and their data access limitations seriously hinder the construction

the origination–confirmation framework survives in settings where platform design differs, such as European retail brokers, options markets, or post-2020 trading data. Finally, I would encourage future research to decompose rigorously the endogenous and exogenous components of attention and herding. Kang (2026) provides an interesting approach derived from physics to model these dynamics.

The broader implication of the thesis is that investor attention should not be treated as a single reduced-form object. The evidence suggests that aggregate retail trading is consistent with a structured attention process: salient market and platform signals originate consideration, search activity provides evidence of active information acquisition, and concentrated participation can be followed by short-run return reversals. This framework helps reconcile why extreme returns, platform visibility, abnormal volume, news, and Google search activity are all related to retail trading, while still playing different empirical roles. As retail investors increasingly encounter markets through interfaces that rank, simplify, and personalize information, the boundary between access to markets and the direction of attention will become increasingly blurred. Understanding how salience becomes information acquisition, how information acquisition becomes coordinated trading, and how coordinated trading feeds back into prices is therefore not a peripheral question from behavioral finance, but a central interrogation for an era in which the investment toolkit of the average retail investor fits in a jeans pocket.

of intraday search measures for a large number of stocks over a long sample period. However, this might change if Google moves forward with their Google Trends API now in alpha testing https://developers.google.com/search/apis/trends. I personally applied for early access but Google did not answer.

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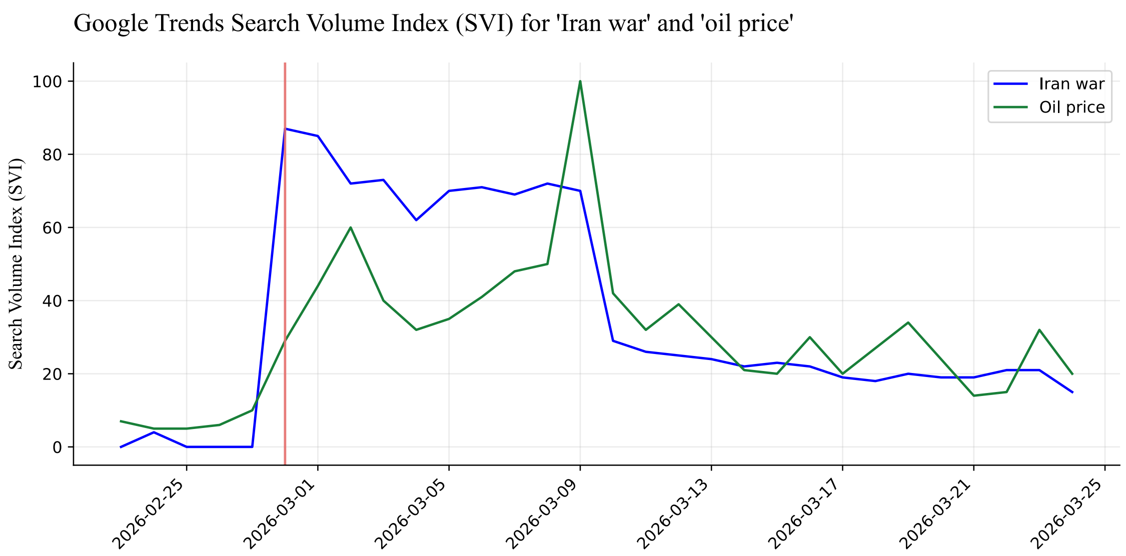
I. Additional Tables and Figures

A.1. Variable Construction Details

A.1.1. Google Trends and the Middle East

To illustrate how raw Google Trends data can be used, I plot in Figure A.1 the search interest for the terms “Iran war” and “oil price” in Switzerland from February 23, 2026 to March 24, 2026. The plot shows that search interest for the two terms spiked on 28 February 2026, when the United States and Israel launched airstrikes against Iranian targets.

Figure A.1. Google Trends search interest for the terms “Iran war” and “oil price” in Switzerland from February 23, 2026 to March 24, 2026. The red vertical line indicates the date of the airstrikes against Iranian targets on February 28, 2026.



A.1.2. RavenPack News Analytics

In this section, I provide additional details on the RavenPack News Analytics dataset, in particular the construction of the three sentiment variables. To build $ess_{i,t}$, $css_{i,t}$, and $anl\_chg_{i,t}$, I follow the methodology outlined in Audrino et al. (2020) and repeated by De Giorgi et al. (2026). I aggregate sentiment scores as weighted sums of the sentiment scores of all news items related to stock i on day t, where the weights are given by the relevance scores of the news items.

Event Sentiment Scores (ESS)

The Event Sentiment Score (ESS) for stock i on day t is originally computed by RavenPack as a score in the interval $[-1, 1]$ with 2 decimal places that represents the news sentiment for a given entity by measuring various proxies sampled from the news—where -1 indicates extremely negative sentiment and 1 indicates extremely positive sentiment. The score is determined by systematically matching stories typically categorized by financial experts as having short-term positive or negative financial or economic impact. The final computation of the score depends on a particular proprietary algorithm that takes into account a significant number of factors. In this case, I simply aggregate the ESS scores of all news items related to stock i on day t as expressed in Equation A.1.

$$ess_{i,t} = \frac{1}{N_{i,t}^{ess}} \sum_{n=1}^{N_{i,t}^{ess}} ess_{n,t} \tag{A.1}$$

Where $N_{i,t}^{ess}$ denotes the number of news items related to stock i on day t, and $ess_{n,t}$ denotes the ESS associated with news item n for stock i on day t.

Composite Sentiment Scores (CSS)

The Composite Sentiment Score (CSS) is also a proprietary RavenPack score—$css_{i,t} \in [-1, 1]$ —stored with up to 2 decimal places. It proxies the news sentiment of a given story by combining various sentiment analysis techniques. The direction of the score is determined by looking at emotionally charged words and phrases and by matching stories typically rated by experts as having short-term positive or negative share price impact. However, since this signal does not account for the relevance of the news items, I follow Audrino et al. (2020) and compute a relevance-weighted CSS as expressed in Equation A.2.

$$css_{i,t} = \frac{\sum_{n=1}^{N_{i,t}^{css}} css_{n,t} \cdot rel_{n,t}}{100 \cdot N_{i,t}^{css}} \tag{A.2}$$

Where $N_{i,t}^{css}$ denotes the number of news items related to stock i on day t for which a CSS score is available, $css_{n,t}$ denotes the CSS associated with news item n for stock i on day t, and $rel_{n,t}$ denotes the relevance score of news item n for stock i on day t.

Analyst Recommendation Changes

This variable is a discrete score computed by RavenPack that captures changes in analyst recommendations. It takes the value of 1 if there is a positive change in the recommendation (e.g., from "hold" to "buy"), -1 if there is a negative change (e.g., from "buy" to "hold"), and 0 if there is no change. I compute the simple average of this score across all news items related to stock i on day t as expressed in Equation A.3.

$$anl\_chg_{i,t} = \frac{\sum_{n=1}^{N_{i,t}^{anl\_chg}} anl\_chg_{n,t}}{100 \cdot N_{i,t}^{anl\_chg}} \tag{A.3}$$

Where $N_{i,t}^{anl\_chg}$ denotes the number of news items related to stock i on day t for which an analyst recommendation change score is available, and $anl\_chg_{n,t}$ denotes the analyst recommendation change score associated with news item n for stock i on day t.

A.2. Differences in Buying and Selling User Flows Regression Results

This section reports the complete regression results for the determinants of Robinhood buy and sell flows. Importantly, the coefficients for the buy-side and sell-side regressions are comparable with caution, since the dependent variable in the sell-side regression is defined as the absolute magnitude of negative user changes, while the dependent variable in the buy-side regression is defined as the positive user changes. Hence, larger values in column (1) correspond to stronger sell-side outflows, while larger values in column (2) correspond to stronger buy-side inflows.

Table A.1. Determinants of Robinhood Buy & Sell Flows

The table reports fixed-effects panel regressions estimated separately for buy-side and sell-side Robinhood user flows. In column (1), the dependent variable is $sellchg_{i,t}$, defined as the absolute magnitude of negative Robinhood user changes, i.e. $sellchg_{i,t} = -userchg_{i,t}$ for observations with $userchg_{i,t} < 0$. Hence, larger values in column (1) correspond to stronger sell-side outflows. In column (2), the dependent variable is $userchg_{i,t}$ restricted to observations with strictly positive user changes, so larger values correspond to stronger buy-side inflows. All regressions include stock and date fixed effects. $asvi_{i,t-1}$ is the lagged measure of abnormal search volume. $top\_20_{i,t-1}$ is an indicator equal to one if the stock's absolute return was ranked in the top 20 on day $t-1$. $abvol_{i,t-1}$ is the logarithm of the ratio of stock market volume on day $t-1$ to the average volume from days $t-21$ to $t-2$. $userchg_{i,t-1}$ is the lagged change in Robinhood users. $num\_news_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$. $num\_news\_relevant_{i,t-1}$ is the lagged number of news articles for stock i on day $t-1$ with RavenPack relevance score of at least 75. $ess_{i,t-1} \in [-1, 1]$ is the lagged Event Sentiment Score. $css_{i,t-1} \in [-1, 1]$ is the lagged relevance-weighted Composite Sentiment Score. $anl\_chg_{i,t-1}$ is the lagged change in analyst recommendations. Robust standard errors, reported in parentheses, are two-way clustered by stock and date. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Dependent variable	
	$sellchg_{i,t}$ Sell-side ($userchg_{i,t} < 0$) (1)	$userchg_{i,t}$ Buy-side ($userchg_{i,t} > 0$) (2)
$top\_20_{i,t-1}$	10.686***	35.639***
	(3.250)	(10.158)
$abvol_{i,t-1}$	10.259***	20.770***
	(2.241)	(7.259)
$userchg_{i,t-1}$	0.035	0.616***
	(0.038)	(0.047)
$asvi_{i,t-1}$	1.259***	−0.651
	(0.172)	(0.699)
$num\_news_{i,t-1}$	−0.0002*	0.0003
	(0.0001)	(0.0003)
$num\_news\_relevant_{i,t-1}$	0.0008	−0.0025
	(0.0006)	(0.0017)
$ess_{i,t-1}$	−4.346*	−17.836***
	(2.361)	(6.480)
$css_{i,t-1}$	7.452	43.554***
	(4.559)	(16.762)
$anl\_chg_{i,t-1}$	−46.976	9.678
	(111.840)	(388.190)
Observations	352,302	385,322
R^2	0.0058	0.3796
Entity FE	Yes	Yes
Date FE	Yes	Yes

A.3. Complete Event-Study Regression Results

This section reports the complete event-study regression results for the timing of the attention process. The main text only reports the coefficients for relative event times from $\tau = -5$ to $\tau = 5$ for brevity, but here I report the full set of coefficients for relative event times from $\tau = -10$ to $\tau = 10$.

Table A.2. Abnormal Search Attention around Extreme Overnight Return Events

The table reports event-time regressions of abnormal Google search intensity around extreme overnight return events. Robust standard errors reported in parentheses are two-way clustered by event and calendar date. Column (1) reports the full event sample. Columns (2) and (3) split the sample according to whether the overnight return is positive or negative. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	$asvi_{i,t}$		
	(1) Overall	(2) Positive Returns	(3) Negative Returns
$\tau = -10$	-0.2100^{***}	-0.2168^{***}	-0.1527^{**}
	(0.0398)	(0.0515)	(0.0766)
$\tau = -9$	-0.2036^{***}	-0.1947^{***}	-0.1623^{**}
	(0.0352)	(0.0475)	(0.0666)
$\tau = -8$	-0.1949^{***}	-0.1954^{***}	-0.1474^{**}
	(0.0313)	(0.0422)	(0.0589)
$\tau = -7$	-0.1868^{***}	-0.1880^{***}	-0.1632^{***}
	(0.0283)	(0.0363)	(0.0526)
$\tau = -6$	-0.1720^{***}	-0.1714^{***}	-0.1406^{***}
	(0.0249)	(0.0300)	(0.0459)
$\tau = -5$	-0.1559^{***}	-0.1495^{***}	-0.1382^{***}
	(0.0198)	(0.0279)	(0.0353)
$\tau = -4$	-0.1273^{***}	-0.1394^{***}	-0.0937^{***}
	(0.0163)	(0.0231)	(0.0286)
$\tau = -3$	-0.1098^{***}	-0.1119^{***}	-0.0883^{***}
	(0.0128)	(0.0172)	(0.0224)
$\tau = -2$	-0.0939^{***}	-0.1030^{***}	-0.0818^{***}
	(0.0122)	(0.0157)	(0.0187)
$\tau = 0$	0.1788^{***}	0.1432^{***}	0.1837^{***}
	(0.0144)	(0.0184)	(0.0193)
$\tau = 1$	0.0463^{***}	0.0397^{*}	0.0252
	(0.0171)	(0.0225)	(0.0226)
$\tau = 2$	−0.0027	−0.0355	−0.0174
	(0.0192)	(0.0272)	(0.0278)
$\tau = 3$	−0.0157	−0.0395	−0.0429
	(0.0239)	(0.0315)	(0.0360)
$\tau = 4$	−0.0018	−0.0322	−0.0396
	(0.0236)	(0.0314)	(0.0430)
$\tau = 5$	0.0205	−0.0043	−0.0150
	(0.0278)	(0.0406)	(0.0486)
$\tau = 6$	0.0289	0.0003	−0.0160
	(0.0342)	(0.0459)	(0.0588)
$\tau = 7$	0.0452	0.0017	0.0047
	(0.0370)	(0.0534)	(0.0637)
$\tau = 8$	0.0425	−0.0125	−0.0051
	(0.0419)	(0.0562)	(0.0748)
$\tau = 9$	0.0627	0.0194	0.0108
	(0.0452)	(0.0630)	(0.0811)
$\tau = 10$	0.0731	0.0293	0.0140
	(0.0493)	(0.0670)	(0.0881)
Observations	131,187	68,670	62,517
R^2	0.0781	0.0843	0.0837
Event FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Reference Date	$\tau = -1$	$\tau = -1$	$\tau = -1$

A.4. Robustness Checks for Regression Discontinuity Analyses

Table A.3. Local-Linear RD Robustness Checks

This table reports local-linear regression-discontinuity estimates around the $300 million market-capitalization cutoff for eligibility for Robinhood's Top Movers list. The dependent variable is either Robinhood intraday user change or Google ASVI. The treatment indicator equals one if market capitalization at the market open exceeds $300 million. All specifications use triangular-kernel weights, which assign greater weight to observations closer to the cutoff. Standard errors, reported in parentheses, are clustered by day. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Bandwidth	Estimate	SE	Observations
Panel A: Robinhood Intraday User Change as Dependent Variable			
$50 million	14.54	(15.42)	2,811
$75 million	21.85*	(11.88)	5,057
$100 million	27.36***	(10.26)	7,834
$125 million	30.24***	(9.12)	11,084
Panel B: Google ASVI as Dependent Variable			
$50 million	0.160**	(0.079)	3,044
$75 million	0.153**	(0.078)	5,488
$100 million	0.146***	(0.077)	8,093
$125 million	0.132***	(0.045)	11,023

A.5. Coefficients for the Top 100 List Regression Discontinuity

This appendix reports the formal regression-discontinuity estimates corresponding to the visual evidence in Figure 9. The purpose of the exercise is to assess whether stocks that are barely inside Robinhood's Top 100 list on day $t-1$ experience higher Robinhood intraday user growth on day t than stocks that are barely outside the list. The design exploits the discontinuity created by the rank-100 boundary. For each day, stocks are ranked by the number of Robinhood users holding them at the close, $users\_close_{i,t-1}$, and the running variable is the lagged centered rank, $100 - Rank_{i,t-1}$.

The estimated specification is:

$$\begin{aligned} userchg_{i,t}^{\text{intra}} = \beta_0 + \beta_1 \mathbf{1}\{Rank_{i,t-1} \leq 100\} + f(100 - Rank_{i,t-1}) \\ + \mathbf{1}\{Rank_{i,t-1} \leq 100\} f(100 - Rank_{i,t-1}) + \varepsilon_{i,t} \end{aligned} \tag{A.4}$$

The coefficient of interest is β_1, which measures the discontinuity in intraday Robinhood user change at the Top 100 boundary. Since the treatment indicator is based on the rank observed on day $t-1$ and the outcome is measured on day t, the specification avoids the mechanical endogeneity that would arise from using the same-day closing rank to explain same-day user changes. The estimates are reported for bandwidths of 15, 20, 25, and 30 ranks around the cutoff and for linear, quadratic, and cubic polynomial specifications.

Standard errors are clustered by day.

Table A.4. Regression Discontinuity in Robinhood Intraday User Change for Top 100 Stocks

This table reports regression-discontinuity estimates around the rank-100 cutoff of Robinhood's Top 100 list. The dependent variable is Robinhood intraday user change on day t, $userchg^{\text{intra}}_{i,t}$. Stocks are ranked on day $t-1$ by the number of Robinhood users holding them at the close, $users\_close_{i,t-1}$. The treatment indicator equals one if stock i is ranked among the 100 most held stocks on day $t-1$ and zero otherwise. The running variable is the lagged centered rank, $100 - Rank_{i,t-1}$, so that the cutoff is zero and positive values correspond to stocks inside the Top 100. Polynomial functions of the running variable are estimated separately on each side of the cutoff, with order $N = 1, 2, 3$. Panel bandwidths vary from 15 to 30 ranks around the cutoff. All specifications use day-level clustered standard errors, reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood Intraday User Change		
	(1)	(2)	(3)
	Panel A: Sample Bandwidth = 15		
$\mathbf{1}\{Rank_{i,t-1} \leq 100\}$	17.11*	27.10*	15.98
	(10.263)	(15.478)	(21.382)
Polynomial order, N	1	2	3
Observations	17,174	17,174	17,174
R^2	0.001	0.001	0.001
	Panel B: Sample Bandwidth = 20		
$\mathbf{1}\{Rank_{i,t-1} \leq 100\}$	16.07*	21.72*	26.40*
	(9.276)	(12.258)	(15.008)
Polynomial order, N	1	2	3
Observations	22,714	22,714	22,714
R^2	0.001	0.001	0.001
	Panel C: Sample Bandwidth = 25		
$\mathbf{1}\{Rank_{i,t-1} \leq 100\}$	15.034*	23.51*	23.51
	(8.206)	(12.831)	(15.837)
Polynomial order, N	1	2	3
Observations	28,254	28,254	28,254
R^2	0.001	0.001	0.001
	Panel D: Sample Bandwidth = 30		
$\mathbf{1}\{Rank_{i,t-1} \leq 100\}$	12.80*	22.10*	24.80*
	(7.427)	(11.785)	(14.819)
Polynomial order, N	1	2	3
Observations	33,794	33,794	33,794
R^2	0.001	0.001	0.001

The estimates in Table A.4 are consistently positive across bandwidths and polynomial specifications. The point estimates imply that stocks barely inside the Top 100 on day $t-1$ experience between approximately 13 and 27 additional Robinhood intraday users on day t, relative to stocks barely outside the list. The estimates are statistically significant at the 10% level in most specifications, although precision is limited and the R^2 values are small, as expected in a local design using high-frequency user-flow outcomes. The evidence is therefore best interpreted as supportive but not definitive: it is consistent with the view that Top 100 visibility generates additional Robinhood buying pressure, but the causal interpretation depends on the local smoothness assumption around the rank cutoff and on the absence of discontinuities in other attention-generating variables, especially returns and absolute returns.

II. Confirmation Signals and the Intensity of Herding Episodes

B.1. Motivation & Estimation Strategy

The distinction between origination and confirmation signals also yields a natural empirical prediction at the stock-day level. If an origination signal brings a stock into retail investors' consideration sets and thereby triggers a herding episode, then the magnitude of that episode should be stronger when the same stock-day also exhibits a confirmation signal in the form of abnormally high Google search intensity. Intuitively, once a stock has become salient enough to trigger herding, the presence of concurrent search activity indicates that investors have indeed been hooked by the origination signals. In that case, one should expect more intense net inflows than during observationally similar herding episodes without a search spike. To evaluate this hypothesis, I estimate the interaction specification in Equation B.1. The dependent variable is the daily Robinhood user change $userchg_{i,t}$, the indicator $rh\_herd_{i,t}$ identifies stock-days classified as herding episodes, and $asvi_{i,t}$ measures contemporaneous abnormal Google search intensity.[1] The coefficient of interest is the interaction term between abnormal search intensity and the herding indicator.

$$\begin{aligned} userchg_{i,t} &= \alpha_i + \delta_t + \beta_1 asvi_{i,t} + \beta_2 rh\_herd_{i,t} \\ &+\beta_3 \left(asvi_{i,t} \times rh\_herd_{i,t}\right) + \theta \mathbf{X}_{i,\tau} + \varepsilon_{i,t} \end{aligned} \tag{B.1}$$

In Equation B.1, α_i and δ_t denote stock and date fixed effects, respectively, and $\mathbf{X}_{i,\tau}$ contains the same controls used in previous regressions to the exception that some variables are not lagged anymore to reflect the contemporaneous nature of the specification. Specifically, $\mathbf{X}_{i,\tau}$ includes: $top\_20_{i,t-1}$, $ab\_vol_{i,t}$, $userchg_{i,t-1}$, $num\_news_{i,t}$, $num\_news\_relevant_{i,t}$, $ess_{i,t}$, $css_{i,t}$, and $anl\_chg_{i,t}$. Standard errors are two-way clustered by stock and date.

The coefficient β_1 captures the association between abnormal search attention and Robinhood user flows on non-herding stock-days. The coefficient β_2 measures the average increment in user flows during herding episodes, conditional on the included controls and fixed effects. Most importantly, β_3 tests whether the attention–flow relation is steeper during herding episodes than outside them.

[1] The use of contemporaneous $asvi_{i,t}$ is deliberate. The purpose of this exercise is not to study whether Google search predicts future herding, but whether heightened search activity coincides with, and is associated with a stronger magnitude of, herding episodes once they occur. In the terminology developed above, the specification therefore targets the role of Google search as a *confirmation signal*.

B.2. Results

Table B.1. Interaction between Abnormal Search Attention and Herding Episodes

This table estimates Equation B.1. The dependent variable is daily Robinhood user change $userchg_{i,t}$. $asvi_{i,t}$ is contemporaneous abnormal Google search intensity. $rh\_herd_{i,t}$ is an indicator equal to one if stock i is in a Robinhood herding episode on day t. The coefficient on the interaction term $asvi_{i,t} \times rh\_herd_{i,t}$ tests whether abnormal search attention is more strongly associated with user inflows during herding episodes than on other stock-days. All specifications include stock and date fixed effects. The table only reports the coefficients of interest. Standard errors, reported in parentheses, are two-way clustered by stock and date. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Robinhood User Change
$asvi_{i,t}$	1.1945***
	(0.2408)
$rh\_herd_{i,t}$	324.7800***
	(72.0570)
$asvi_{i,t} \times rh\_herd_{i,t}$	465.2200***
	(94.5360)
Stock FE	Yes
Date FE	Yes
Observations	933,008
R^2	0.3778

Table B.1 reports the estimates. The coefficient on $rh\_herd_{i,t}$ is large and positive, indicating that user inflows increase sharply on stock-days classified as herding episodes. More importantly, the interaction term between $asvi_{i,t}$ and $rh\_herd_{i,t}$ is positive, economically large, and precisely estimated. The point estimate implies that the association between abnormal search intensity and Robinhood user flows is substantially stronger during herding episodes than outside them. Quantitatively, the slope of $userchg_{i,t}$ with respect to $asvi_{i,t}$ equals $\hat{\beta}_1 = 1.1945$ on non-herding stock-days, but rises to $\hat{\beta}_1 + \hat{\beta}_3 = 466.4145$ when $rh\_herd_{i,t} = 1$. Thus, conditional on the same stock and date effects and the same information set captured by the controls, abnormal search intensity is associated with a much steeper increase in user inflows precisely when retail trading is already concentrated in a herding state.

Overall, Table B.1 supports our postulate. Origination signals such as extreme returns or platform visibility may place a stock into the consideration sets of retail investors and thereby trigger the onset of herding. Conditional on that onset, however, the intensity of the episode is significantly greater when the same stock-day also exhibits abnormally high search activity. In that sense, search (a confirmation signal) does not merely accompany salience; it identifies the subset of salient episodes in which retail attention appears to be more broadly engaged and, consequently, in which user inflows become especially pronounced. Finally, the results presented in this Appendix should be treated carefully due to the mechanical correlation between the dependent variable $userchg_{i,t}$ and the independent variable $rh\_herd_{i,t}$, which may lead to an upward bias in the estimate of β_3.

III. Asset Pricing Consequences in the Attention Sample

C.1. Event-Study around Herding Events

This section reports the event-time asset-pricing results for the attention sample, defined as those observations for which Google Search Volume Index data are available. The empirical specification mirrors the event-study analysis in the main text. Herding events are again identified as the top 0.5% of stocks with positive user change ratios on day 0 and at least 100 Robinhood users on the prior trading day. Relative to the baseline sample, however, this exercise is conducted on a more restricted universe of stocks, since Google search data are available primarily for Russell 3000 constituents. The attention sample therefore contains 2,005 herding events.

Table C.1 shows that the return pattern in the attention sample is directionally similar to the baseline evidence, but substantially weaker in magnitude. The herding day remains associated with a positive and statistically significant abnormal return of 2.51%, and the corresponding BHAR reaches 2.82%. This confirms that herding episodes in the attention sample still occur around salient price movements. However, the event-day price increase is much smaller than in the unrestricted sample, consistent with the fact that the attention sample is tilted toward larger, more liquid stocks. In such stocks, retail order flow is less likely to generate large price dislocations, a point that is consistent with Barber et al. (2022), who find that return reversals after Robinhood herding are most pronounced among smaller firms, and with Da et al. (2011), who show that retail-attention price pressure is strongest where retail trading is more likely to matter for marginal prices.

The postevent evidence also points to attenuation rather than disappearance. Daily abnormal returns are not consistently negative immediately after the herding day, and the first four postevent BHARs remain close to zero. The reversal emerges only after several trading days: BHARs become negative after day 5 and statistically significant from day 7 onward. By day 20, the postevent BHAR equals -1.22%, compared with -3.71% in the baseline sample. The percentage of positive BHARs also falls below one-half throughout the postevent window, reaching 44% by day 20. Thus, the attention sample does not overturn the baseline conclusion; rather, it shows that the return reversal is materially weaker in the larger-stock universe for which search attention can be observed.

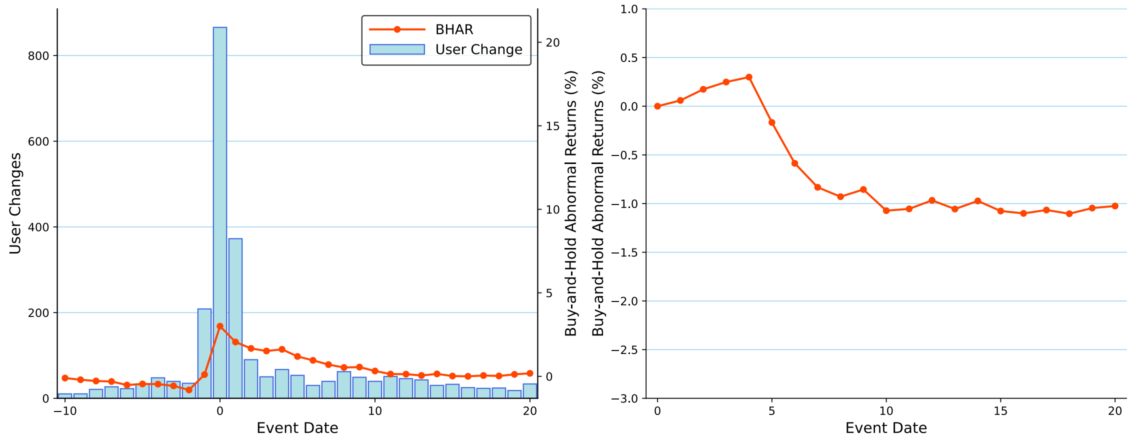
Figure C.1 illustrates the same attenuation visually. The left panel shows that user changes are sharply concentrated on the herding day, confirming that the event definition continues to isolate days of unusually intense Robinhood inflows. The associated BHAR response, however, is modest relative to the baseline sample. The right panel shows that postevent abnormal performance initially increases slightly before declining after the first

Table C.1. Event-Time Abnormal Returns in the Attention Sample

This table reports abnormal returns around Robinhood user herding events in the attention sample. Abnormal returns (AR) are computed as the raw return minus the CRSP value-weighted market return. Abnormal returns are averaged across all events. Buy-and-hold abnormal returns ($BHAR$) are computed as the product of one plus the stock's return through event day t less the product of one plus the market return over the same period. Standard errors are computed by clustering on event day. *% Positive* is the percentage of returns that are positive. Herding events are defined as the top 0.5% of stocks with positive user change ratio on day 0 and a minimum of 100 users on the prior day. The sample contains 2,005 herding events. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Event Day	AR (%)	Std. Err. (%)	% Positive	BHAR (%)	Std. Err. (%)	% Positive
			Preevent			
-10	−0.09	0.13	49%	−0.09	0.13	49%
-9	−0.17	0.14	48%	−0.24	0.19	48%
-8	−0.01	0.16	48%	−0.32	0.22	47%
-7	−0.02	0.16	48%	−0.38	0.27	48%
-6	−0.23	0.14	47%	−0.59 *	0.31	46%
-5	−0.02	0.15	49%	−0.52	0.36	47%
-4	−0.16	0.16	49%	−0.53	0.47	47%
-3	−0.19	0.17	49%	−0.65	0.47	48%
-2	−0.46 **	0.22	49%	−0.97 *	0.51	48%
-1	0.75	0.58	52%	−0.10	0.74	48%
0	2.51 ***	0.66	55%	2.82 ***	1.08	47%
			Postevent			
1	0.04	0.29	48%	0.04	0.29	48%
2	0.00	0.23	47%	0.14	0.48	47%
3	0.05	0.19	49%	0.18	0.53	47%
4	0.11	0.19	48%	0.22	0.50	45%
5	−0.30 *	0.15	47%	−0.21	0.49	46%
6	−0.37 ***	0.14	46%	−0.67	0.44	45%
7	−0.23	0.14	48%	−0.95 **	0.42	45%
8	0.06	0.20	48%	−0.99 **	0.43	45%
9	0.08	0.14	47%	−0.94 **	0.45	46%
10	−0.17	0.14	45%	−1.21 ***	0.44	46%
11	−0.07	0.15	44%	−1.24 **	0.52	44%
12	0.11	0.12	49%	−1.12 **	0.55	44%
13	−0.05	0.12	48%	−1.16 **	0.55	44%
14	0.10	0.13	48%	−1.11 **	0.55	44%
15	−0.13	0.11	47%	−1.22 **	0.56	43%
16	−0.04	0.12	49%	−1.22 **	0.55	43%
17	−0.01	0.11	49%	−1.24 **	0.58	44%
18	−0.05	0.11	48%	−1.32 **	0.56	44%
19	0.16	0.17	48%	−1.17 **	0.59	45%
20	−0.03	0.13	49%	−1.22 **	0.57	44%

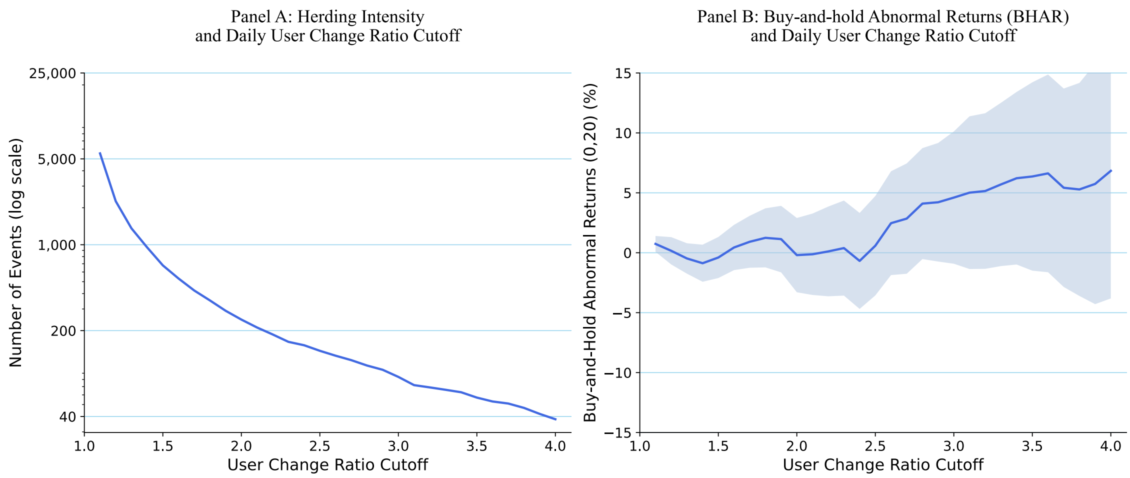
Figure C.1. Returns around Herding Events in the Attention Sample. The panel on the left depicts mean buy-and-hold abnormal returns (BHARs) and Robinhood user changes from 10 days before to 20 days after herding events in the attention sample. The red line represents BHARs, whereas the blue bars represent user changes. The panel on the right displays postevent mean BHARs starting from day 0. Herding events are defined as the top 0.5% of stocks with positive user change ratio on day 0 and a minimum of 100 users on the prior day.



few trading days and stabilizing around a negative level close to one percent. This delayed and smaller reversal is consistent with the interpretation that attention-induced buying still contains a transitory component in the attention sample, but that the price impact of such buying is less pronounced when the analysis is restricted to larger firms.

Figure C.2 further underscores the limits of the attention sample. Panel A shows the same mechanical trade-off as in the baseline sample: raising the user change ratio cutoff sharply reduces the number of identified herding episodes. Panel B, however, does not display the clear monotonic relation between herding intensity and subsequent underperformance observed in the unrestricted sample. The estimates become more erratic at higher cutoffs, where the number of events is small, and the magnitude of the reversal varies substantially across thresholds. This instability is informative. It suggests that, within the larger-stock universe for which Google search data are available, variation in Robinhood herding intensity is not sufficient to recover the strong intensity–reversal relation documented in the baseline sample.

Figure C.2. Price Reversal & Herding Intensity in the Attention Sample. The panels display how herding intensity and 20-day buy-and-hold abnormal returns (BHARs) vary with the daily user change ratio cutoff in the attention sample. Panel A plots the number of herding episodes on a log scale against the user change ratio cutoff. Panel B plots 20-day BHARs (%) against the user change ratio cutoff. The daily user change ratio cutoff varies from 1.1 to 4.0.



IV. Resources & Aids Declaration

D.1. Code

The code used for data processing and analysis in this thesis is available on my `Github` repository at https://github.com/YVYX04/Bachelor_Thesis. The repository is not completely exhaustive, but it contains the main scripts for data cleaning, variable construction, and analyses. The code is written in `Python 3.14.3`, and it is organized into folders corresponding to different stages of the research process. I also relied on various open-source libraries, including `pandas`, `numpy`, `statsmodels`, and `matplotlib` for data manipulation, statistical analysis, and visualization. Finally, I used LaTeX for document preparation.

D.2. Directory of Auxiliary Aids

In addition to the code and software mentioned above, I used the following resources and aids during the research and writing process.

Table D.1. Directory of Auxiliary Aids

Resource/Aid	Purpose
`ChatGPT 5.5`	Rewording and grammar checks. Importantly, it was not used to rephrase substantial content or generate original ideas. However, it assisted in providing mutliple feedback rounds on the overall structure and clarity of the text.
`Claude Code`	Used for code debugging and troubleshooting. I used it to identify and fix specific coding errors, or implement code blueprints. I did not rely on it for writing original code.
`GitHub Copilot`	Used for code suggestions and autocompletion.
`Jupyter Notebooks`	Used for interactive data analysis and visualization.
`Python 3.14.3`	Used for data manipulation, statistical analysis, and modeling.
`Visual Studio Code`	Used for code editing and development. LaTeX code was also written and compiled using this IDE.
`HSG Swisscovery`	Used for accessing academic journals and books.
`Google Scholar`	Used for literature search and review.
`GitHub`	Used for version control and sharing code.
`Peers and Advisors`	Used for feedback, discussion, and guidance throughout the research process. In particular, Prof. Dr. Xing Huang, as a co-author of Barber et al. (2022), provided insights on implementation details used in their research.

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Declaration of Authorship

I hereby declare,

- that I have written this thesis independently,
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been declared;
- that I have handled both input and output responsibly when using AI. I confirm that I have therefore only relied in public data or data released with consent and that I have checked, declared and comprehensibly referenced all results and/or other forms of AI assistance in the required form and that I am aware that I am responsible if incorrect content, violations of data protection law, copyright law or scientific misconduct (e.g. plagiarism) have also occurred unintentionally;
- that I have mentioned all sources used and cited them correctly according to established academic citation rules;
- that I have acquired all immaterial rights to any materials I may have used, such as images or graphics, or that these materials were created by me;
- that the topic, the thesis or parts of it have not already been the object of any work or examination of another course, unless this has been expressly agreed with the faculty member in advance and is stated as such in the thesis;
- that I am aware of the legal provisions regarding the publication and dissemination of parts or the entire thesis and that I comply with them accordingly;
- that I am aware that my thesis can be electronically checked for plagiarism and for third-party authorship of human or technical origin and that I hereby grant the University of St. Gallen the copyright according to the Examination Regulations as far as it is necessary for the administrative actions;
- that I am aware that the University will prosecute a violation of this Declaration of Authorship and that disciplinary as well as criminal consequences may result, which may lead to expulsion from the University or to the withdrawal of my title.

By submitting this thesis, I confirm through my conclusive action that I am submitting the Declaration of Authorship, that I have read and understood it, and that it is true.

Yvan Richard
St. Gallen, May 17, 2026

Signature: Richard.